Bellman-sufficient Information Complexity

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Abstract

We develop Bellman-sufficient information complexity, a representation-level framework for the information-theoretic minimax analysis of sequential decision making. The theory covers interactive environments unfolding over long streams of experience and benchmarks all nonanticipating algorithms. A Bellman-sufficient state closes the controlled recursion, while an index $Y = \chi(\Omega)$ identifies the decision-relevant information. Upper bounds arise from a log-penalized Bellman program, and lower bounds from a Bellman–Fano comparison along a reference trajectory. When the two sides exhibit matching information growth at a common localization scale, they yield an information–risk sandwich. Within this framework, UCB, E2D, and AMS/EBO arise through calibration, one-step offsets, and robust belief optimization, respectively.

As a major application, we give a negative resolution to a canonical, widely studied form of the GP–UCB minimax-optimality question. For every $0 < \alpha < 1/4$, there exists a single bounded continuous kernel whose minimax regret is $\Theta(T^{1-\alpha})$ along an infinite sequence of horizons. On the same problem, a finite-marginal action-index AIR Bellman policy attains this order, whereas both an anytime maximal-information GP–UCB rule and the unit-ball specialization of the original RKHS GP–UCB exploration schedule incur linear regret. This separates realized information acquisition from the cost of uniform optimism and explains why localization can be essential within the Bellman recursion. A reproducible finite-scale experiment illustrates the predicted cloud-exploration mechanism for both headline GP–UCB calibrations.

Contents

1 Introduction

Information-theoretic minimax theory asks how much risk remains when an experiment can reveal only limited information. In classical noninteractive estimation the experiment is fixed before the data are observed, and sharp rates are obtained by matching an upper information-risk construction with a lower entropy calculation: KL information measures what the experiment can distinguish, while local prior mass or packing entropy measures how many alternatives remain statistically indistinguishable. This is the perspective behind the information-risk upper and lower bounds of Zhang (2006), the entropy characterization of minimax rates in Yang and Barron (1999), and the classical Fano–Assouad method (Yu, 1997). In interactive decision making and dynamic control, the learner's actions change the experiment itself. The learner chooses actions by a nonanticipating policy, so the experiment itself is adapted to the past. The main challenge is therefore that decisions, observations, and information accumulation are not separate objects; they are produced jointly by a single controlled stochastic process.

Algorithmic Fano's method is one way to respect this adaptivity. It compares the real history with an algorithm-dependent reference, or "ghost," history rather than with a fixed nonadaptive sampling scheme. This viewpoint appears in recent interactive Fano frameworks for interactive decision making (Chen et al., 2024) and in algorithmic upper and lower bounds for representation learning (Li and Xu, 2026; Xu, 2026). It is also closely related to the decision-estimation coefficient (DEC) approach of Foster et al. (2021) and its constrained or localized refinements (Foster et al., 2023), as well as to the earlier information-ratio approach (Russo and Van Roy, 2014; Lattimore and György, 2021). One central difficulty is that a sharp lower bound may not collapse a T-round adaptive trajectory into a one-round comparison too early. Without proper localization, such a collapse can erase the geometry of the evolving posterior, confidence set, or reference state, precisely the geometry on which a matching upper bound may rely.

Kernel bandits provide a major application of this principle. GP–UCB, introduced by Srinivas et al. (2010), is an influential algorithm for kernelized bandit optimization: the original paper received the ICML 2020 Test of Time Award.[1] Whether its suboptimal regret bounds reflect the analysis or the algorithm has been posed explicitly as a COLT open problem and studied extensively since then (Vakili et al., 2021; Whitehouse et al., 2023; Wang and Zhang, 2026). The classical GP–UCB analysis combines a confidence coefficient with maximal information gain; the modern anytime calibration in Chowdhury and Gopalan (2017) has the same structural dependence. Our construction shows that maximal information gain can affect the sampling rule itself, rather than merely loosen its final analysis: individually negligible directions inflate the global confidence

[1] https://icml.cc/virtual/2020/test-of-time/7965

multiplier and induce unnecessary exploration. The geometric starting point is the weighted-basis hub–cloud construction of Xu (2026); the present argument reverses its amplitude ordering and embeds it in a sequential RKHS bandit, making the cloud negligible for minimax regret while retaining its effect on the GP–UCB exploration multiplier. This yields a full-RKHS polynomial separation, first for finite kernels and then for one fixed bounded continuous kernel on a compact countable action space. The separation applies both to a canonical anytime maximal-information rule and to the unit-ball specialization of the original RKHS GP–UCB exploration schedule. On the same fixed construction, an epochwise finite-marginal action-index AIR Bellman policy recovers the minimax order under a predetermined truncation of directions whose total decision value is small. The fixed Matérn result of Wang and Zhang (2026) gives a complementary separation for the polynomial-effective-optimism schedules covered by that theorem. These are negative resolutions for precise and commonly studied forms of GP–UCB optimality; the broader question for arbitrary localized or data-dependent tuning remains open.

The broader motivation is that existing information-theoretic minimax theory is not yet well suited to dynamic planning and reinforcement learning across episodes (Barto, Sutton, and Watkins, 1989; Kaelbling, 1993; Russell and Norvig, 1995; Littman, 1996; Silver and Sutton, 2025). The present paper suggests that closing gaps in episodic interactive decision making and developing theory for across-episode reinforcement learning share a common missing diagnostic: Bellman recursion and dynamic programming should appear explicitly on both the upper- and lower-bound sides. We frame this perspective within a unified sequential decision-making framework.

The thesis of this paper is that the intrinsic object for sequential decision making is an indexed Bellman-sufficient representation. The frequentist environment space Ω, together with the unrestricted nonanticipating benchmark, identifies the primitive sequential decision-making objective. A Bellman-sufficient state representation turns the problem into a dynamic program over a sufficient state: a sufficient statistic, model posterior, frequentist estimator, or other representation that summarizes the environmental and historical information needed to close the Bellman recursion. The index $Y = \chi(\Omega)$ specifies the information target whose acquisition is charged. It may be the optimal action, an optimal policy, a value object, an active finite marginal, or the full environment.

The same formalism also explains why information potentials are not limited to statistical bandits. In planning, search, or reasoning, the latent instance may be a world model, problem instance, proof environment, or task specification, while the index may be a plan, answer, policy, value function, or proof object. Whenever the process admits a Bellman-sufficient state and a maintained marginal on that index, the logarithmic mass potential is again the generic information potential. This paper keeps most of the technical development in statistical bandits and reinforcement learning, where the upper and lower bounds can be stated cleanly.

When a sufficient state representation closes the Bellman recursion with a smaller information index, the learner need not pay for estimating irrelevant features of the model. This is the sense in which learning is a special dynamic program: unlike ordinary computational dynamic programming, where the value function is problem-specific, the statistical potential is generically logarithmic in the index belief. The exact log-penalized Bellman program is the upper object that can match the Bellman–Fano lower comparison; popular algorithms including UCB, E2D, and AMS/EBO are tractable relaxations of this same dynamic information principle.

The basic identity on the upper-bound side is formulated at a fixed frequentist truth and expressed through Bellman recursion and the chosen information index. Fix a truth $\omega^\star$, let $y^\star = \chi(\omega^\star)$, and let q_t be the retained reference marginal on the index. The specialized logarithmic potential

$$\phi_t(s; y^\star) = \gamma[-\log q_t(y^\star)]$$

turns the one-step fixed-truth log gain into a Bellman information potential. The corresponding indexed Algorithmic Information Ratio (AIR) bracket is

$$\ell_{\omega^\star}(S_t, p_t) + \mathbb{E}_{\omega^\star}[\phi_{t+1}(S_{t+1}; y^\star) \mid S_t, p_t] - \phi_t(S_t; y^\star),$$

and its sum telescopes to a regret bound of order $\gamma \log(1/q_1(y^\star))$ plus the accumulated bracket errors (Xu and Zeevi, 2025). Bayesian posterior averaging of the same coordinate identity gives, when the initial packet is deterministic or independent of the index, the chain rule

$$I_\mu(Y; H_T) = \mathbb{E}_{H'_T} \sum_{t=1}^{T} \mathcal{I}_\chi(S'_t, p'_t),$$

where S'_t is the reference posterior state and p'_t is the algorithm's action distribution at that reference history. The key algorithmic principle is that the logarithmic mass penalty should be propagated through a controlled Bellman recursion.

The lower-bound side is organized by the Bellman–Fano method. The index specifies which uncertainty is charged, and the reference history specifies which low-regret ghost trajectories are counted. The algorithm-specific quantile theorem, Theorem 6.1, is an application of the interactive Fano method (Chen et al., 2024) to Bellman-sufficient representations; Theorem 6.2 gives its algorithm-uniform Bellman recursion. In its simplest algorithm-specific form one obtains

$$\text{if} \qquad p_r^{\mathsf{Alg}}(\mu, \chi) \le \frac{1}{2} \quad \text{and} \quad T\bar{C}_\chi^{\mathsf{Alg}}(\mu) \le \mathrm{kl}\left(\frac{1}{2}, p_r^{\mathsf{Alg}}(\mu, \chi)\right), \qquad \text{then} \qquad \mathbb{E}_{\Omega\sim\mu}\mathbb{E}_{\mathbb{P}_\Omega^{\mathsf{Alg}}} L_\Omega(H_T) \ge \frac{Tr}{2}.$$

Here r is an average-regret threshold, p_r^{Alg} is the posterior-reference ghost probability of average regret at most r, and $\bar{C}_\chi^{\mathsf{Alg}} = T^{-1}\mathbb{E}\sum_t \mathcal{I}_\chi(S'_t, p'_t)$ is average indexed information. The critical balance is

$$T\bar{C}_\chi \asymp \log \frac{1}{p_r}.$$

When the ghost-good mass is roughly $\exp(-d_{\mathrm{eff}})$, the admissible total information is d_{eff}-level. This is the Fano/local-entropy radius, not the constant two-point radius. Theorem 6.2 converts the algorithmic lower-bound approach into a frequentist minimax statement. The Bellman–Fano method may use the exact Bellman-recursion values for information capacity and ghost mass, or valid supersolutions that upper-bound those exact values.

Theorem 3.8 states the resulting minimax information-risk sandwich. After a state, index, reference update, and calibration mechanism have been fixed, the upper side pays a log-penalized Bellman value with coordinate cost $\log(1/q_1(\chi(\omega)))$; the lower side pays a Bellman–Fano ghost entropy $\log(1/p_r)$. The central comparison is therefore the objective ratio between these two logarithmic quantities. When the ghost entropy has the same order as the upper code length, and the Bellman upper value has regular growth in its information budget, the upper and lower regret bounds match at the same radius. This is the interactive analogue of noninteractive information-risk matching: fixed-design local prior mass is replaced by ghost-good mass along an adaptive reference history, and fixed-sample KL is replaced by a controlled Bellman information telescope. We use "Bellman-sufficient information complexity" for this compatible state/index accounting. It becomes a matching complexity statement only after the upper and lower bounds are separately established and shown to agree up to the stated constants or logarithmic factors.

The word "Bellman" in the title is deliberate. The exact value of an interactive problem is a dynamic program, following Bellman's principle of optimality (Bellman, 1957). An upper bound is a Bellman supersolution, or admissible relaxation in the sense of Rakhlin et al. (2012): it lies above the dynamic value and gives an algorithm. A lower bound is an exact Bellman–Fano value comparison, or a computable bound obtained by upper-bounding the exact information-capacity and ghost-good-mass values. The canonical information-theoretic upper algorithm is therefore the AIR information-potential Bellman program, together with its model-index specialization, MAIR: it keeps the continuation value on the indexed state and optimizes the Bellman bracket before any one-step relaxation is imposed. UCB (Auer et al., 2002; Abbasi-Yadkori et al., 2011; Srinivas et al., 2010), E2D (Foster et al., 2021, 2023), and AMS/EBO (Lattimore and György, 2021; Foster et al., 2022; Xu and Zeevi, 2025; Liu et al., 2025, 2026) are useful because they control, relax, or robustify this Bellman bracket; they should not be confused with the exact Bellman program itself.

1.1 Main contributions.

1. We formulate *sequential decision making with Bellman-sufficient representations*, a formal representation-level framework for reinforcement learning, optimal control, and interactive decision making. In this framework, one studies the unrestricted nonanticipating benchmark through Bellman recursion, while a chosen index $Y = \chi(\Omega)$ specifies the decision-relevant information whose acquisition is charged. The state must close feasible actions, fixed-truth prediction, loss evaluation, and updating; when posterior-reference information is used, it must also close posterior predictive and conditional-index laws. Several bandit and reinforcement learning examples illustrate the framework with states given by sufficient statistics, model posteriors, frequentist estimators, or other Bellman-sufficient representations.

2. We prove the indexed posterior-reference chain rule and the generic indexed AIR regret identity. AIR is recovered by taking the decision index $Y = A^\star(\Omega)$, while MAIR is recovered by taking the model or environment index $Y = \Omega$. We then record simplified statements and alternative proofs, show how the gradient bracket unifies different update rules, and explain the role of Danskin's theorem in the original AIR proof of Xu and Zeevi (2025).

3. We isolate the main information-complexity sandwich. The upper value pays an initial coordinate code length, while the Bellman–Fano lower value is governed by ghost entropy. We prove entropy-gap and regret-gap comparison statements under regularity conditions.

4. We give a unified "one identity, four algorithm families" account centered on a log-penalized Bellman upper theorem. The first family is the exact logarithmic information-potential Bellman program; this is the tightest upper object that can match the Bellman–Fano lower value. UCB methods control the fixed-truth log-potential bracket through self-normalized calibration and optimism. E2D methods optimize a one-step offset obtained by replacing the continuation value by a local statistical separation penalty. AMS/EBO methods optimize KL-regularized robust belief relaxations, equivalently dual log-partition relaxations, of the same indexed AIR bracket.

5. We formulate the algorithmic quantile theorem and the Bellman–Fano method as minimax lower bounds uniform over algorithms. The lower-bound argument uses the same stepwise indexed information, state representation, and Bellman recursion to control both information gain and ghost mass. Bandit examples show that this approach recovers matching lower bounds.

6. We organize the examples as information-complexity sandwich tests. For kernel bandits we prove a full-RKHS separation: on a bounded diagonal-kernel family, maximal-information-calibrated GP–UCB incurs $\Omega(T)$ regret while the minimax value is $\Theta(T^{1-\alpha})$, $0 < \alpha < 1/4$. A

gluing argument realizes the gap along an infinite subsequence for one fixed bounded continuous kernel on a compact countable domain, where a single epochwise finite-marginal action-index AIR Bellman policy attains the minimax order. The GP–UCB lower bound covers both a canonical modern anytime maximal-information calibration and the unit-ball exploration schedule of the influential original RKHS GP–UCB algorithm. This gives a negative answer to a widely studied form of the GP–UCB minimax-optimality question, while leaving arbitrary localized or data-dependent tuning open. A reproducible finite-scale experiment compares these two calibrations, the finite-horizon prototype of the anytime calibration, and localized AIR/AMS under a common truth. We then relate this mechanism to the fixed-kernel Matérn lower bound under the scaling and uniform-confidence assumptions of Wang and Zhang (2026). Gaussian multi-armed bandits give a finite-index minimax-rate matching example, and hypercube linear bandits give a high-dimensional indexed lower bound with standard upper bounds matching up to logarithmic factors.

1.2 Organization.

Section 2 defines sequential decision making, information indices, and Bellman-sufficient representations. Section 3 states the information-complexity sandwich and the entropy/regret comparison principles. Section 4 gives the indexed information telescope and the exact AIR/MAIR regret identities. Section 5 gives the log-penalized Bellman upper theorem and upper algorithmic relaxations including UCB families, E2D, and AMS/EBO. Section 6 gives the quantile lower theorem and the Bellman–Fano method. Section 7 develops the finite-marginal AIR bound and the GP–UCB separation for kernel bandits, presents a short numerical illustration, and then summarizes the Gaussian multi-armed-bandit and hypercube linear-bandit minimax-rate examples. Supplementary state representations, technical comparisons, complete examples, and all proofs are collected in the appendices.

2 Sequential decision making with Bellman-sufficient representations

2.1 Sequential decision making and Bellman state compression

The primitive object is sequential decision making over histories. This viewpoint encompasses bandits in rich environments (Lattimore and Szepesvári, 2020), episodic and sequential reinforcement learning (Barto, Sutton, and Watkins, 1989; Littman, 1996; Sutton and Barto, 2018), as well as planning, search, reasoning, and optimal control (Silver and Sutton, 2025; Cormen et al., 2001; Bertsekas, 2017). The central point is simple: from basic online-learning models to dynamic planning and reasoning models, future losses, interactions, and observations generally depend on the history (Cesa-Bianchi and Lugosi, 2006; Recht, 2019; Chen et al., 2021; Hazan et al., 2025). Our sequential decision-making model is designed to capture this dependence. Both the terminology of sequential decision making and the goal of developing a formal model follow the long tradition of Barto, Sutton, and Watkins (1989); Kaelbling (1993); Russell and Norvig (1995); Littman (1996). This formulation is particularly close to Section 1.2 of Littman (1996), which contrasts episodic and sequential environments, describes episodic environments as a degenerate type of sequential environment, and emphasizes the latter as the more general setting in which an agent makes a sequence of interrelated decisions without necessarily being reset to a starting state.

A formal model of the sequential decision-making problem. There is an environment class Ω, and performance is required for every fixed truth $\omega^\star \in \Omega$. A prior μ is introduced only when we form a Bayes value, a Yao lower bound, a posterior-reference trajectory, or an algorithmic belief. The primitive history used in the paper is the full pre-action history, denoted H_{t-1}. To avoid separate notation for contexts, physical states, public randomization, and protocol variables, we regard observations as packets. There may be an initial packet O_0 before the first decision, and recursively

$$H_0 = O_0, \qquad H_t = (H_{t-1}, A_t, O_t).$$

The packet O_t contains whatever is publicly revealed after the round-t action and before the next decision: feedback, the next context, the next physical state, the next stage marker, public randomization, or other protocol information. The learner's decision rule is not part of the history; it is the policy being optimized. Private randomization may be represented by the decision kernel, or conditioned on when analyzing a fixed randomized implementation. Whenever we use an unconditional posterior-reference chain rule or a Bellman–Fano theorem below, we assume explicitly that O_0 is deterministic or independent of the chosen index Y. If O_0 is informative about Y, the stepwise identities are conditional on O_0 and the unconditional information budget acquires the initial term $I(Y; O_0)$. Fixed-truth Bellman recursions do not require this convention. Unless a theorem states otherwise, displayed Bellman values at s_1 and Bellman–Fano statements are conditional on a fixed initial packet and hence a fixed initial state. If S_1 is random, one applies the result conditionally and averages the entire conditional bound. In particular, a random-state ghost mass must be averaged before taking its negative logarithm; it is not generally valid to average the conditional ghost entropies. Independence of O_0 from Y alone does not make S_1 deterministic.

At time t, after $h \in \mathcal{H}_{t-1}$, the learner may choose an action in a measurable set $\mathcal{A}_t(h)$. An unrestricted nonanticipating algorithm is a sequence of kernels

$$\pi_t^{\mathsf{Alg}}(\cdot \mid h) \in \Delta(\mathcal{A}_t(h)), \qquad h \in \mathcal{H}_{t-1},$$

with deterministic algorithms included as degenerate kernels. Let $\mathfrak{A}_T^{\mathrm{na}}$ denote the class of all such algorithms, and write

$$p_t = \pi_t^{\mathsf{Alg}}(\cdot \mid H_{t-1}), \qquad A_t \sim p_t.$$

The most general one-step primitives allow the environment or an adaptive adversary to depend on the announced mixed action as a current control:

$$O_t \sim P_{\omega^\star,t}(\cdot \mid H_{t-1}, p_t, A_t), \qquad \ell_t(\omega^\star, H_{t-1}, p_t, A_t). \tag{1}$$

When the law and loss depend only on the realized action, as in ordinary stochastic bandits and Markov decision processes, we suppress the argument p_t and write $P_{\omega,t}(\cdot \mid H_{t-1}, A_t)$ and $\ell_t(\omega, H_{t-1}, A_t)$. The increment may be signed, provided the cumulative target used in a lower-bound theorem is integrable and has the stated lower bound, usually $L_\omega(H_T) \geq 0$. The Bellman clock t is abstract. In episodic reinforcement learning, it may represent either a macro-episode, with the action given by a policy governing all within-episode stages, or a micro-time pair consisting of an outer episode index and an inner-stage index.

The exact frequentist objective in this paper is

$$\mathfrak{R}_T^\star(\Omega) := \inf_{\mathsf{Alg} \in \mathfrak{A}_T^{\mathrm{na}}} \sup_{\omega \in \Omega} \mathbb{E}_\omega^{\mathsf{Alg}} \sum_{t=1}^{T} \ell_t(\omega, H_{t-1}, p_t, A_t).$$

This is the reinforcement-learning and interactive-decision-making problem in its general sequential risk-minimization form: choose a nonanticipating algorithm whose Bellman risk is small uniformly over environments.

Bellman state and Bellman sufficiency. Our theory centers on the specification of an analytical Bellman state for sequential decision making. This object should not be confused with the physical state often used in Markov decision process models. A compressed Bellman state is a dynamic sufficient statistic for decision making,

$$S_t = \phi_t(H_{t-1}).$$

Because the observation packets already contain contexts, physical states, public randomization, and other pre-action variables, no separate raw-history notation is needed. If an implementation retains an additional memory object, that object is either a measurable function of the history, included in the observation packets, or conditioned on in the analysis. Such a compression may be used in Bellman recursions only if the available actions, observation law, loss, and update are measurable with respect to the retained state and the current control. Thus there must exist an action set $\mathcal{A}_t(s)$, a kernel $P_{\omega,t,s,p,a}$, and a loss $\ell_{\omega,t}(s,p,a)$ such that, whenever $S_t(h) = s$,

$$\begin{aligned} \mathcal{A}_t(h) &= \mathcal{A}_t(s), \\ P_{\omega,t}(\cdot \mid h, p, a) &= P_{\omega,t,s,p,a}, \\ \ell_t(\omega, h, p, a) &= \ell_{\omega,t}(s,p,a). \end{aligned}$$

When the environment does not observe the announced mixed action, we suppress p and write $P_{\omega,t,s,a}$ and $\ell_{\omega,t}(s,a)$. There must also be an update map or controlled transition, written in the general notation as

$$S_{t+1} = \tau_t(S_t, p_t, A_t, O_t),$$

and shortened to $\tau_t(S_t, A_t, O_t)$ when p_t is irrelevant. If any of these conditions fails, the retained state is incomplete and must be augmented; see Definition 2.1 for the formal statement. The full history $S_t = H_{t-1}$ always satisfies the fixed-truth Bellman closure, because all current contexts, physical states, public randomization, and other pre-action variables have been included in the observation-packet history. As illustrated in Section A.2, a full environment posterior together with the current time/context/physical state also provides a fallback sufficient state for broad classes of Bayesian reference experiments.

The notation $P_{\omega,t,s,a}$ and $\ell_{\omega,t}(s,a)$ means only that all observed variables needed to determine the next observation law and the loss have been included in the Bellman state s. In applications this state may include an estimator or posterior object, the relevant physical state, an adversary state, or a relaxation variable. It does not mean that a bandit arm or an MDP transition kernel literally depends on an estimator alone. Once a Bellman-sufficient state is fixed, all main Bellman operators optimize over $p \in \Delta(\mathcal{A}_t(s))$. Throughout, the mixed-action loss is the action average

$$\ell_{\omega,t}(s,p) := \mathbb{E}_{a\sim p}\ell_{\omega,t}(s,p,a),$$

with the argument p suppressed from the integrand when the environment does not observe the announced distribution. In stationary examples, or when the stage t is included in s, we sometimes write $\mathcal{A}(s)$ for $\mathcal{A}_t(s)$.

Reference dynamic program and minimax lower bound. On a sufficient Bellman state, with update $S_{t+1} = \tau_t(S_t, p_t, A_t, O_t)$, or with p_t suppressed when irrelevant, the Bayes/reference dynamic program induced by a posterior b_s is

$$V_{T+1}(s) = 0, \qquad V_t(s) = \inf_{p\in\Delta(\mathcal{A}_t(s))} \left\{\ell(s,p) + \mathbb{E}_{a\sim p,\, o\sim P_{s,p,a}} V_{t+1}(\tau_t(s,p,a,o))\right\}, \tag{2}$$

where

$$\ell(s,p) = \mathbb{E}_{a\sim p} \int \ell_\omega(s,p,a) b_s(d\omega), \qquad P_{s,p,a} = \int P_{\omega,s,p,a} b_s(d\omega).$$

Equation (2) is not the definition of the frequentist problem; it is the Bellman recursion induced by a chosen reference representation. Minimax lower bounds are connected to such reference recursions through Yao's principle,

$$\mathfrak{R}_T^\star(\Omega) \geq \sup_{\mu\in\Delta(\Omega)} \inf_{\mathsf{Alg}\in\mathfrak{A}_T^{\mathrm{na}}} \mathbb{E}_{\Omega\sim\mu}\mathbb{E}_{\mathbb{P}_\Omega^{\mathsf{Alg}}} \sum_{t=1}^T \ell_t(\Omega, H_{t-1}, p_t, A_t).$$

This offers a lower-bound perspective on information complexity at the level of broad environment classes, rather than only for particular classes.

Further remarks on adversarial and estimated losses, dynamic-policy comparators, and across-episode scope are collected in Appendix A.

2.2 Information index, conditional loss, and entropy accounting

The latent environment space is Ω, but the information charged by the upper and lower bounds may be a coarser coordinate. An information index is a measurable map

$$Y = \chi(\Omega) \in \mathcal{Y}.$$

The index is chosen by the analyst and should reflect the decision-relevant object: an optimal arm, an optimal policy, a value function, a finite active marginal, or, when no compression is justified, the full environment. The fixed-truth Bellman recursion still uses $\omega^\star$. The index enters through the logarithmic potential, the posterior/reference information accounting, and the lower ghost event.

The loss need not be a function of Y alone. For Bayes or Fano calculations we therefore use the conditional index loss, assuming the relevant regular conditional laws exist. In the general convention with a current mixed-action control,

$$\ell_t^\chi(y,h,p,a) := \mathbb{E}_\mu[\ell_t(\Omega,h,p,a) \mid \chi(\Omega) = y, H_{t-1} = h]\,. \tag{3}$$

When the one-step primitives do not depend on p, the argument is suppressed. When the retained state is an exact posterior-indexed lift in Definition 2.1, this conditional loss and the corresponding conditional predictive law are state-measurable and may be written as $\ell_{t,y}^\chi(s,p,a)$ and $P_{t,s,p,a}^y$. These conditional-index objects are not required for ordinary fixed-truth Bellman control; they are required for indexed posterior averaging and for the Bellman–Fano lower-bound method.

We use the cumulative and average losses, with the same suppression convention,

$$L_\omega(H_T) = \sum_{t=1}^T \ell_t(\omega, H_{t-1}, p_t, A_t), \qquad \bar{L}_\omega(H_T) = T^{-1} L_\omega(H_T),$$

and

$$L_\chi(y,H_T) = \sum_{t=1}^T \ell_t^\chi(y, H_{t-1}, p_t, A_t), \qquad \bar{L}_\chi(y,H_T) = T^{-1} L_\chi(y,H_T).$$

Under the Bayesian mixture generated by $\Omega \sim \mu$,

$$\mathbb{E}_{\Omega\sim\mu, H_T\sim\mathbb{P}_\Omega^{\mathsf{Alg}}} L_\Omega(H_T) = \mathbb{E}_{Y,H_T} L_\chi(Y,H_T). \tag{4}$$

For the upper identities, the increments may be any integrable costs for which the displayed expectations exist. For the quantile lower bounds, the object that must be nonnegative is the cumulative indexed loss, not necessarily each one-step increment. Throughout the clean lower-bound statements we assume $L_\chi(Y, H_T) \geq 0$ almost surely under the true mixture. If a problem has a known lower bound $L_\chi \geq -B$, the same statements apply to the shifted loss $L_\chi + B$, with the corresponding shift subtracted at the end.

The two logarithmic quantities compared later are the upper coordinate code length and the lower ghost entropy. If q_1 is the initial reference marginal on $\mathcal{Y}$, the fixed-truth upper telescope pays

$$L_0(\Omega_0; q_1, \chi) := \sup_{\omega \in \Omega_0} \log \frac{1}{q_1(\chi(\omega))}.$$

If the upper comparison is over a finite index set of size M and q_1 is uniform on exactly those indices, then $L_0 = \log M$. At a regret radius r, a hard prior μ with $q_1 = \chi_\# \mu$ gives the algorithm-uniform Bellman–Fano ghost mass and entropy

$$p^\star_{\mu,r} = \sup_{\mathsf{Alg} \in \mathfrak{A}^{\mathrm{na}}_T} \mathbb{P}_{Y \sim q_1,\, H'_T \sim \bar{\mathbb{P}}^{\mathsf{Alg}}_\mu} \{\bar{L}_\chi(Y, H'_T) \leq r\}, \qquad E^\star_{\mu,r} := \log \frac{1}{p^\star_{\mu,r}}.$$

Thus the primitive entropy comparison is $L_0 / E^\star_{\mu,r}$, not the size of the full model class. The detailed comparison and its conversion into a regret gap are stated in Section 3.

2.3 Bellman-sufficient representation

We now give the formal representation axioms used throughout the rest of the paper. The frequentist truth $\omega^\star$ remains fixed. Priors, posteriors, confidence objects, exponential-weights distributions, and algorithmic beliefs are reference objects: they may define algorithms or analytical bounds, but they are not substitutes for the fixed environment unless a theorem explicitly takes a Bayesian average.

Definition 2.1 (Indexed Bellman-sufficient representation)**.** *Fix an index map $\chi : \Omega \to \mathcal{Y}$. A retained state process*

$$S_t = \phi_t(H_{t-1})$$

is a fixed-truth Bellman-sufficient representation *if the following objects are determined by the current time-state pair (t, s).*

(i) Admissible actions. *There is a state-measurable action set $\mathcal{A}_t(s)$ such that $\mathcal{A}_t(h) = \mathcal{A}_t(s)$ whenever $S_t(h) = s$.*

(ii) Predictive sufficiency. *For every environment ω, mixed action $p \in \Delta(\mathcal{A}_t(s))$, and realized action $a \in \mathcal{A}_t(s)$, there is a kernel $P_{\omega,t,s,p,a}$ such that, for every history h with $S_t(h) = s$,*

$$P_{\omega,t}(\cdot \mid h, p, a) = P_{\omega,t,s,p,a}.$$

When the environment does not react to the announced mixed action, the argument p is suppressed.

(iii) Loss sufficiency. *There is an integrable state-measurable cost function $\ell_{\omega,t}(s, p, a) \in \mathbb{R}$ such that*

$$\ell_t(\omega, h, p, a) = \ell_{\omega,t}(s, p, a) \qquad \textit{whenever } S_t(h) = s.$$

The lower-bound quantile theorem does not require every increment to be nonnegative; it requires the cumulative indexed loss to be lower bounded, and in the displayed lower bounds we assume $L_\chi(Y, H_T) \geq 0$ almost surely.

(iv) Update sufficiency. *There is an update map or controlled kernel τ_t such that the next retained state is generated from the retained state and the current interaction. In deterministic-update notation,*

$$S_{t+1} = \tau_t(S_t, p_t, A_t, O_t),$$

with p_t suppressed when irrelevant. Random next contexts, physical next states, stage markers, and public randomization are part of the observation packet O_t.

When these clauses hold, Bellman recursion is closed under each fixed truth. A compressed state is therefore a dynamic sufficient statistic for the Bellman primitives: feasible actions, prediction, loss evaluation, and continuation.

If, in addition, a prior or reference law μ is fixed and one wants exact state-based posterior-reference information accounting, the full-history posterior objects must descend to the state. On the full history define

$$b_{t,h} := \mathcal{L}_\mu(\Omega \mid H_{t-1} = h), \qquad q_{t,h} := \chi_\# b_{t,h},$$

and, for $q_{t,h}$-almost every y,

$$b^y_{t,h} := \mathcal{L}_\mu(\Omega \mid H_{t-1} = h, \chi(\Omega) = y).$$

The full-history index-conditional predictive law and loss are

$$P^y_{t,h,p,a} := \int P_{\omega,t}(\cdot \mid h, p, a)\, b^y_{t,h}(d\omega),$$

$$\ell^\chi_{t,h,p,a}(y) := \int \ell_t(\omega, h, p, a)\, b^y_{t,h}(d\omega).$$

An exact posterior-indexed lift on the state *is the additional fiber-invariance requirement that, whenever $S_t(h) = s$, there exist state-measurable objects satisfying*

$$q_{t,h} = q_{t,s}, \tag{5}$$

and, for $q_{t,s}$-almost every y,

$$P^y_{t,h,p,a} = P^y_{t,s,p,a}, \qquad \ell^\chi_{t,h,p,a}(y) = \ell^\chi_{t,y}(s,p,a), \tag{6}$$

together with a state-measurable reference update for $q_{t,s}$. These posterior and index-conditional clauses are not needed for the fixed-truth Bellman recursion itself. They are needed when the posterior-averaged chain rule, the Bayes information-potential program, or the Bellman–Fano lower bound is run on the compressed state.

The notation will usually suppress the explicit time subscript when the stage is part of the state, and it will suppress the mixed-action argument when the environment does not react to the announced distribution. Thus $P_{\omega,s,a}$, $\ell_\omega(s,a)$, and $\mathcal{A}(s)$ mean the appropriate special cases of $P_{\omega,t,s,p,a}$, $\ell_{\omega,t}(s,p,a)$, and $\mathcal{A}_t(s)$ at the current Bellman time. This convention is harmless only after the state and current control include the variables that make actions, losses, observation laws, and updates state-measurable.

Bellman sufficiency versus classical sufficiency and Bellman rank. Classical sufficiency provides the static prototype for Bellman-state sufficiency (Fisher, 1922): a statistic $T(X)$ is sufficient for a family $\{P_\theta\}$ when the conditional distribution of the data given T is independent of θ. Bellman sufficiency is a closely related but conceptually distinct controlled analogue. The retained state must preserve everything needed for future actions, predictions, losses, information coordinates, and updates under admissible controls. In particular, one need not estimate all of $\omega^\star$ if a smaller state closes these recursions.

As explained above, Bellman sufficiency is a closedness condition for the controlled recursion, not a requirement of classical exact sufficiency expressed through conditional independence. The additional exact posterior-indexed lift conditions in equations (5) and (6) are state-level requirements and should be distinguished from full-history posterior sufficiency. Full-history posterior objects are tautological on H_{t-1}. A compressed state satisfies the exact indexed lift only when the quantities needed by the indexed Bellman recursion are constant on the fibers of S_t: the indexed marginal, the index-conditional predictive law, and the index-conditional loss. Thus full model-posterior sufficiency implies the lift, but the lift can be weaker. It should be read as an indexed conditional-moment closure condition: the state is sufficient for the conditional Bellman experiment given the index, not necessarily for the full environment; see Recht (2019, Section 3.3) for a possible connection between this viewpoint and the classical reinforcement-learning and control literature. This distinction is important for Section 2.4, which explains why sufficient states matter as representation-level objects without asserting that exact compression reduces the information in a fixed index.

The index $Y = \chi(\Omega)$ is the coordinate whose logarithmic mass is charged by the upper information potential and whose ghost-good mass appears in the lower Fano bound. It need not identify the whole environment. In bandits it may be the optimal arm; in reinforcement learning it may be an optimal policy or value object; in planning or reasoning it may be a plan, proof, or answer. The mathematics is the same whenever a Bellman-sufficient state and a reference marginal q_s for the index are available.

The formalism is aligned with the representation-level philosophy behind low Bellman rank (Jiang et al., 2017), but it is deliberately stricter. Bellman rank is a factorization of expected Bellman errors relative to a function class, roll-in distribution, and Bellman-error evaluation family. It is not, by itself, a state: it does not determine the realized next observation, the fixed-truth loss, the posterior or reference update, or the conditional laws for an index. It becomes an indexed Bellman-sufficient representation only after those missing objects are supplied and shown to close the recursions above.

Appendix A collects basic examples involving sufficient statistics, posterior states, and finite marginals, and shows how the general framework encompasses the standard Decision Making with Structured Observations (DMSO) setting (Foster et al., 2021, 2023).

2.4 Why sufficient states and compression matter

The full history H_{t-1} is always a valid Bellman state, so state compression is not needed for the formal existence of a dynamic program. The theory developed in this paper could also be stated using the full history H_{t-1} together with full-history information indices. Indeed, any state-measurable information index is automatically an index on the full history: if $S_t = \phi_t(H_{t-1})$, then a state-dependent index $\chi_{S_t}(\Omega)$ lifts to the full-history index $\chi_{\phi_t(H_{t-1})}(\Omega)$. Conversely, a full-history index descends to the compressed state only if it is constant on the fibers of the compression. The role of sufficient states is therefore representational rather than existential: they identify a level of compression at which the relevant Bellman recursions and information accounting close. Such a state need not be unique, nor must it be minimal.

It is already clear that sufficient states are useful at a practical level: specifying them makes it tractable to identify the relevant information index, construct upper-bound algorithms with clear links to popular design principles, and formulate lower-bound methods. The question in this subsection is more conceptual: whether sufficient states offer any information-theoretic advantage beyond the full history. The answer is yes. By data processing, compression can only reduce the information budget; when the retained state is Bellman-sufficient in the appropriate controlled sense, this reduction need not lose any of the Bellman structure required for upper or lower bounds. The key is to distinguish Bellman sufficiency from classical conditional-independence sufficiency. Otherwise, if the relevant information gain were always invariant under compression, sufficient states would be more a methodological convenience than an intrinsic representation-level object. A general formal claim that every proof device operating on the full history, but not passing through the constructed sufficient state, must lead to coarser bounds would be too strong. We do not pursue such a claim here.

It is helpful to separate three notions. Classical model sufficiency is the static conditional-independence condition

$$\Omega \perp H_{t-1} \mid S_t,$$

which says that the state retains all posterior information about the full environment. Index-marginal sufficiency for $Y = \chi(\Omega)$ is weaker:

$$Y \perp H_{t-1} \mid S_t.$$

It says only that the state retains the full-history posterior marginal of the index. Bellman sufficiency, as used in Definition 2.1, is different: it is the dynamic closure of the primitives needed for decision making, namely feasible actions, fixed-truth predictive kernels, losses or surrogate losses, updates, and, when needed, the indexed reference objects used by the logarithmic potential and the Bellman–Fano method. Thus a Bellman-sufficient state need not be sufficient for the full model, and a state-based reference process may be coarser than the full-history posterior; its validity comes from the Bellman inequalities that close on that state.

Proposition 2.2 (State compression and information monotonicity)**.** *Let $S_t = \phi_t(H_{t-1})$ be a fixed-truth Bellman-sufficient representation. Suppose a one-step Bellman cost c_ω and a continuation value F_{t+1} are state-measurable, so that*

$$c_\omega(h,p) = c_\omega(s,p), \qquad F_{t+1}(H_t) = f_{t+1}(S_{t+1}), \qquad s = S_t(h).$$

Then the full-history Bellman operator factors through S_t: for every h with $S_t(h) = s$,

$$\inf_{p\in\Delta(\mathcal{A}_t(h))} \sup_{\omega\in\mathcal{C}_t(h)} \{c_\omega(h,p) + \mathbb{E}_{\omega,h,p}F_{t+1}(H_t)\} = \inf_{p\in\Delta(\mathcal{A}_t(s))} \sup_{\omega\in\mathcal{C}_t(s)} \{c_\omega(s,p) + \mathbb{E}_{\omega,s,p}f_{t+1}(S_{t+1})\},$$

whenever the comparison set is also state-measurable. Moreover, for any reference law and any fixed index $Y = \chi(\Omega)$,

$$I(Y; S_t) \le I(Y; H_{t-1}).$$

Equality in the information display holds precisely under index-marginal sufficiency, $Y \perp H_{t-1} \mid S_t$.

Proof. See Appendix H.1.

This proposition is the formal reason sufficient states are useful. State compression preserves the Bellman problem whenever the primitives close on the state, while data processing ensures that the state-based information accounting for a fixed index is no larger than the full-history accounting. If exact index-marginal sufficiency holds, the two information quantities are equal; if the state intentionally uses a coarser algorithmic or reference belief, the information quantity may be strictly smaller. That strict reduction is not a free theorem about the original unrestricted problem: it is valid only when the upper Bellman recursion and the lower Bellman–Fano recursion are both proved on the same coarsened state. In this sense the framework is representational. One searches for a state and index on which the dynamic program and the entropy accounting simultaneously close; when they do, the resulting sandwich can be tighter and more intrinsic than a bound written directly on the full history, even for the same global index Y.

3 Information complexity sandwich

This section isolates the comparison used throughout the paper. After a Bellman-sufficient state, an index, a reference update, and a calibration mechanism have been fixed, the upper and lower bounds are expressed in the same units. The upper logarithmic Bellman theorem pays an initial coordinate code length for the realized index. The lower Bellman–Fano theorem is governed by the ghost entropy of reference histories that are already good for the sampled index. The comparison is therefore an information-risk sandwich: an upper Bellman value at an index-code budget is compared with a lower Bellman–Fano value at a ghost-entropy budget. This section makes the finite-index logarithmic gap, the constant-gap localization condition, and the growth/fixed-point requirements explicit.

3.1 Code length, hard priors, and ghost entropy

Let $\Omega_0 \subseteq \Omega$ be the problem class being lower and upper bounded, let $\chi : \Omega \to \mathcal{Y}$ be the index, and let q_1 be an initial reference marginal on $\mathcal{Y}$. The fixed-truth upper telescope pays the worst-case initial coordinate code length

$$L_0(\Omega_0; q_1, \chi) := \sup_{\omega\in\Omega_0} \log \frac{1}{q_1(\chi(\omega))}, \tag{7}$$

with the convention that the value is infinite if $q_1(\chi(\omega)) = 0$ for some $\omega \in \Omega_0$.

For a hard prior μ supported on Ω_0, write $q_1 = \chi_{\#}\mu$. At an average-loss radius r, the ghost-good probability of an algorithm Alg is

$$p^{\mathsf{Alg}}_{\mu,r} := p^{\mathsf{Alg}}_r(\mu, \chi) = \mathbb{P}_{Y\sim q_1,\, H'_T\sim\bar{\mathbb{P}}^{\mathsf{Alg}}_\mu}\{\bar{L}_\chi(Y, H'_T) \le r\}, \tag{8}$$

where Y and the algorithm-induced reference history H'_T are independent under the ghost law. The algorithm-uniform ghost mass and its entropy are

$$p^\star_{\mu,r} := \sup_{\mathsf{Alg}\in\mathfrak{A}^{\mathrm{na}}_T} p^{\mathsf{Alg}}_{\mu,r} = \Gamma^{r,\star}_1(q_1, s_1, 0), \qquad E^\star_{\mu,r} := \log\frac{1}{p^\star_{\mu,r}}. \tag{9}$$

A ghost-mass supersolution gives $p^\star_{\mu,r} \le \overline{\Gamma}^r_1(q_1, s_1, 0)$ and hence the computable entropy lower bound $-\log\overline{\Gamma}^r_1(q_1, s_1, 0) \le E^\star_{\mu,r}$.

Definition 3.1 (Bellman–Fano admissible hard prior)**.** *Fix $r > 0$. A prior μ supported on Ω_0 is* Bellman–Fano admissible at radius r *if the lower Bellman recursions, or valid supersolutions of* *them, provide a nonnegative indexed-information upper bound C_μ and a valid algorithm-uniform ghost-entropy lower bound $\underline{E}_{\mu,r} \le E^\star_{\mu,r} = \log(1/p^\star_{\mu,r})$ such that*

$$C_\mu + \log 2 \le \frac{1}{2}\underline{E}_{\mu,r}. \tag{10}$$

The intrinsic choice is $C_\mu = C^\star_1(s_1)$ and $\underline{E}_{\mu,r} = E^\star_{\mu,r} = -\log \Gamma^{r,\star}_1(q_1, s_1, 0)$, where $C^\star$ and $\Gamma^{r,\star}$ are the exact information-capacity and ghost-good Bellman values from Section 6.3. Computable proofs may use the conservative choices $C_\mu = \overline{C}_1(s_1)$ and $\underline{E}_{\mu,r} = -\log \overline{\Gamma}^r_1(q_1, s_1, 0)$; since $\overline{\Gamma}^r_1$ upper-bounds the algorithm-uniform ghost mass, this is indeed no larger than $E^\star_{\mu,r}$.

The condition $p^\star_{\mu,r} \le \varrho < 1$ sometimes appears as a convenient nondegeneracy diagnostic: it prevents the lower entropy denominator from vanishing. It plays the same mathematical role as the nondegeneracy requirements in offset or constrained DEC comparisons (Foster et al., 2021, 2023): without positive decision separation at the chosen information scale, the denominator of the rate comparison is zero and no lower bound can be extracted. In the Bellman–Fano formulation this condition need not be imposed separately once the hard prior is defined through admissibility.

Lemma 3.2 (Admissibility implies nondegenerate ghost entropy)**.** *If μ is Bellman–Fano admissible at radius r, then*

$$E^\star_{\mu,r} \ge \underline{E}_{\mu,r} \ge 2\log 2, \qquad p^\star_{\mu,r} \le \frac{1}{4}.$$

Consequently a separate assumption $p^\star_{\mu,r} \le \varrho < 1$ is unnecessary for any admissible hard prior. Conversely, if one starts from a proposed prior and only knows $p^\star_{\mu,r} \le \varrho < 1$, then the entropy denominator is at least $\log(1/\varrho)$, but this by itself does not prove Bellman–Fano admissibility because the information upper bound C_μ must also satisfy (10).

Proof. See Appendix H.2.

Proposition 3.3 (Finite-index entropy comparison with an explicit hard prior)**.** *Let $\mathcal{Y}_h \subseteq \chi(\Omega_0)$ be finite with $|\mathcal{Y}_h| = M \ge 2$. Choose one representative environment $\omega^y \in \Omega_0$ for each $y \in \mathcal{Y}_h$ such that $\chi(\omega^y) = y$, and let μ_h be the uniform prior on $\{\omega^y : y \in \mathcal{Y}_h\}$. Then q_1 is uniform on $\mathcal{Y}_h$, and*

$$L_0(\{\omega^y : y \in \mathcal{Y}_h\}; q_1, \chi) = \log M.$$

If μ_h is Bellman–Fano admissible at radius r, then the entropy mismatch between the upper coordinate code length and the lower ghost entropy satisfies

$$\frac{L_0}{E^\star_{\mu_h,r}} \le \frac{\log M}{2\log 2}. \tag{11}$$

Thus, without pursuing constant matching, admissibility alone gives an at-most-logarithmic finite-index entropy gap. More sharply, if for every reference history h',

$$G_r(h') := \{y \in \mathcal{Y}_h : \bar{L}_\chi(y, h') \le r\}$$

has cardinality at most m_r, for some integer $1 \le m_r < M$, then

$$E^\star_{\mu_h,r} \ge \log \frac{M}{m_r}, \qquad \frac{L_0}{E^\star_{\mu_h,r}} \le \frac{\log M}{\log(M/m_r)}. \tag{12}$$

In particular, the one-sided constant-factor entropy comparison used here follows from $E^\star_{\mu_h,r} \ge c\log M$, for example through $m_r \le M^{1-\alpha}$ for some constant $\alpha > 0$. A two-sided entropy equivalence requires the stronger statement $E^\star_{\mu_h,r} \asymp \log M$. A mere nondegeneracy bound $p^\star_{\mu_h,r} \le \varrho < 1$ gives only $L_0/E^\star_{\mu_h,r} \le \log M/\log(1/\varrho)$.

Proof. See Appendix H.3.

3.2 From entropy comparison to regret comparison

An entropy comparison alone is not a regret comparison. The missing ingredient is a regularity property of the upper Bellman value as its information budget changes. This is the same structural reason that constrained DEC bounds are stated through a coefficient, localization, or fixed-point condition rather than through a bare logarithmic-model-cardinality entropy term (Foster et al., 2023): cardinality controls the amount of information charged, but a risk modulus is still needed to turn that information budget into a regret radius.

For a fixed state/index representation and the log-penalized Bellman program in Section 5.1, write

$$\mathsf{U}_T(L) := \inf_{\gamma>0} \left\{ W_1^\gamma(s_1) + \gamma L + \Delta_T^\gamma \right\}, \tag{13}$$

where W^γ is the log-penalized Bellman value and $\Delta_T^\gamma \geq 0$ collects explicit statewise bracket, approximation, and optimization errors for that value as in Theorem 5.2. For the exact surely calibrated Bellman dynamic program, $\Delta_T^\gamma = 0$. We assume U_T is nondecreasing in L, as holds for the optimized upper value in the applications considered here.

Assumption 3.4 (Bellman upper-growth condition). *There exist constants $C_{\mathrm{gr}} \geq 1$ and $\beta \in [0, 1]$ such that, for all $L > 0$ and $a \geq 1$,*

$$\mathsf{U}_T(aL) \leq C_{\mathrm{gr}} a^\beta \mathsf{U}_T(L).$$

This is a regularity condition on the chosen upper value. It is not a consequence of finite index cardinality alone; it is the Bellman analogue of the sub-root or growth condition used to convert entropy or localized information radii into risk radii in fixed experiments and in constrained DEC analyses.

Lemma 3.5 (Why a growth condition is needed). *No regret-ratio comparison follows from an entropy-ratio comparison for an arbitrary nondecreasing upper value U_T. More precisely, fix $0 < E < L$ and any $B > 0$. There is a nondecreasing function U such that $\mathsf{U}(L)/\mathsf{U}(E) \geq B$.*

Proof. See Appendix H.4.

Definition 3.6 (Upper-at-lower-entropy calibration). *Let μ be Bellman–Fano admissible at radius r with algorithm-uniform ghost entropy $E^\star_{\mu,r}$. The upper value is* calibrated at the lower entropy scale *if*

$$\mathsf{U}_T(E^\star_{\mu,r}) \leq C_{\mathrm{base}}\, Tr \tag{14}$$

for a universal or problem-controlled constant C_{base}. Equivalently, the phrase "the upper and lower values close at the same radius" means (14); it is not an additional philosophical assumption. It can be verified directly, by a localized or constrained information-gain estimate, or by choosing r as a fixed point of the rate equation $\mathsf{U}_T(E^\star_{\mu,r}) \asymp Tr$.

Proposition 3.7 (Three scale-closing diagnostics). *For a Bellman–Fano admissible pair (μ, r), each of the following conditions implies upper-at-lower-entropy calibration (14).*

(i) Direct budget control: $\mathsf{U}_T(E^\star_{\mu,r}) \leq C_{\mathrm{base}} Tr$.

(ii) Coefficient control: there exists $\gamma > 0$ such that

$$W_1^\gamma(s_1) + \gamma E^\star_{\mu,r} + \Delta_T^\gamma \leq C_{\mathrm{base}} Tr.$$

(iii) Fixed-point control: r is chosen so that $\mathsf{U}_T(E^\star_{\mu,r})/T \leq C_{\mathrm{base}} r$.

Condition (ii) is the form closest to constrained DEC and localized information-gain analyses: the same localized comparison that controls the information budget also bounds the decision term at radius r. Condition (iii) packages the same requirement as a rate fixed point.

Proof. See Appendix H.5.

Theorem 3.8 (Frequentist Bellman information-risk sandwich)**.** *Fix $\Omega_0 \subseteq \Omega$, a horizon T, an index χ, a Bellman-sufficient state/reference update, and an initial index law q_1 with finite $L_0 = L_0(\Omega_0; q_1, \chi)$. Suppose the upper side admits a calibrated log-penalized Bellman bound whose budget value is U_T in (13). Then*

$$\mathfrak{R}_T^\star(\Omega_0) \le \mathsf{U}_T(L_0). \tag{15}$$

Conversely, if a prior μ supported on Ω_0 is Bellman–Fano admissible at radius r, then

$$\mathfrak{R}_T^\star(\Omega_0) \ge \frac{Tr}{2}. \tag{16}$$

If, in addition, Assumption 3.4 holds and the upper value is calibrated at the lower entropy scale in the sense of Definition 3.6, then

$$\mathfrak{R}_T^\star(\Omega_0) \le C_{\mathrm{gr}} C_{\mathrm{base}} \left(\max\left\{1, \frac{L_0}{E^\star_{\mu,r}}\right\}\right)^\beta Tr, \tag{17}$$

and hence the ratio between the displayed upper value and the Bellman–Fano lower bound $Tr/2$ is at most

$$2C_{\mathrm{gr}} C_{\mathrm{base}} \left(\max\left\{1, \frac{L_0}{E^\star_{\mu,r}}\right\}\right)^\beta .$$

If, in addition, $\mu = \mu_h$ is the uniform representative prior of Proposition 3.3, the upper reference law is the same marginal $q_1 = \chi_\# \mu_h$, and $\chi(\Omega_0)$ consists exactly of those M indices, then $L_0 = \log M$, and admissibility alone gives the crude logarithmic comparison

$$\frac{L_0}{E^\star_{\mu,r}} \le \frac{\log M}{2\log 2},$$

while the multiplicity/localization condition $|G_r(h')| \le m_r$ gives

$$\frac{L_0}{E^\star_{\mu,r}} \le \frac{\log M}{\log(M/m_r)}.$$

The same statement applies after replacing Ω_0 throughout by the representative subfamily used by a uniform hard prior. Thus logarithmic-gap matching on a common finite comparison class requires only Bellman–Fano admissibility, upper growth, and calibration at the lower entropy scale. Constant-factor matching follows from the stronger one-sided localization $E^\star_{\mu,r} \ge c \log M$, or from an equivalent constraint or fixed-point argument; a two-sided entropy characterization additionally requires $E^\star_{\mu,r} = O(\log M)$.

Proof. See Appendix H.6.

Relation to offset and constrained DEC. The condition $p^{\star}_{\mu,r} \leq \varrho < 1$ is not a separate structural assumption in the final sandwich; it is a quick diagnostic that a proposed hard prior has nonzero algorithm-uniform ghost entropy. Bellman–Fano admissibility is the cleaner replacement because it simultaneously checks nonzero ghost entropy and small indexed information. This is analogous to offset or constrained DEC theory in the following nondegeneracy and localization sense: a rate comparison is meaningful only after the decision-separation term at the chosen scale is nondegenerate and the information term is localized or constrained at the same scale (Foster et al., 2021, 2023).

Constrained DEC incorporates the nondegeneracy and localization checks into the coefficient or fixed-point definition, while the accompanying upper-growth condition controls the information-to-regret conversion (Foster et al., 2023). For a finite model-index class, this can still leave a worst-case logarithmic gap in the upper and lower sandwich, measured in the model cardinality. A tight theory of DEC is technically challenging partly because it seeks a one-step and state-uniform coefficient that simultaneously captures decision loss, statistical indistinguishability, localization, and adaptive information accumulation. This is a stronger requirement than assigning a single global dimension to the model class. Even in classical statistical learning, sharp rates are often governed by localized, distribution-dependent, data-dependent, or metric-dependent quantities, such as empirical or local Rademacher complexities and chaining functionals for the relevant induced metric. Thus a worst-case cardinality or global-dimension gap under regularity conditions is a rigorous universal fallback, but it can be too coarse for precise applications. This is one motivation for complementing a global coefficient with a dynamic, state-dependent Bellman information bound.

The Bellman sandwich makes explicit three checks that coefficient-based theories often package together: (i) hard-prior admissibility on the lower side, which combines information control with nonvacuous ghost entropy and localization; (ii) upper-at-lower-entropy calibration, which checks that the log-penalized Bellman upper value is of the right order at the entropy scale supplied by the hard prior; and (iii) upper growth, which converts an entropy mismatch into a regret mismatch. This separation allows the lower side to be proved by Bellman–Fano ghost entropy and the upper side by exact log-potential Bellman programming, UCB, E2D/DEC, AMS/EBO, or another relaxation. DEC theory provides a complementary and often more self-contained route: its indistinguishability game and E2D-type algorithm are controlled through the same coefficient, so the corresponding calibration is built into the coefficient analysis. For a common finite model-index comparison class, either analysis may retain a logarithmic dependence on cardinality. Under a uniform hard prior, Bellman–Fano admissibility, and an upper comparison over those same indices, the entropy gap is at most logarithmic; localization giving $E^{\star}_{\mu,r} \geq c \log M$ makes its one-sided ratio constant. The examples in Section 7 illustrate the lower entropy calculation and minimax-rate matching, while explicitly distinguishing this from verification of a particular Bellman upper program.

4 Indexed information and exact AIR/MAIR identities

4.1 Posterior-reference histories and index information

The mutual-information identities in this subsection use the exact posterior-reference version of Definition 2.1. The fixed-truth log-gain regret identity in the next subsection should not be substituted into the posterior chain rule unless the two objects are induced by a coherent reference law.

For a prior μ and algorithm Alg, define the Bayesian posterior-reference trajectory law

$$\bar{\mathbb{P}}_{\mu}^{\mathsf{Alg}} := \int \mathbb{P}_{\omega}^{\mathsf{Alg}}\, \mu(d\omega).$$

A reference history is denoted

$$H_T' \sim \bar{\mathbb{P}}_{\mu}^{\mathsf{Alg}}, \qquad S_t' = \phi_t(H_{t-1}'), \qquad b_t' = \mathcal{L}_{\mu}(\Omega \mid H_{t-1}'),$$

and the algorithm's reference decision kernel is

$$p_t' = p_t(\cdot \mid H_{t-1}').$$

Definition 4.1 (Indexed information gain)**.** *At a current time-state (t, s), let q_s be the posterior law of $Y = \chi(\Omega)$ and let $P_{s,p,a}^{y}$ and $P_{s,p,a}$ be the conditional and marginal predictive laws from Definition 2.1; we suppress t in these laws when the stage is part of s. For $p \in \Delta(\mathcal{A}_t(s))$, define*

$$\mathcal{I}_{\chi}(s,p) := \mathbb{E}_{a\sim p} \int D_{\mathrm{KL}}(P_{s,p,a}^{y} \| P_{s,p,a}) q_s(dy). \tag{18}$$

When the environment does not respond to the announced mixed action, the argument p is suppressed. When $Y = \Omega$, this is model-index information. When $Y = \pi^{\star}(\Omega)$, this is action-index information. In the action-index case, the belief is still a belief over environments; only the information target is the optimal action.

Define the cumulative and average indexed information of an algorithm by

$$C_{\chi}^{\mathsf{Alg}}(\mu) := \mathbb{E}_{H_T' \sim \bar{\mathbb{P}}_{\mu}^{\mathsf{Alg}}} \sum_{t=1}^{T} \mathcal{I}_{\chi}(S_t', p_t'), \qquad \bar{C}_{\chi}^{\mathsf{Alg}}(\mu) := \frac{1}{T} C_{\chi}^{\mathsf{Alg}}(\mu). \tag{19}$$

Proposition 4.2 (Reference-history indexed chain rule)**.** *Assume that the retained state is an exact posterior-indexed lift in the sense of Definition 2.1, that the displayed KL terms are well defined and integrable, and that the initial packet O_0 is deterministic or independent of Y. For every prior μ, index map χ, and algorithm* Alg,

$$I_{\mu}(Y; H_T) = \mathbb{E}_{H_T' \sim \bar{\mathbb{P}}_{\mu}^{\mathsf{Alg}}} \sum_{t=1}^{T} \mathcal{I}_{\chi}(S_t', p_t') = T \bar{C}_{\chi}^{\mathsf{Alg}}(\mu). \tag{20}$$

If O_0 may carry information about Y, the right-hand side equals $I_{\mu}(Y; H_T \mid O_0)$ and the unconditional identity has the additional term $I_{\mu}(Y; O_0)$.

Proof. See Appendix H.7.

4.2 Fixed-truth indexed AIR bracket

The posterior-averaged information gain in (18) is the correct object for Bayesian chain rules and Bellman–Fano lower-bound method. The upper Bellman program uses the corresponding fixed-truth coordinate identity. The coordinate identity is most cleanly stated in AIR form: a current reference marginal on the index is scored by the logarithmic posterior of the index after the current action and observation. The point-coordinate formulas below are literal for finite or countable indices with positive masses. For a dominated non-atomic index, $q(y)$ and $q_{\nu}^{+}(y \mid \cdot)$ denote Radon–Nikodym densities with respect to a fixed dominating measure, and the identities require positivity and integrability of the displayed log-density ratios. The later step that discards the terminal coordinate log loss is restricted to a discrete index.

Coefficient convention. Throughout the upper-bound identities and Bellman programs, the information multiplier is denoted by $\gamma > 0$: a one-step log gain G is charged as γG, and an initial coordinate code length is charged as $\gamma \log(1/q)$. Some AIR/MAIR/DEC papers use a temperature parameter η with multiplier $1/\eta$ (Xu and Zeevi, 2025; Liu et al., 2025, 2026); in the present notation this is simply $\gamma = 1/\eta$.

Fix a state s, an action distribution $p \in \Delta(\mathcal{A}_t(s))$, and a pair belief

$$\nu \in \Delta(\Omega \times \mathcal{Y})$$

over environments and index values. Let q_ν be the $\mathcal{Y}$-marginal of ν, and let q be a current reference marginal that is positive on every coordinate evaluated by the logarithmic score. The pair belief induces, for each realized action a, the predictive mixture

$$P_{\nu,s,p,a}(\cdot) := \int P_{\omega,t}(\cdot \mid s, p, a)\, \nu(d\omega, dy),$$

and the posterior index marginal

$$q_\nu^+(B \mid s, p, a, o) := \nu(Y \in B \mid s, p, a, O = o), \qquad B \subseteq \mathcal{Y}.$$

When the observation laws are dominated by a common measure and have densities $p_{\omega,t,s,p,a}(o)$, this update is explicitly

$$q_\nu^+(B \mid s, p, a, o) = \frac{\int_{\Omega \times B} p_{\omega,t,s,p,a}(o)\, \nu(d\omega, dy)}{\int_{\Omega \times \mathcal{Y}} p_{\omega,t,s,p,a}(o)\, \nu(d\omega, dy)}. \tag{21}$$

In ordinary stochastic bandits and MDPs the dependence on p is suppressed.

Let $\ell_{\omega,y}(s,p)$ denote the comparison loss associated with pair (ω, y); for a valid fixed truth one takes $y = \chi(\omega)$ and $\ell_{\omega,\chi(\omega)} = \ell_\omega$. Define the general indexed AIR functional

$$\mathfrak{A}_{q,\gamma}(s,p,\nu) := \mathbb{E}_{(\Omega,Y)\sim\nu} \ell_{\Omega,Y}(s,p) - \gamma \mathbb{E}_{(\Omega,Y)\sim\nu,\, a\sim p,\, O\sim P_{\Omega,t}(\cdot\mid s,p,a)} \log \frac{q_\nu^+(Y \mid s,p,a,O)}{q(Y)}. \tag{22}$$

Equivalently,

$$\mathfrak{A}_{q,\gamma}(s,p,\nu) = \mathbb{E}_\nu \ell_{\Omega,Y}(s,p) - \gamma \mathcal{I}_\nu(Y; O \mid s,p) - \gamma D_{\mathrm{KL}}(q_\nu \| q), \tag{23}$$

where

$$\mathcal{I}_\nu(Y; O \mid s,p) := \mathbb{E}_{a\sim p} \int D_{\mathrm{KL}}(P^y_{\nu,s,p,a} \| P_{\nu,s,p,a}) q_\nu(dy)$$

is the one-step posterior-averaged information computed under the candidate pair belief. Thus q is the current reference index marginal, while q_ν^+ is constructed from the algorithmic pair belief ν and the current likelihood model. In an exact Bayesian specialization, ν is the current posterior over $(\Omega, \chi(\Omega))$ and $q = q_\nu$. In AIR/AMS/EBO, ν may be an optimized or robust candidate pair belief, and the KL term in (23) accounts for moving its index marginal away from the current reference q.

Lemma 4.3 (AIR gradient bracket)**.** *Assume a finite or countable index with positive reference masses and finite or dominated observation laws, or use the dominated-index density convention above. Assume also that the displayed Gateaux derivatives and log ratios exist and are integrable. Up to an additive constant on the probability simplex,*

$$\frac{\partial}{\partial \nu(\omega, y)} \mathfrak{A}_{q,\gamma}(s,p,\nu) = \ell_{\omega,y}(s,p) - \gamma \mathbb{E}_{a\sim p,\, O\sim P_{\omega,t}(\cdot\mid s,p,a)} \log \frac{q_\nu^+(y \mid s,p,a,O)}{q(y)}. \tag{24}$$

Consequently, for every fixed pair (ω, y)*,*

$$B^{\mathrm{AIR}}_{q,\gamma}(s,p,\nu;\omega,y) := \mathfrak{A}_{q,\gamma}(s,p,\nu) + \left\langle \nabla_\nu \mathfrak{A}_{q,\gamma}(s,p,\nu), \delta_{(\omega,y)} - \nu \right\rangle$$
$$= \ell_{\omega,y}(s,p) - \gamma \mathbb{E}_{a\sim p,\, O\sim P_{\omega,t}(\cdot|s,p,a)} \log \frac{q^+_\nu(y \mid s,p,a,O)}{q(y)}. \tag{25}$$

Proof. See Appendix H.8.

4.3 Exact AIR identity and model-index specialization

We now present simplified statements and alternative proofs of several results that are central to the AIR/MAIR literature (Xu and Zeevi, 2025; Liu et al., 2025, 2026). These results are regret identities, not merely upper bounds, but the identity has a specific hypothesis: the next index marginal must be the posterior index marginal generated by the pair belief ν_t that appears in the AIR functional. Under this ν-generated update, Lemma 4.3 gives the one-step AIR bracket and the coordinate logarithm telescopes. Other reference updates, such as confidence-to-belief or exponential-weights updates, may yield useful log-potential bounds, but they are not the AIR identity unless they can be represented by an admissible pair belief and the corresponding posterior-index update. Thus AIR is the exact posterior-scoring algebra inside the Bellman program; more general log-potential bounds require separate calibration or relaxation arguments.

Theorem 4.4 (Exact indexed AIR regret identity)**.** *Let an algorithm generate, at each reached state* S_t*, a current reference index marginal* q_t *that is positive on the coordinate being evaluated, a pair belief* $\nu_t \in \Delta(\Omega \times \mathcal{Y})$*, and a decision law* $p_t \in \Delta(\mathcal{A}_t(S_t))$*. After sampling* $A_t \sim p_t$ *and observing* O_t*, update the reference marginal by*

$$q_{t+1}(\cdot) = q^+_{\nu_t}(\cdot \mid S_t, p_t, A_t, O_t). \tag{26}$$

Fix a pair $(\omega^\star, y^\star)$ *such that* $q_t(y^\star) > 0$ *and* $q_{t+1}(y^\star) > 0$ *almost surely along the analyzed trajectory, and assume that the displayed losses and log ratios are integrable. Then, for every* $\gamma > 0$*,*

$$\mathbb{E}^{\mathsf{Alg}}_{\omega^\star} \sum_{t=1}^{T} \ell_{\omega^\star,y^\star}(S_t,p_t) = \gamma \mathbb{E}^{\mathsf{Alg}}_{\omega^\star} \log \frac{q_{T+1}(y^\star)}{q_1(y^\star)} + \mathbb{E}^{\mathsf{Alg}}_{\omega^\star} \sum_{t=1}^{T} B^{\mathrm{AIR}}_{q_t,\gamma}(S_t,p_t,\nu_t;\omega^\star,y^\star). \tag{27}$$

For the original fixed-truth regret, take $y^\star = \chi(\omega^\star)$ *and* $\ell_{\omega^\star,y^\star} = \ell_{\omega^\star}$*. The expectation is over action randomization, observations, and any exogenous algorithmic randomness.*

Proof. See Appendix H.9.

The variational interpretation through posterior scoring and Danskin's theorem is given in Appendix B.

Identity versus lower bound. The identity is tightest when the final coordinate potential is retained. Rearranging (27) gives

$$\mathbb{E}^{\mathsf{Alg}}_{\omega^\star} \sum_{t=1}^{T} \ell_{\omega^\star,y^\star}(S_t,p_t) = \gamma \log \frac{1}{q_1(y^\star)} - \gamma \mathbb{E}^{\mathsf{Alg}}_{\omega^\star} \log \frac{1}{q_{T+1}(y^\star)} + \mathbb{E}^{\mathsf{Alg}}_{\omega^\star} \sum_{t=1}^{T} B^{\mathrm{AIR}}_{q_t,\gamma}(S_t,p_t,\nu_t;\omega^\star,y^\star).$$

For a finite or countable index, dropping the nonpositive terminal term $-\gamma \mathbb{E} \log(1/q_{T+1}(y^\star))$ gives the usual upper value and loses exactly the remaining coordinate code length. For a non-atomic index represented by densities, that term need not be nonpositive and must be retained or controlled separately. This observation does not itself give a minimax lower bound; the lower side still requires a Bellman–Fano or ghost-quantile value. It explains why the same logarithmic coordinate is the right quantity on both sides: the upper proof pays the initial code length only after discarding the final unresolved code length, while the lower proof bounds how much indexed entropy must remain under ghost histories.

Stationary-posterior specialization. In the exact posterior-indexed lift of Definition 2.1, the pair belief is the current posterior law of $(\Omega, \chi(\Omega))$ and q_t is its index marginal. Then $q^+_{\nu_t}$ is the exact posterior index update and the one-step fixed-truth log gain can also be written in predictive-law form. For $y = \chi(\omega)$,

$$J_\chi(s, p; \omega) := \mathbb{E}_{a\sim p,\, O\sim P_{\omega,t}(\cdot|s,p,a)} \log \frac{q^+_{\nu_s}(y \mid s, p, a, O)}{q_s(y)}. \tag{28}$$

Equivalently, when the relevant laws are dominated,

$$J_\chi(s, p; \omega) = \mathbb{E}_{a\sim p}\left[D_{\mathrm{KL}}(P_{\omega,s,p,a} \| P_{s,p,a}) - D_{\mathrm{KL}}(P_{\omega,s,p,a} \| P^{\chi(\omega)}_{s,p,a})\right]. \tag{29}$$

Posterior averaging of J_χ over ω conditional on $Y = y$, and then over $y \sim q_s$, recovers $\mathcal{I}_\chi(s, p)$. Thus J_χ is a fixed-truth coordinate log gain, whereas $\mathcal{I}_\chi$ is its posterior-averaged information counterpart.

For $\gamma > 0$, the exact-posterior fixed-truth indexed bracket is

$$B_{\chi,\gamma}(s, p; \omega) := \ell_\omega(s, p) - \gamma J_\chi(s, p; \omega). \tag{30}$$

It is the specialization of (25) obtained by taking the pair belief to be the current posterior over $(\Omega, \chi(\Omega))$ and the reference marginal to be q_s.

Environment-index specialization. If $Y = \Omega$, the pair belief is concentrated on the graph $y = \omega$. Write the current environment belief as μ and the reference model marginal as ρ. Point masses below are interpreted literally on a finite or countable model class and as densities relative to a fixed dominating measure otherwise; all displayed density ratios must be positive at the evaluated truth and integrable. Then (22) becomes

$$\mathrm{MAIR}_{\rho,\gamma}(s, p, \mu) := \mathbb{E}_{\omega\sim\mu}\ell_\omega(s, p) - \gamma\mathbb{E}_{a\sim p} I_\mu(\Omega; O \mid s, p, a) - \gamma D_{\mathrm{KL}}(\mu\|\rho). \tag{31}$$

The fixed-truth bracket is

$$B^{\mathrm{MAIR}}_{\rho,\gamma}(s, p, \mu; \omega^\star) = \ell_{\omega^\star}(s, p) - \gamma\mathbb{E}_{a\sim p} D_{\mathrm{KL}}(P_{\omega^\star,s,p,a}\|P_{\mu,s,p,a}) - \gamma \log \frac{\mu(\omega^\star)}{\rho(\omega^\star)}. \tag{32}$$

If ρ_{t+1} is the Bayes posterior update of μ_t after (S_t, p_t, A_t, O_t), the required point masses or densities remain positive at $\omega^\star$, and the losses and log ratios are integrable, then

$$\mathbb{E}^{\mathsf{Alg}}_{\omega^\star} L_{\omega^\star}(H_T) = \gamma\mathbb{E}^{\mathsf{Alg}}_{\omega^\star} \log \frac{\rho_{T+1}(\omega^\star)}{\rho_1(\omega^\star)} + \mathbb{E}^{\mathsf{Alg}}_{\omega^\star} \sum_{t=1}^{T} B^{\mathrm{MAIR}}_{\rho_t,\gamma}(S_t, p_t, \mu_t; \omega^\star). \tag{33}$$

The stationary or current-posterior choice $\mu_t = \rho_t$ removes the explicit density-ratio correction in (32) and gives the KL-DEC offset. In this specialization the potential is the model-coordinate log posterior. In action-index AIR it is instead the log posterior of the optimal action or policy. The latter is the sharper choice whenever the decision problem only requires the index and not the whole environment.

Lemma 4.5 (MAIR gradient bracket and three canonical specializations)**.** *Assume the observation laws are dominated and the displayed derivatives exist. For fixed s, p, ρ, μ, up to an additive constant on the probability simplex,*

$$\frac{\partial}{\partial\mu(\omega)}\mathrm{MAIR}_{\rho,\gamma}(s,p,\mu) = \ell_\omega(s,p) - \gamma\log\frac{\mu(\omega)}{\rho(\omega)} - \gamma\mathbb{E}_{a\sim p}D_{\mathrm{KL}}(P_{\omega,s,p,a}\|P_{\mu,s,p,a}).$$

Consequently, for every $\omega^\star$,

$$\begin{aligned}&\mathrm{MAIR}_{\rho,\gamma}(s,p,\mu) + \langle\nabla_\mu\mathrm{MAIR}_{\rho,\gamma}(s,p,\mu), \delta_{\omega^\star} - \mu\rangle \\ &\quad= \ell_{\omega^\star}(s,p) - \gamma\mathbb{E}_{a\sim p}D_{\mathrm{KL}}(P_{\omega^\star,s,p,a}\|P_{\mu,s,p,a}) - \gamma\log\frac{\mu(\omega^\star)}{\rho(\omega^\star)}.\end{aligned} \tag{34}$$

This bracket has three common specializations.

1. If $\mu = \rho$, the logarithmic correction vanishes and the bracket becomes the KL-DEC offset

$$\ell_{\omega^\star}(s,p) - \gamma\mathbb{E}_{a\sim p}D_{\mathrm{KL}}(P_{\omega^\star,s,p,a}\|P_{\mu,s,p,a}). \tag{35}$$

2. If $\bar\mu \in \arg\max_{\mu\in\Delta(\Omega)}\mathrm{MAIR}_{\rho,\gamma}(s,p,\mu)$ is an interior maximizer, then

$$\mathrm{MAIR}_{\rho,\gamma}(s,p,\bar\mu) + \langle\nabla_\mu\mathrm{MAIR}_{\rho,\gamma}(s,p,\bar\mu), \delta_{\omega^\star} - \bar\mu\rangle \le \mathrm{MAIR}_{\rho,\gamma}(s,p,\bar\mu). \tag{36}$$

This is the model-index AMS/EBO bracket control.

3. Let $p_{\omega,a} = dP_{\omega,s,p,a}/d\nu_{s,p,a}$ and define, for an auxiliary pair $(\bar a, \bar o)$,

$$\frac{d\mu^{1/2}_{\bar a,\bar o}}{d\rho}(\omega) := \frac{\sqrt{p_{\omega,\bar a}(\bar o)}}{\int\sqrt{p_{\omega',\bar a}(\bar o)}\,\rho(d\omega')}. \tag{37}$$

Let $h^2(P,Q) := 1 - \int\sqrt{dP\,dQ}$. If $\bar a \sim p$ and $\bar o \sim P_{\omega^\star,s,p,\bar a}$, then

$$\begin{aligned}&\mathbb{E}_{\bar a,\bar o}\Big[\mathrm{MAIR}_{\rho,\gamma}(s,p,\mu^{1/2}_{\bar a,\bar o}) + \langle\nabla_\mu\mathrm{MAIR}_{\rho,\gamma}(s,p,\mu^{1/2}_{\bar a,\bar o}), \delta_{\omega^\star} - \mu^{1/2}_{\bar a,\bar o}\rangle\Big] \\ &\quad\le \ell_{\omega^\star}(s,p) - \gamma\mathbb{E}_{a\sim p}\mathbb{E}_{\omega\sim\rho}h^2(P_{\omega^\star,s,p,a}, P_{\omega,s,p,a}).\end{aligned} \tag{38}$$

Thus the square-root posterior turns the MAIR bracket into a Hellinger DEC offset.

Proof. See Appendix H.10.

5 Upper bounds: one identity and four algorithm families

We first isolate the fixed-truth log-potential telescope and then specify when it is an AIR/MAIR identity. The potential is the logarithmic score of the maintained index marginal. For AIR/MAIR identities, the next marginal is not arbitrary: it is the posterior index marginal produced by a specified pair belief ν over (Ω, Y). Exact Bayes and stationary-posterior MAIR are special cases of this construction. Confidence-to-belief and exponential-weights procedures can also be used as logarithmic reference updates, but then the displayed Bellman bracket is a bound for that update rather than the AIR gradient identity unless an admissible pair-belief representation has been verified.

Fix a truth $\omega^\star$ and write $y^\star = \chi(\omega^\star)$. At state s, the log potential uses the current reference marginal q_s on the index. A local algorithmic control u specifies an action law $p_u \in \Delta(\mathcal{A}_t(s))$ and a state-measurable reference-index update

$$q_u^+(\cdot \mid s, a, o) \in \Delta(\mathcal{Y}).$$

In the AIR/MAIR specialization, this update is generated by a pair belief $\nu_u \in \Delta(\Omega \times \mathcal{Y})$, namely

$$q_u^+(\cdot \mid s, a, o) = q_{\nu_u}^+(\cdot \mid s, p_u, a, o),$$

with q_ν^+ defined in (21). Exact Bayes and stationary-posterior MAIR are ν-generated updates. Confidence-to-belief and exponential-weights procedures may also define logarithmic reference updates, but then the theorem uses the corresponding state-measurable update directly; such a bound should not be called the AIR gradient identity unless an admissible pair-belief representation has been verified.

The fixed-truth reference log gain used by the Bellman program is

$$\mathsf{G}_\chi(s, u; \omega^\star) := \mathbb{E}_{a \sim p_u,\, O \sim P_{\omega^\star,t}(\cdot|s,p_u,a)} \log \frac{q_u^+(y^\star \mid s, a, O)}{q_s(y^\star)}. \tag{39}$$

When $u = (p, \nu)$ is represented by an AIR pair belief, we also write $\mathsf{G}_\chi(s, p, \nu; \omega^\star)$. In that case $q_u^+ = q_\nu^+$, and G_χ is the fixed-truth AIR posterior-coordinate log gain. When ν is the exact posterior pair law, this is $J_\chi(s, p; \omega^\star)$ from (28). Thus the log potential is always the coordinate log score of a specified reference index marginal, but the AIR/MAIR regret identity additionally requires that the next marginal be the ν-posterior coordinate associated with the same pair belief used in the AIR functional.

For any predictable continuation potential V_t on states and any coefficient $\gamma > 0$, define the specialized fixed-truth logarithmic potential

$$\Phi_t^{V,\gamma}(s; y^\star) := V_t(s) + \gamma \log \frac{1}{q_s(y^\star)}. \tag{40}$$

The corresponding log-potential Bellman bracket is

$$\mathfrak{B}_\chi^{V,\gamma}(s, u; \omega^\star) := \ell_{\omega^\star}(s, p_u) + \mathbb{E}_{\omega^\star}[V_{t+1}(S^+) \mid s, u] - V_t(s) - \gamma \mathsf{G}_\chi(s, u; \omega^\star). \tag{41}$$

Here $A \sim p_u$, $O \sim P_{\omega^\star,t}(\cdot \mid s, p_u, A)$, the reference part of the next state is updated by $q_u^+(\cdot \mid s, A, O)$, and any other state variables are updated by the rule specified by u. In the AIR specialization $u = (p, \nu)$, this bracket is written $\mathfrak{B}_\chi^{V,\gamma}(s, p, \nu; \omega^\star)$. The identity behind the upper section is the telescope

$$\begin{aligned}\mathbb{E}_{\omega^\star} L_{\omega^\star}(H_T) = V_1(s_1) - \mathbb{E}_{\omega^\star} V_{T+1}(S_{T+1}) + \gamma \log \frac{1}{q_1(y^\star)} - \gamma \mathbb{E}_{\omega^\star} \log \frac{1}{q_{T+1}(y^\star)} \\ + \mathbb{E}_{\omega^\star} \sum_{t=1}^{T} \mathfrak{B}_\chi^{V,\gamma}(S_t, u_t; \omega^\star).\end{aligned} \tag{42}$$

Retaining the terminal term gives the sharp identity for the analyzed algorithm. For a finite or countable index, the common upper bound drops the nonpositive term $-\gamma \mathbb{E} \log(1/q_{T+1}(y^\star))$; the overestimate from this simplification is exactly the remaining coordinate code length. For a non-atomic index represented by densities, the terminal term need not have this sign and requires separate control. This is an upper identity, not a minimax lower bound. A Bellman–Fano lower bound is still a separate argument, but it is naturally matched to the same coordinate because it controls how much index entropy can remain under low-regret ghost histories.

Thus an upper bound is a Bellman supersolution for this log-potential bracket. The exact indexed AIR Bellman program optimizes dynamically over the state, action law, and any algorithmically chosen reference-update rule. If that update rule is represented by a pair belief, the pair belief is an algorithmic prediction variable. By contrast, AMS/EBO places candidate beliefs under a robust maximization and uses a saddle or first-order condition to control the same fixed-truth AIR bracket. UCB controls the bracket by calibration and optimism, while E2D drops or bounds the continuation to obtain a one-step offset.

Theorem 5.1 (Generic fixed-truth indexed AIR/MAIR upper bound)**.** *Fix $\omega^\star$ and $y^\star = \chi(\omega^\star)$. Assume $q_1(y^\star) > 0$ and that the reference update remains positive on $y^\star$ along the analyzed trajectory. Assume also that the point-coordinate or dominated-density convention applies and that the losses, log ratios, continuation potentials, and displayed error sum are integrable. If an algorithm chooses a decision law p_t and a state-measurable prediction belief ν_t so that the reference update is $q_{t+1} = q^+_{\nu_t}(\cdot \mid S_t, p_t, A_t, O_t)$ and, for every reached state,*

$$\mathfrak{B}^{V,\gamma}_{\chi}(S_t, p_t, \nu_t; \omega^\star) \leq \varepsilon_t,$$

then

$$\mathbb{E}^{\mathsf{Alg}}_{\omega^\star} L_{\omega^\star}(H_T) \leq V_1(s_1) + \gamma \log \frac{1}{q_1(y^\star)} - \mathbb{E}^{\mathsf{Alg}}_{\omega^\star} V_{T+1}(S_{T+1}) - \gamma \mathbb{E}^{\mathsf{Alg}}_{\omega^\star} \log \frac{1}{q_{T+1}(y^\star)} + \mathbb{E}^{\mathsf{Alg}}_{\omega^\star} \sum_{t=1}^{T} \varepsilon_t. \tag{43}$$

Proof. See Appendix H.11.

It is useful to keep one concrete state specialization in mind. A fixed estimation procedure maps histories to a reference update. In AIR form the update is represented by a pair belief ν_t over (Ω, Y); in an exact model-posterior specialization it is a belief over environments with $Y = \chi(\Omega)$. The retained state includes the current reference index marginal

$$q_t \in \Delta(\mathcal{Y}), \tag{44}$$

and, when the update is posterior based, the next marginal is $q_{t+1} = q^+_{\nu_t}(\cdot \mid S_t, p_t, A_t, O_t)$. In a well-specified Bayes analysis, ν_t is the exact posterior over $(\Omega, \chi(\Omega))$. In a frequentist analysis, the state may instead contain a calibrated algorithmic belief, confidence object, or exponential-weights density. Such an object is valid in the AIR identity only when it induces an admissible pair belief and posterior-coordinate update; otherwise it gives a separate log-score bound whose validity must be proved through calibration or domination. Posterior averaging of the coordinate potential gives

$$\mathbb{E}_{Y \sim \nu_\chi} \gamma \log \frac{1}{q_s(Y)} = \gamma H(\nu_\chi) + \gamma D_{\mathrm{KL}}(\nu_\chi \| q_s), \tag{45}$$

and for a finite index set the same logarithmic scoring rule has the KL-dual log-partition form

$$\Psi_\gamma(q, z) := \gamma \log \sum_{y \in \mathcal{Y}} q(y) e^{z_y/\gamma} = \sup_{r \in \Delta(\mathcal{Y})} \left\{ \langle r, z \rangle - \gamma D_{\mathrm{KL}}(r \| q) \right\}. \tag{46}$$

The coordinate log loss, its posterior average, and the dual log partition are three representations of the same KL/logarithmic scoring rule. They are not interchangeable in a proof: fixed-truth guarantees use (40), posterior-averaged Bayes identities use (45), and AMS/EBO relaxations use (46).

In the model-index specialization, write the posterior/reference belief as μ_t. If the maintained reference update is exact Bayes, then $\mathsf{G}_\chi = J_\chi$ and the fixed-truth MAIR information increment is

$$J_t^{\mathrm{MAIR}}(p;\omega^\star) := \mathbb{E}_{a\sim p} D_{\mathrm{KL}}(P_{\omega^\star,t}(\cdot \mid S_t, a) \,\|\, P_{\mu_t,t}(\cdot \mid S_t, a)) \,. \tag{47}$$

The action-index AIR version is identical with the index $Y = A^\star$, the reference marginal q_t, and the AIR posterior coordinate $q^+(a^\star \mid a, o)$ in place of the model posterior.

5.1 Information-potential Bellman programming

The canonical frequentist upper object is an indexed AIR log-penalized Bellman program on calibrated state/index representations. It is an exact dynamic program for the specified state, comparison sets, information potential, and control family, but it is not asserted to equal the unrestricted minimax value. Let $\mathfrak{C}_t(s)$ be the set of environments that the algorithmic state regards as admissible at state s. Let $\mathfrak{U}_t(s)$ be the local set of admissible algorithmic controls. A control $u \in \mathfrak{U}_t(s)$ specifies an action law $p_u \in \Delta(\mathcal{A}_t(s))$ and a reference-update rule. When this update is represented by a pair belief, write that belief as ν_u and set $q^+ = q^+_{\nu_u}$. If the pair belief is fixed by the state, then u is simply the action law together with this fixed update. The robust AMS/EBO belief variable is not this algorithmic control; it is introduced later under a supremum to control the bracket. For $\gamma > 0$, define

$$W_{T+1}^\gamma(s) = 0, \qquad W_t^\gamma(s) := \inf_{u\in\mathfrak{U}_t(s)} \sup_{\omega\in\mathfrak{C}_t(s)} \left\{\ell_\omega(s, p_u) + \mathbb{E}_\omega[W_{t+1}^\gamma(S^+) \mid s, u] - \gamma \mathsf{G}_\chi(s, u; \omega)\right\}. \tag{48}$$

A measurable selector attaining the infimum in (48) is the information-penalized Bellman dynamic-programming algorithm. In the common AIR specialization, $u = (p, \nu)$ and

$$\mathsf{G}_\chi(s, u; \omega) = \mathsf{G}_\chi(s, p, \nu; \omega).$$

In exact Bayes, ν is fixed by the posterior state; in a non-AIR log-score bound, u specifies the state-measurable reference update directly. This construction builds on the principle of admissible relaxations (Rakhlin et al., 2012), which organize learning through dynamic programming over loss states, and is particularly close to the partial-information relaxation viewpoint of Rakhlin and Sridharan (2016), where estimator-like state variables and potential terms quantify uncertainty. It can also be viewed as a sequential analogue of classical minimum-complexity, information-risk, and Gibbs-type estimation in fixed experiments (Barron and Cover, 1991; Yang and Barron, 1999; Zhang, 2006). The key new feature is that a generic logarithmic mass penalty is propagated through a controlled Bellman recursion, rather than applied only once to a terminal estimator, and the recursion is carried by a Bellman-sufficient state together with an explicit information index.

Theorem 5.2 (Frequentist log-penalized Bellman upper bound). *Fix $\Omega_0 \subseteq \Omega$ and $\gamma > 0$. Suppose the reference update and the comparison sets $\mathfrak{C}_t$ are surely calibrated for Ω_0: for every $\omega^\star \in \Omega_0$, one has $\omega^\star \in \mathfrak{C}_t(S_t)$ almost surely whenever round t is reached. Suppose the algorithm chooses a local control $u_t \in \mathfrak{U}_t(S_t)$, writes $p_t = p_{u_t}$ for its action law, updates the index marginal by the reference-update rule contained in u_t, and satisfies*

$$\sup_{\omega\in\mathfrak{C}_t(S_t)} \left\{\ell_\omega(S_t, p_{u_t}) + \mathbb{E}_\omega[W_{t+1}^\gamma(S^+) \mid S_t, u_t] - \gamma \mathsf{G}_\chi(S_t, u_t; \omega)\right\} \le W_t^\gamma(S_t) + \varepsilon_t.$$

Then, for every fixed truth $\omega^\star \in \Omega_0$ with $y^\star = \chi(\omega^\star)$ and $q_1(y^\star) > 0$, provided the index is finite or countable, the reference update remains positive on $y^\star$ along the trajectory, and the losses, coordinate log ratios, continuation values, and displayed error sum are integrable,

$$\mathbb{E}_{\omega^\star}^{\mathsf{Alg}} L_{\omega^\star}(H_T) \le W_1^\gamma(s_1) + \gamma \log \frac{1}{q_1(y^\star)} + \mathbb{E}_{\omega^\star}^{\mathsf{Alg}} \sum_{t=1}^{T} \varepsilon_t. \tag{49}$$

Proof. See Appendix H.12.

Remark 5.3 (High-probability calibration)**.** *The preceding theorem deliberately assumes sure state-wise calibration. A future-dependent event $\mathcal{E}$ cannot simply be inserted into the conditional Bellman brackets and charged afterward by $\mathbb{E}[L^{+}\mathbf{1}\{\mathcal{E}^c\}]$: conditioning a continuation value and then restricting to $\mathcal{E}$ does not preserve the displayed Bellman inequality. A high-probability version requires a separate stopped-process or supermartingale argument, or explicit predictable failure brackets. In the kernel-UCB result below we therefore prove the deterministic regret inequality on its confidence event directly and only then take expectations.*

The posterior-averaged Bayesian recursion is a specialization rather than the primitive fixed-truth statement. If the robust comparison set is replaced by the current posterior law and one averages the coordinate identity over $Y \sim q_s$, then

$$\mathbb{E}[H_\chi(S_{t+1}) \mid s, p] - H_\chi(s) = -\mathcal{I}_\chi(s, p).$$

For a fixed multiplier $\gamma > 0$, define the Bayes information-regularized value

$$U^\gamma_{T+1} \equiv 0, \qquad U^\gamma_t(s) := \inf_{p \in \Delta(\mathcal{A}_t(s))} \left\{ \ell(s,p) + \mathbb{E}[U^\gamma_{t+1}(S_{t+1}) \mid s, p] - \gamma \mathcal{I}_\chi(s,p) \right\}. \tag{50}$$

Equivalently,

$$\Phi^\gamma_t(s) := U^\gamma_t(s) + \gamma H_\chi(s) \tag{51}$$

solves the admissible-relaxation recursion

$$\Phi^\gamma_t(s) = \inf_{p \in \Delta(\mathcal{A}_t(s))} \left\{ \ell(s,p) + \mathbb{E}[\Phi^\gamma_{t+1}(S_{t+1}) \mid s, p] \right\}, \qquad \Phi^\gamma_{T+1}(s) = \gamma H_\chi(s). \tag{52}$$

Theorem 5.4 (Posterior-averaged information-potential upper bound)**.** *Assume the indexed Bellman representation is exact under the Bayesian mixture law, Y is finite or countable, and $H_\mu(Y) < \infty$. If a policy chooses p_t satisfying*

$$\ell(S_t, p_t) + \mathbb{E}[U^\gamma_{t+1}(S_{t+1}) \mid S_t, p_t] - \gamma \mathcal{I}_\chi(S_t, p_t) \le U^\gamma_t(S_t) + \varepsilon_t,$$

then

$$\mathbb{E} L_\Omega(H_T) \le U^\gamma_1(s_1) + \gamma I_\mu(Y; H_T) + \mathbb{E}\sum_{t=1}^T \varepsilon_t \le U^\gamma_1(s_1) + \gamma H_\mu(Y) + \mathbb{E}\sum_{t=1}^T \varepsilon_t. \tag{53}$$

This is the posterior average of the fixed-truth coordinate telescope, not a pointwise replacement for Theorem 5.2.

Proof. See Appendix H.13.

The minimax information-risk sandwich and the entropy/regret comparison are stated in Section 3. The rest of this upper-bound section develops the upper side of that sandwich: the fixed-truth coordinate telescope, the exact log-penalized Bellman program, and the main tractable relaxations.

5.2 UCB families: calibration plus optimism

An upper-confidence-bound (UCB) algorithm maintains a prediction and an uncertainty scale, builds a confidence band from them, and chooses an action that is optimistic within that band (Auer et al., 2002; Abbasi-Yadkori et al., 2011; Srinivas et al., 2010). We use the following notation throughout. The unmultiplied uncertainty scale is denoted by $\sigma_t(a)$ in the generic discussion and by $s_t(x)$ in the GP specialization below. The confidence half-width is

$$w_t(a) = \beta_t \sigma_t(a),$$

where β_t is the confidence multiplier. Calibration says that the fixed truth is contained in the band of radius $w_t(a)$; optimism says that the selected action maximizes the upper envelope. Together, in reward-maximization problems, these two facts give

$$r_{\omega^\star}(s, a^\star) \le m_t(a^\star) + w_t(a^\star) \le m_t(a_t) + w_t(a_t) \le r_{\omega^\star}(s, a_t) + 2w_t(a_t),$$

and hence the instantaneous regret is at most $2\beta_t\sigma_t(a_t)$. An elliptical-potential/log-determinant lemma then controls the realized sum of $\sigma_t^2(a_t)$. Thus UCB does not solve the AIR/MAIR bracket optimization directly; it controls the relevant fixed-truth bracket after the optimistic action has been chosen.

Lemma 5.5 (Calibration and optimism imply bracket control)**.** *Fix a truth $\omega^\star$ and condition on an event $\mathcal{E}$ on which the algorithm's state is calibrated for that truth. Suppose that, for every reached round t, the action a_t chosen by the algorithm and a predictable uncertainty scale $\sigma_t(a) \ge 0$ satisfy*

$$\ell_{\omega^\star}(S_t, a_t) \le c_{\mathrm{opt}}\, \beta_t \sigma_t(a_t), \tag{54}$$

where c_{opt} is a numerical optimism constant; in the usual two-sided reward-confidence proof above, $c_{\mathrm{opt}} = 2$. Then for every $\gamma > 0$,

$$\ell_{\omega^\star}(S_t, a_t) - \gamma\sigma_t^2(a_t) \le \frac{c_{\mathrm{opt}}^2 \beta_t^2}{4\gamma}. \tag{55}$$

Consequently, for the deterministic-action fixed-truth bracket in (30),

$$B_{\chi,\gamma}(S_t, \delta_{a_t}; \omega^\star) \le \frac{c_{\mathrm{opt}}^2 \beta_t^2}{4\gamma} + \gamma\left(\sigma_t^2(a_t) - J_\chi(S_t, \delta_{a_t}; \omega^\star)\right), \tag{56}$$

where the fixed-truth index log gain is defined in (28). If, in addition, the log-determinant/elliptical-potential argument gives a realized bound

$$\sum_{t=1}^{T} \sigma_t^2(a_t) \le c_{\mathrm{sn}} G_T,$$

for an information telescope G_T, then summing (56) reduces UCB regret control to the comparison between G_T and the fixed-truth log-gain telescope.

Proof. See Appendix H.14.

The correction $\sigma_t^2(a_t) - J_\chi(S_t, \delta_{a_t}; \omega^\star)$ is not a second estimation cost. It is the algebraic difference between the tractable self-normalized variance proxy and the fixed-truth posterior log gain appearing in the bracket. In linear-UCB and GP–UCB proofs, the log-determinant or elliptical-potential argument shows that the realized sum of $\sigma_t^2(a_t)$ is controlled by an information telescope. The confidence multiplier β_t is separate: it is the price of converting calibrated uncertainty into regret through optimism. Therefore

$$\text{self-normalized calibration} + \text{optimism} \quad \Longrightarrow \quad \text{fixed-truth AIR/MAIR bracket control.}$$

In this organizing perspective, the round-heterogeneous summation of unmultiplied uncertainty scales, usually controlled by an elliptical-potential or log-determinant argument, plays the role of a realized estimation-complexity telescope. The confidence multiplier enters instead through the regret-to-uncertainty conversion, and hence through the coefficient in the fixed-truth bracket. This separates two mechanisms that are often blurred in informal UCB discussions: posterior or confidence widths quantify the "estimation complexity" of the underlying parameter, as explained by Lattimore (2023), while optimism and calibration determine how that geometry pays for regret. The resulting unification reveals a common posterior-telescoping structure while keeping distinct the estimation telescope and the coefficient mechanisms behind UCB, E2D, and AMS/EBO.

5.3 E2D: robust one-step offset optimization

The estimation-to-decision (E2D) algorithms (Foster et al., 2021) choose p by solving a one-round robust DEC offset at the current time-state pair. In model-index notation the schematic KL form is

$$\inf_{p\in\Delta(\mathcal{A}_t(s))} \sup_{\omega\in\mathfrak{C}_t(s)} \left\{\ell_\omega(s,p) - \gamma\mathbb{E}_{a\sim p} D_{\mathrm{KL}}(P_{\omega,s,p,a} \| P_{\rho,s,p,a})\right\}, \tag{57}$$

where ρ is a reference belief or reference model and $\mathfrak{C}_t(s)$ is a localized comparison set that contains the fixed truth on the calibration event. The notation is intentionally aligned with the log-penalized Bellman program. As Lemma 4.5 illustrates, the E2D objective can be recovered from the one-step MAIR bracket: it uses the current feasible action set and localized comparison set, but replaces the dynamic continuation value by a one-step separation penalty. The same one-round minimax principle applies to the original Hellinger formulation and to constrained variants (Foster et al., 2023).

What E2D keeps is the immediate regret and a one-step statistical-separation penalty. What it drops, freezes, or upper bounds is the Bellman continuation term $\mathbb{E}_\omega[V_{t+1}(S^+) \mid s,p] - V_t(s)$, together with the exact coordinate reference log gain G_χ in (41). The stationary-posterior recovery (35) and the stationary-square-root-posterior recovery (38) make this reduction from the MAIR bracket explicit. Thus E2D is not an optimizer of the full Bellman potential unless one proves that the omitted continuation is dominated by the displayed one-step penalty. In an exact Bayesian calculation, posterior averaging can turn the KL penalty into a one-step mutual-information term. In a fixed-truth or frequentist calculation, the corresponding statement is the reference posterior-ratio identity for G_χ, or its exact-Bayes specialization J_χ, plus whatever calibration or localization is needed to compare $P_{\rho,s,p,a}$ with the reference predictive law in that identity. The safe interpretation is therefore: E2D is a one-step relaxation, re-solved at each state, of the fixed-truth log-posterior Bellman bracket.

5.4 AMS/EBO: robust convex belief optimization

The Exploration by Optimization (EBO) algorithm (Lattimore and György, 2021; Foster et al., 2022) and the Adaptive Minimax Sampling (AMS) algorithm (Xu and Zeevi, 2025; Liu et al., 2025, 2026) should be understood as tractable robust relaxations of the log-penalized Bellman program, not as the same minimization over the algorithmic update variable in (48). AMS may be viewed as a constructive implementation of the EBO principle, replacing abstract functional-estimator optimization with an executable sampling rule. Work with an action or policy index $Y = \chi(\Omega)$. For clarity, this subsection states the coordinate argument for finite $\mathcal{Y}$; a dominated extension requires explicit positivity, absolute-continuity, and differentiability hypotheses. At state s, let $q \in \Delta(\mathcal{Y})$ be positive on every coordinate evaluated by the logarithmic score and let $\mathfrak{B}_t(s)$ be a calibrated convex set of candidate pair beliefs $\tilde{\nu}$ on (ω, y), supported on $y = \chi(\omega)$. In this subsection the candidate belief is a robust comparison variable controlled by the inner maximization; after the maximizer $\bar{\nu}_t$ is selected, its posterior index update is used to control the fixed-truth AIR bracket. The one-step robust AIR/EBO objective is the indexed AIR functional from (22), equivalently

$$\mathfrak{A}_{q,\gamma}(s, p, \nu) = \ell_\nu(s, p) - \gamma \mathcal{I}_\chi(s, p; \nu) - \gamma D_{\mathrm{KL}}(\nu_\chi \| q), \tag{58}$$

where ν_χ is the index marginal, $\ell_\nu(s, p) = \mathbb{E}_{(\omega,y)\sim\nu} \ell_\omega(s, p)$, and $\mathcal{I}_\chi(s, p; \nu) = \mathbb{E}_{a\sim p} I_\nu(Y; O \mid s, p, a)$. The robust maximization over beliefs is essential:

$$p_t \in \operatorname*{arg\,min}_{p\in\Delta(\mathcal{A}_t(s))} \max_{\tilde{\nu}\in\mathfrak{B}_t(s)} \mathfrak{A}_{q_t,\gamma}(s, p, \tilde{\nu}), \qquad \bar{\nu}_t \in \operatorname*{arg\,max}_{\tilde{\nu}\in\mathfrak{B}_t(s)} \mathfrak{A}_{q_t,\gamma}(s, p_t, \tilde{\nu}). \tag{59}$$

Without the inner $\max_{\tilde{\nu}}$, the display is only a posterior AIR calculation for a chosen belief, not a robust AMS/EBO rule.

The convex-analytic interpretation is the following. The fixed-truth proof uses the coordinate log loss $\gamma \log(1/q_t(y^\star))$. Averaging this coordinate under a candidate belief gives $\gamma H(\nu_\chi) + \gamma D_{\mathrm{KL}}(\nu_\chi \| q_t)$. When the next reference marginal is the posterior coordinate induced by ν, the expected change of this averaged potential is $-\gamma \mathcal{I}_\chi(s, p; \nu) - \gamma D_{\mathrm{KL}}(\nu_\chi \| q_t)$. Equivalently, optimizing over possible next index laws yields the log-partition dual (46). Hence the EBO potential is the KL-dual log partition, while the fixed-truth AIR argument uses the coordinate posterior log loss; they coincide only through KL duality and the chosen update, not as identical scalar functions.

For the AIR functional in (58), concavity in $\tilde{\nu}$ follows from the posterior-scoring representation in Appendix B: it is the infimum over prediction rules Q of an objective affine in $\tilde{\nu}$, provided the prediction class and support convention are fixed across beliefs. Thus the concavity hypothesis in Lemma 5.6 is automatic in this setting. For modified belief relaxations or restricted prediction classes, concavity must be verified directly or supplied by a valid concave upper bound.

Lemma 5.6 (AMS/EBO saddle control of the AIR fixed-truth bracket)**.** *Assume the finite or dominated differentiability setting of Lemma 4.3. Fix $(\omega^\star, y^\star)$ with $y^\star = \chi(\omega^\star)$. Suppose $\tilde{\nu} \mapsto \mathfrak{A}_{q_t,\gamma}(s, p_t, \tilde{\nu})$ is concave on the convex set $\mathfrak{B}_t(s)$, $\bar{\nu}_t$ maximizes it as in (59), and the comparison direction $\delta_{(\omega^\star,y^\star)} - \bar{\nu}_t$ is admissible for $\mathfrak{B}_t(s)$. If*

$$\max_{\tilde{\nu}\in\mathfrak{B}_t(s)} \mathfrak{A}_{q_t,\gamma}(s, p_t, \tilde{\nu}) \le \varepsilon_t,$$

then the fixed-truth AIR bracket with coefficient γ satisfies

$$\ell_{\omega^\star}(s, p_t) - \gamma \mathbb{E}_{a\sim p_t, O\sim P_{\omega^\star,t}(\cdot|s,p_t,a)} \log \frac{\bar{\nu}_t(Y = y^\star \mid s, p_t, a, O)}{q_t(y^\star)} \le \varepsilon_t. \tag{60}$$

If the reference update is $q_{t+1}(\cdot) = \bar{\nu}_t(Y \in \cdot \mid s, p_t, A_t, O_t)$ and remains positive at $y^\star$, and if all preceding hypotheses hold almost surely along the trajectory, then Theorem 5.1 with $V_t \equiv 0$ gives

$$\mathbb{E}_{\omega^\star} L_{\omega^\star}(H_T) \le \gamma \log \frac{1}{q_1(y^\star)} + \mathbb{E}_{\omega^\star} \sum_{t=1}^{T} \varepsilon_t \tag{61}$$

for finite or countable index sets. If the comparison direction is only approximately admissible, its effect must be bounded explicitly in the one-step bracket and added to ε_t; a future-dependent containment event cannot be appended after the telescope. Averaging this coordinate bound under a Bayesian prior replaces $\log(1/q_1(Y))$ by the corresponding cross-entropy or by $H_\mu(Y)$ when $q_1 = \mu_\chi$.

Proof. See Appendix H.15.

A dynamic AMS/EBO relaxation is obtained by adding a continuation term before the robust maximization:

$$\mathfrak{A}^{V}_{t,q,\gamma}(s,p,\tilde{\nu}) = \ell_{\tilde{\nu}}(s,p) + \mathbb{E}_{\tilde{\nu}} V_{t+1}(S^{+}) - V_t(s) - \gamma \mathcal{I}_{\chi}(s,p;\tilde{\nu}) - \gamma D_{\mathrm{KL}}(\tilde{\nu}_{\chi} \| q).$$

This is a valid relaxation of the robust Bellman recursion (48) only after the same first-order or direct bracket bound is proved. The inner maximization over $\tilde{\nu}$ is a robust upper-bounding device; it does not contradict the Bellman-program minimization over algorithmic controls, because these variables play different roles. The standard one-step EBO display is the case $V_t \equiv 0$; the dynamic version is an admissible relaxation of the central log-penalized Bellman program, not an automatic consequence of the one-step objective.

Remark 5.7 (The one identity and the four families)**.** *The AIR identity is the fixed-truth log-potential telescope* (42) *with a ν-generated posterior index update. The exact information-risk Bellman program is* (48)*, where the minimization is over algorithmic controls and update rules. Posterior-averaged Bayes analysis is obtained by averaging the same coordinate identity, giving* (50)*. UCB controls the bracket through confidence and optimism. E2D replaces the continuation by a one-step separation penalty. AMS/EBO optimizes a KL-dual robust belief relaxation, with the required* $\max_{\tilde{\nu}}$ *over candidate beliefs appearing explicitly. Thus the common object is not a universal one-step coefficient but the logarithmic Bellman bracket on a sufficient state and a chosen index, with algorithmic prediction variables and robust comparison beliefs kept distinct.*

6 Lower bounds: reference histories and quantile indices

6.1 Ghost probability and true good probability

Fix an average-regret threshold $r \geq 0$. The posterior-reference ghost-good probability is

$$p_r^{\mathsf{Alg}}(\mu,\chi) := \mathbb{P}_{Y \sim \mu_{\chi},\, H'_T \sim \bar{\mathbb{P}}_{\mu}^{\mathsf{Alg}}} \left(\bar{L}_{\chi}(Y, H'_T) \leq r \right), \tag{62}$$

where Y and H'_T are independent under the ghost law and μ_χ is the marginal law of Y. The true good probability is

$$q_r^{\mathsf{Alg}}(\mu,\chi) := \mathbb{P}_{Y,H_T} \left(\bar{L}_{\chi}(Y, H_T) \leq r \right), \tag{63}$$

where (Y, H_T) are generated by $\Omega \sim \mu$ and $H_T \sim \mathbb{P}_{\Omega}^{\mathsf{Alg}}$.

The ghost probability is algorithm-dependent. That dependence is intentional: it preserves the reference trajectory geometry instead of replacing it with a worst-case static small ball.

6.2 Reference-history quantile theorem

Let

$$\mathrm{kl}(q,p) = q \log \frac{q}{p} + (1-q) \log \frac{1-q}{1-p}$$

be binary KL divergence. For $p \in [0,1]$ and $c \geq 0$, define

$$\psi(p,c) := \sup\{q \in [0,1] : \mathrm{kl}(q,p) \leq c\}.$$

Theorem 6.1 (Reference-history quantile indexed-information lower bound)**.** *For every prior μ, index map χ, threshold $r \geq 0$, and algorithm Alg, assume that the posterior-reference process satisfies the exact posterior-indexed lift in the sense of Definition 2.1, that O_0 is deterministic or independent of Y, and that the cumulative indexed loss satisfies $L_\chi(Y, H_T) \geq 0$ almost surely under the true mixture law. Then*

$$\mathrm{kl}\big(q_r^{\mathsf{Alg}}(\mu,\chi), p_r^{\mathsf{Alg}}(\mu,\chi)\big) \leq C_\chi^{\mathsf{Alg}}(\mu) = T\bar{C}_\chi^{\mathsf{Alg}}(\mu). \tag{64}$$

Consequently,

$$\mathbb{E}_{\Omega\sim\mu}\mathbb{E}_{\mathbb{P}_\Omega^{\mathsf{Alg}}} L_\Omega(H_T) \geq Tr\left[1 - \psi\big(p_r^{\mathsf{Alg}}(\mu,\chi), T\bar{C}_\chi^{\mathsf{Alg}}(\mu)\big)\right]. \tag{65}$$

In particular, if $p_r^{\mathsf{Alg}}(\mu,\chi) \leq 1/2$ and

$$T\bar{C}_\chi^{\mathsf{Alg}}(\mu) \leq \mathrm{kl}\left(\frac{1}{2}, p_r^{\mathsf{Alg}}(\mu,\chi)\right), \tag{66}$$

then

$$\mathbb{E}_{\Omega\sim\mu}\mathbb{E}_{\mathbb{P}_\Omega^{\mathsf{Alg}}} \bar{L}_\Omega(H_T) \geq \frac{r}{2}, \qquad \mathbb{E}_{\Omega\sim\mu}\mathbb{E}_{\mathbb{P}_\Omega^{\mathsf{Alg}}} L_\Omega(H_T) \geq \frac{Tr}{2}. \tag{67}$$

A convenient sufficient condition is

$$T\bar{C}_\chi^{\mathsf{Alg}}(\mu) + \log 2 \leq \frac{1}{2}\log\frac{1}{p_r^{\mathsf{Alg}}(\mu,\chi)}. \tag{68}$$

If O_0 may be informative about Y, the same conclusions hold after replacing every occurrence of $T\bar{C}_\chi^{\mathsf{Alg}}(\mu)$ in the information budget by

$$I_\mu(Y; O_0) + T\bar{C}_\chi^{\mathsf{Alg}}(\mu).$$

Proof. See Appendix H.16.

The relation to interactive Fano and the posterior-reference chain rule is detailed in Appendix C.

6.3 Bellman–Fano lower-bound method

Theorem 6.1 is algorithm-specific. We now give an algorithm-uniform lower-bound method with two components: the first bounds the information of any algorithm, and the second bounds the ghost probability of success of any algorithm.

Exact information Bellman value and supersolutions. The intrinsic algorithm-uniform information quantity is an exact Bellman value. For a function F on states, define

$$(\mathsf{C}_t F)(s) := \sup_{p\in\Delta(\mathcal{A}_t(s))} \left\{\mathcal{I}_\chi(s,p) + \mathbb{E}_{a\sim p, o\sim P_{s,p,a}} F(\tau(s,p,a,o))\right\}.$$

Set

$$C_{T+1}^\star(s) = 0, \qquad C_t^\star(s) = (\mathsf{C}_t C_{t+1}^\star)(s).$$

Then $C_1^\star(s_1)$ is the exact posterior-reference indexed-information capacity on the retained state. A sequence $\overline{C}_t$ is a valid information supersolution if

$$\overline{C}_{T+1} \geq 0, \qquad \overline{C}_t(s) \geq (\mathsf{C}_t \overline{C}_{t+1})(s) \quad \forall t, s.$$

For every algorithm starting from s_1,

$$C_\chi^{\mathsf{Alg}}(\mu) \leq C_1^\star(s_1) \leq \overline{C}_1(s_1). \tag{69}$$

Thus the exact Bellman recursion is the tight intrinsic object; supersolutions are used only as computable or analytically convenient upper bounds.

Exact ghost-good Bellman value and supersolutions. Let $m \in \Delta(\mathcal{Y})$ be the target-index distribution used to evaluate success, let s be the posterior-reference state, and let $g : \mathcal{Y} \to \mathbb{R}$ be an accumulated average-loss profile. For threshold r, define the exact ghost-good Bellman value by

$$\Gamma^{r,\star}_{T+1}(m, s, g) = m\{y : g(y) \le r\},$$

$$\Gamma^{r,\star}_{t}(m, s, g) = \sup_{p\in\Delta(\mathcal{A}_t(s))} \mathbb{E}_{a\sim p, o\sim P_{s,p,a}} \Gamma^{r,\star}_{t+1}\left(m, \tau(s, p, a, o), g + \frac{1}{T}\ell^{\chi}_{\cdot}(s, p, a)\right),$$

where $\ell^{\chi}_{\cdot}(s, p, a)$ is the function $y \mapsto \ell^{\chi}_{y}(s, p, a)$. A function sequence $\overline{\Gamma}^{r}_{t}$ is a valid ghost-mass supersolution if it dominates the terminal condition and the same Bellman right-hand side with $\overline{\Gamma}^{r}_{t+1}$ in place of $\Gamma^{r,\star}_{t+1}$. We henceforth take ghost-mass supersolutions to be $[0, 1]$-valued. Any nonnegative supersolution may be clipped at one, since the ghost Bellman operator is monotone and maps $[0, 1]$-valued functions into $[0, 1]$; this clipping preserves the supersolution inequality. Then, for every algorithm,

$$p^{\mathsf{Alg}}_{r}(\mu, \chi) \le \Gamma^{r,\star}_{1}(\mu_\chi, s_1, 0) \le \overline{\Gamma}^{r}_{1}(\mu_\chi, s_1, 0). \tag{70}$$

The exact ghost entropy and its computable lower bound are therefore

$$\mathcal{E}^{r}_{\star,1}(\mu, \chi) := -\log \Gamma^{r,\star}_{1}(\mu_\chi, s_1, 0), \qquad \underline{\mathcal{E}}^{r}_{1}(\mu, \chi) := -\log \overline{\Gamma}^{r}_{1}(\mu_\chi, s_1, 0). \tag{71}$$

The entropy lower bound is conservative: $\underline{\mathcal{E}}^{r}_{1} \le \mathcal{E}^{r}_{\star,1}$.

Theorem 6.2 (Bellman–Fano quantile-index lower bound)**.** *Assume that the retained state is an exact posterior-indexed lift under μ, that the indexed information terms are well defined and integrable, and that a fixed jointly measurable version of $\ell^{\chi}_{y}(s, p, a)$ is used in the ghost product experiment. The theorem is stated conditionally on a fixed initial packet and resulting state s_1; an unconditional result for a random initial state follows by applying the full conditional bound and then averaging, as described in Section 2. Assume the cumulative indexed loss is nonnegative as in Theorem 6.1. Suppose $\overline{C}_t$ is an information supersolution and $\overline{\Gamma}^{r}_{t}$ is a $[0, 1]$-valued ghost-good-mass supersolution; the exact choices $\overline{C} = C^\star$ and $\overline{\Gamma} = \Gamma^{r,\star}$ give the tight intrinsic Bellman–Fano values. The statement is for the unrestricted nonanticipating class; for a Bellman-compatible restricted class, replace each local supremum in the exact recursions and in the supersolution recursions by the corresponding local feasible set. Then*

$$\inf_{\mathsf{Alg}\in\mathfrak{A}^{\mathrm{na}}_{T}} \mathbb{E}_{\Omega\sim\mu} \mathbb{E}_{\mathbb{P}^{\mathsf{Alg}}_{\Omega}} L_\Omega(H_T) \ge Tr\left[1 - \psi\left(e^{-\underline{\mathcal{E}}^{r}_{1}(\mu,\chi)}, \overline{C}_1(s_1)\right)\right]. \tag{72}$$

In particular, if

$$\overline{C}_1(s_1) + \log 2 \le \frac{1}{2}\underline{\mathcal{E}}^{r}_{1}(\mu, \chi), \tag{73}$$

then

$$\inf_{\mathsf{Alg}\in\mathfrak{A}^{\mathrm{na}}_{T}} \mathbb{E}_{\Omega\sim\mu} \mathbb{E}_{\mathbb{P}^{\mathsf{Alg}}_{\Omega}} L_\Omega(H_T) \ge \frac{Tr}{2}. \tag{74}$$

Consequently,

$$\mathfrak{R}^{\star}_{T} \ge \sup_{\mu,\chi,r} \left\{\frac{Tr}{2} : \overline{C}_1(s_1) + \log 2 \le \frac{1}{2}\underline{\mathcal{E}}^{r}_{1}(\mu, \chi)\right\}.$$

Proof. See Appendix H.17.

The critical average-regret radius is

$$r_\star(\mu, \chi) := \sup\left\{r : \overline{C}_1(s_1) + \log 2 \le \frac{1}{2}\underline{\mathcal{E}}^{r}_{1}(\mu, \chi)\right\}. \tag{75}$$

The lower bound is order $Tr_\star$. A two-point Le Cam proof corresponds to the special case in which $\underline{\mathcal{E}} = O(1)$. A Fano or local-prior-mass lower bound has $\underline{\mathcal{E}}$ of order dimension or effective dimension.

Relation to DEC. The Bellman formulation keeps hard-prior admissibility, evaluation of the upper recursion at the lower entropy scale, and growth of the upper value separate, while DEC derives upper and lower guarantees from a one-step coefficient. Appendix D gives the formal comparison and discusses when a one-step reduction preserves the dynamic rate.

7 Applications: kernel bandits and information-complexity sandwiches

This section contains complementary kinds of applications. Kernel bandits are the major algorithmic application: they illustrate the Bellman decomposition and its consequences for algorithm design through a full-RKHS separation between finite-marginal AIR Bellman programming and canonical maximal-information GP–UCB rules, including the unit-ball exploration schedule of the original RKHS GP–UCB algorithm, together with a comparison to fixed-domain Matérn results.

The Gaussian multi-armed bandit and hypercube linear bandit examples instantiate the lower information–risk construction of Section 3 and yield minimax rates matched by standard algorithms. The detailed analyses are deferred to Appendices F and G.

7.1 Kernel bandits: Bellman decompositions and algorithms

This subsection specializes the indexed Bellman language to kernel bandits on a finite active action set. The point is not to make the unknown function finite-dimensional. The truth remains an element of an RKHS; only the marginal experiment observed through the active actions is finite. This marginal is enough to define the GP posterior update, the model-index MAIR information, and the action-index AIR law of the optimal active action. We present the finite-marginal indexed Bellman program, prove its action-index AIR bound, and then compare it with GP–UCB. Additional kernel details, including GP–E2D and AMS/EBO specializations and the Matérn comparison, are given in Appendix E.

Problem and finite active marginal. Let $\mathcal{H}_k$ be a reproducing kernel Hilbert space (RKHS) on a possibly infinite domain $\mathcal{X}_{\text{all}}$, with kernel k satisfying $k(x,x) \le \kappa^2$. The learner is evaluated on a finite active set

$$\mathcal{X} = \{x_1, \dots, x_n\} \subset \mathcal{X}_{\text{all}}.$$

The unknown reward function $f^\star \in \mathcal{H}_k$ satisfies $\|f^\star\|_{\mathcal{H}_k} \le B$, and at round t the learner chooses $X_t \in \mathcal{X}$ and observes

$$Y_t = f^\star(X_t) + \varepsilon_t, \qquad \mathbb{E}[\varepsilon_t \mid H_{t-1}, X_t] = 0, \tag{76}$$

where the noise is conditionally R-sub-Gaussian. For the Gaussian calculations below we use the algorithmic reference model with variance parameter $\lambda > 0$; the frequentist confidence event is obtained by the usual self-normalized argument.

For an action set $\mathcal{X}$ and reward function f, define cumulative regret by

$$\mathrm{Reg}_T(f; \mathsf{Alg}) := \sum_{t=1}^{T} \left\{ \max_{x \in \mathcal{X}} f(x) - f(X_t) \right\}. \tag{77}$$

We write $\mathrm{Reg}_T(f)$ when the algorithm is clear and $\mathrm{Reg}_T(\mathsf{Alg})$ when the environment is carried by the expectation.

The exact posterior and AIR/MAIR chain-rule statements in this subsection are made under the well-specified Gaussian observation model $\varepsilon_t \overset{\text{i.i.d.}}{\sim} N(0, \lambda)$. The pathwise GP–UCB confidence argument continues to hold for conditionally sub-Gaussian noise, but a Gaussian pseudo-posterior under non-Gaussian noise is only an algorithmic state and does not satisfy the exact Bayesian chain rule without a separate transfer argument.

Let $F = (f(x_1), \ldots, f(x_n)) \in \mathbb{R}^n$ be the active reward vector and let $K_{\mathcal{X}}$ be the $n \times n$ kernel matrix. The GP reference prior is

$$F \sim N(0, K_{\mathcal{X}}), \qquad Y_t \mid F, X_t = x_i \sim N(F_i, \lambda).$$

After history H_{t-1}, the algorithmic posterior on F is Gaussian,

$$F \mid H_{t-1} \sim N(m_t, \Sigma_t).$$

For $x_i \in \mathcal{X}$, write $m_t(x_i) = e_i^\top m_t$ and $s_t^2(x_i) = e_i^\top \Sigma_t e_i$. If $X_t = x_i$, the exact posterior update is

$$m_{t+1} = m_t + \frac{\Sigma_t e_i}{\lambda + s_t^2(x_i)}\{Y_t - m_t(x_i)\}, \tag{78}$$

$$\Sigma_{t+1} = \Sigma_t - \frac{\Sigma_t e_i e_i^\top \Sigma_t}{\lambda + s_t^2(x_i)}. \tag{79}$$

This is a finite marginal posterior update. It should not be read as an assumption that $f^\star$ has n parameters. The RKHS function may be infinite-dimensional; only the observations and decisions in this finite experiment depend on the vector F.

Model-index information and action-index information. For the model-index specialization, take the index to be the active reward vector $Y_{\mathrm{MAIR}} = F$. The one-step GP/MAIR information at state H_{t-1} is

$$\mathcal{I}_t^{\mathrm{GP}}(p) := \mathbb{E}_{x\sim p} \frac{1}{2} \log\left(1 + \frac{s_t^2(x)}{\lambda}\right). \tag{80}$$

Along a realized action sequence,

$$C_T^{\mathrm{GP}} := \sum_{t=1}^{T} \mathcal{I}_t^{\mathrm{GP}}(\delta_{X_t}) = \frac{1}{2} \log\det\left(I + \lambda^{-1} K_{X_{1:T}, X_{1:T}}\right), \tag{81}$$

where repeated actions are included in the Gram matrix. This is the realized information gain of the algorithmic trajectory. It can be much smaller than the maximal information gain that maximizes over all length-T designs.

For the action-index specialization, assume ties are broken by a fixed rule and set

$$A^\star(F) \in \arg\max_{x\in\mathcal{X}} F_x, \qquad q_t(a) := \mathbb{P}(A^\star = a \mid H_{t-1}).$$

Let ν_t^a be the exact conditional law of F given H_{t-1} and $A^\star = a$. It is generally a truncated Gaussian on the cone where a is optimal. Define the conditional predictive law

$$P_{t,a,x}(\cdot) := \int N(F_x, \lambda)(\cdot)\, \nu_t^a(dF),$$

for the observation at action x given the event $A^\star = a$. The marginal predictive is

$$P_{t,x} = \sum_{a\in\mathcal{X}} q_t(a) P_{t,a,x} = N(m_t(x), \lambda + s_t^2(x)).$$

The AIR information increment is

$$\mathcal{I}_t^{\mathrm{AIR}}(p) := \mathbb{E}_{x\sim p} \sum_{a\in\mathcal{X}} q_t(a) D_{\mathrm{KL}}(P_{t,a,x} \| P_{t,x}). \tag{82}$$

The conditional AIR regret is

$$\Delta_t^{\mathrm{AIR}}(p) := \sum_{a\in\mathcal{X}} q_t(a) \mathbb{E}_{F\sim\nu_t^a} \mathbb{E}_{x\sim p}[F_a - F_x]. \tag{83}$$

The exact indexed chain rule gives

$$\sum_{t=1}^{T} \mathbb{E}\mathcal{I}_t^{\mathrm{AIR}}(p_t) = I(A^\star; H_T) \le H(A^\star) \le \log n.$$

Thus AIR places the entropy telescope on the decision index rather than on the ambient RKHS. The price is that the coefficient converting action-index information into regret must be controlled by an algorithmic bracket.

Algorithm 1: finite-marginal indexed Bellman DP. On the finite active marginal, the canonical frequentist log-potential upper benchmark is the Bellman program of Theorem 5.2. Its action-index branch is AIR, while its model-index branch is MAIR. The program is relative to the displayed state, comparison sets, and reference update, and need not equal the unrestricted minimax value. The state contains (m_t, Σ_t, q_t), where (m_t, Σ_t) is the algorithmic GP marginal on F and q_t is the maintained reference law of the index, either $A^\star(F)$ for AIR or F for MAIR. Let $\mathfrak{C}_t(m, \Sigma, q)$ be a surely calibrated local set of candidate active reward vectors containing the fixed truth $F^\star$ at every reached state. A local control $u \in \mathfrak{U}_t(m, \Sigma, q)$ specifies an action law p_u and a reference update. In AIR form the update is generated by a pair belief ν_u on active reward vectors and their optimal-action indices, so that $q^+ = q^+_{\nu_u}$. For the robust saddle calculation, candidate pair beliefs may be supported on the closed optimal relation $\{(f, y) : y \in \arg\max_x f(x)\}$; the true index continues to use the prescribed tie-breaking rule $A^\star(f)$. In the action-index version define

$$y(f) = A^\star(f), \qquad \mathsf{G}_t^{\mathrm{AIR}}(u; f) := \mathbb{E}_{x\sim p_u,\, O\sim N(f_x,\lambda)} \log \frac{q^+(y(f) \mid m, \Sigma, q, x, O)}{q(y(f))},$$

where q^+ is the chosen reference update for the optimal-action marginal. For a candidate reward vector f, write

$$\Delta_f(p) := \max_{x\in\mathcal{X}} f(x) - \mathbb{E}_{x\sim p} f(x).$$

The robust finite-marginal Bellman recursion is

$$W_t^\gamma(m, \Sigma, q) = \inf_{u\in\mathfrak{U}_t(m,\Sigma,q)} \sup_{f\in\mathfrak{C}_t(m,\Sigma,q)} \left\{ \Delta_f(p_u) + \mathbb{E}_f W_{t+1}^\gamma(m^+, \Sigma^+, q^+) - \gamma \mathsf{G}_t^{\mathrm{AIR}}(u; f) \right\}.$$

The model-index MAIR version replaces $y(f)$ by the finite reward vector f and uses the corresponding model-coordinate posterior log gain. For the finite action index, Theorem 5.2 implies that an approximate minimizer of this recursion has regret at most

$$W_1^\gamma(m_1, \Sigma_1, q_1) + \gamma \log \frac{1}{q_1(A^\star(F^\star))} + \text{explicit bracket/optimization errors.}$$

For the continuous model index $F \in \mathbb{R}^n$, one cannot drop the terminal log-density term by treating it as a discrete code length: a posterior density may exceed one. A valid continuous-index statement must retain that terminal term or replace point coordinates by neighborhoods, a finite cover, or a PAC–Bayes comparison. The finite action index $A^\star$ avoids this issue and is the clean frequentist specialization used below. If the maintained belief is the exact Bayesian posterior and one averages this coordinate recursion over F, the robust supremum is replaced by posterior expectation and the dynamic program reduces to the posterior-averaged AIR recursion

$$U_t^\gamma(m,\Sigma,q) = \inf_{p\in\Delta(\mathcal{X})} \left\{\Delta_t^{\mathrm{AIR}}(p) + \mathbb{E}U_{t+1}^\gamma(m^+,\Sigma^+,q^+) - \gamma\mathcal{I}_t^{\mathrm{AIR}}(p)\right\}.$$

Thus the finite-marginal Bellman DP is the clean upper object for comparison with the lower theorem: the frequentist version uses the coordinate log potential and calibrated candidate truths, while the Bayesian version is its posterior average. GP–UCB is treated next; two further algorithmic specializations are collected in Appendix E.

The next theorem gives a uniform frequentist rate for the action-index branch. For a finite retained action set $\mathcal{R}$, write

$$\mathrm{Reg}_T^{\mathrm{all}}(f) := \sum_{t=1}^T \left\{\sup_{x\in\mathcal{X}_{\mathrm{all}}} f(x) - f(X_t)\right\}$$

for regret against the full decision domain, and define

$$\varepsilon_{\mathcal{R}} := \sup_{f\in\mathcal{F}} \left\{\sup_{x\in\mathcal{X}_{\mathrm{all}}} f(x) - \max_{x\in\mathcal{R}} f(x)\right\}.$$

Theorem 7.1 (Finite-marginal action-index AIR Bellman bound)**.** *Let $|\mathcal{R}| = K \geq 2$, suppose the retained reward-vector class*

$$\Theta_{\mathcal{R}} := \{(f(x))_{x\in\mathcal{R}} : f\in\mathcal{F}\}$$

is compact and contained in $[-L,L]^K$, and let the observations have independent $N(0,\lambda)$ noise with $\lambda > 0$. For every $\gamma > 0$, the finite-marginal action-index AIR Bellman program admits a policy, confined to $\mathcal{R}$ and initialized with the uniform law on the attainable optimal-action indices, such that

$$\sup_{f\in\mathcal{F}} \mathbb{E}_f \mathrm{Reg}_T^{\mathrm{all}} \leq T\varepsilon_{\mathcal{R}} + \frac{(\lambda+L^2)KT}{2\gamma} + \gamma\log K. \tag{84}$$

Consequently, the choice

$$\gamma = \sqrt{\frac{(\lambda+L^2)KT}{2\log K}}$$

gives

$$\sup_{f\in\mathcal{F}} \mathbb{E}_f \mathrm{Reg}_T^{\mathrm{all}} \leq T\varepsilon_{\mathcal{R}} + \sqrt{2(\lambda+L^2)KT\log K}. \tag{85}$$

The robust AIR/AMS saddle supplies the control for an explicit supersolution of the exact Bellman program and itself achieves the same bound. Under the minimax and KL-duality identification stated above, the corresponding EBO formulation has the same guarantee. More precisely, the robust action-index control is the finite-marginal specialization of the AIR/AMS saddle of Xu and Zeevi (2025); its EBO interpretation follows the optimization principle of Lattimore and György (2021), rather than identifying the two algorithms line by line.

Proof. See Appendix H.19.

The theorem is deliberately stated for the finite-valued optimal-action index. Since $A^\star$ is a function of F, data processing gives the analogous posterior-averaged one-step information estimate with the model index F. A uniform frequentist MAIR statement, however, must retain the terminal log-density term for this continuous index or replace point coordinates by neighborhoods, a finite cover, or a PAC–Bayes comparison. Finite marginality alone does not turn the model coordinate into a discrete code length.

Algorithm 2: GP–UCB. GP–UCB uses the Gaussian marginal posterior (m_t, Σ_t) on the finite active marginal. In this paragraph, e_x denotes the coordinate vector associated with $x \in \mathcal{X}$, and $s_t(x) = (e_x^\top \Sigma_t e_x)^{1/2}$ denotes the unmultiplied posterior standard deviation, i.e., the generic uncertainty scale $\sigma_t(x)$ from Lemma 5.5. The confidence half-width is

$$w_t(x) := \beta_t s_t(x),$$

where β_t is the confidence multiplier. The algorithm chooses

$$X_t \in \arg\max_{x \in \mathcal{X}} \{m_t(x) + \beta_t s_t(x)\}. \tag{86}$$

The following event is the frequentist calibration:

$$\mathcal{E}_{\mathrm{GP}} := \{|f^\star(x) - m_t(x)| \le w_t(x) = \beta_t s_t(x) \quad \text{for all } t \le T, x \in \mathcal{X}\}. \tag{87}$$

For finite $\mathcal{X}$, standard RKHS self-normalized concentration gives $\mathbb{P}(\mathcal{E}_{\mathrm{GP}}) \ge 1 - \delta$ for the usual choice of β_t. For example, up to universal constants one may take

$$\beta_t \asymp B + \frac{R}{\sqrt{\lambda}} \sqrt{\Gamma_{t-1}^{\mathrm{GP}} + \log(1/\delta)}, \qquad \Gamma_m^{\mathrm{GP}} := \sup_{x_{1:m} \in \mathcal{X}^m} \frac{1}{2} \log\det\bigl(I + \lambda^{-1} K_{x_{1:m}, x_{1:m}}\bigr),$$

with the precise form depending on the normalization of k and the ridge parameter. Here $K_{x_{1:m}, x_{1:m}}$ is the Gram matrix of the possibly repeated design sequence $(x_1, \dots, x_m)$. The quantity Γ_m^{GP} is a worst-case calibration device for the confidence event; it is different from the realized information gain C_T^{GP} paid in the regret bound. The next theorem isolates the deterministic part of the GP–UCB proof.

Theorem 7.2 (GP–UCB realized-information bound)**.** *Assume that $(\beta_t)_{t \le T}$ is a deterministic nondecreasing sequence. On the event $\mathcal{E}_{\mathrm{GP}}$, the GP–UCB rule (86) satisfies*

$$\mathrm{Reg}_T(f^\star) \le 2\beta_T \sum_{t=1}^T s_t(X_t) \le 2\beta_T \sqrt{T\, c_{\kappa,\lambda}\, C_T^{\mathrm{GP}}}, \tag{88}$$

where

$$c_{\kappa,\lambda} := \frac{2\kappa^2}{\log(1 + \kappa^2/\lambda)}$$

and C_T^{GP} is the realized information gain in (81). Consequently, if rewards are bounded in $[0,1]$ and $\mathbb{P}(\mathcal{E}_{\mathrm{GP}}) \ge 1 - \delta$, then

$$\mathbb{E}\,\mathrm{Reg}_T(f^\star) \le 2\beta_T \sqrt{T\, c_{\kappa,\lambda}\, \mathbb{E} C_T^{\mathrm{GP}}} + T\delta.$$

Proof. See Appendix H.18.

Calibration and realized information. The theorem is Lemma 5.5 with the generic uncertainty scale $\sigma_t(X_t) = s_t(X_t)$ and the realized posterior-reference log-determinant telescope

$$\sum_t \frac{1}{2} \log\left(1 + s_t^2(X_t)/\lambda\right) = C_T^{\mathrm{GP}}.$$

The confidence half-width is $w_t(X_t) = \beta_t s_t(X_t)$, but the multiplier β_t is not part of the information telescope; it is the price of calibration and optimism. The proof therefore uses s_t and w_t for different purposes: the log-determinant argument sums s_t^2, while w_t bounds instantaneous regret. Under posterior averaging, the log determinant is mutual information. At a fixed truth, however, the MAIR log gain is a KL divergence from the true predictive law to the posterior predictive law and is not equal term by term to $\frac{1}{2}\log(1 + s_t^2/\lambda)$. The theorem combines calibration with an elliptical-potential bound; it does not identify these two quantities. GP–UCB pays the realized C_T^{GP} in its regret analysis, even when the multiplier used to select actions is calibrated by the worst-case Γ_T^{GP}.

Kernel-specific GP–E2D and AMS/EBO specializations are given in Appendix E.2.

7.2 A full-RKHS separation for maximal-information GP–UCB

The finite-scale construction underlying the fixed-kernel result is given in Appendix E.1. Building on the hub–cloud construction of Xu (2026), our construction uses a small-amplitude cloud that an unrestricted minimax policy can afford to ignore, even though the cloud's multiplicity inflates the maximal-information exploration multiplier. The separation is also conceptually related to the minimax adaptivity gap studied by Maiti et al. (2026), which likewise translates a spatial trap into a minimax separation.

For a bounded kernel k and unit Gaussian observation noise, define the maximal information gain

$$\overline{\Gamma}_s(k) := \sup_{x_{1:s} \in \mathcal{X}^s} \frac{1}{2} \log\det(I + K_{x_{1:s}, x_{1:s}}), \qquad \overline{\Gamma}_0(k) := 0. \tag{89}$$

Starting from the zero-mean GP with covariance k and unit ridge parameter, let m_t, s_t be the posterior mean and standard deviation. The single anytime rule studied below is

$$\overline{\beta}_t := 1 + \sqrt{2\left\{\overline{\Gamma}_{t-1}(k) + 2\log(t \vee 2)\right\}}, \qquad X_t \in \arg\max_{x \in \mathcal{X}} \left\{m_t(x) + \overline{\beta}_t s_t(x)\right\}, \tag{90}$$

with a fixed tie-breaking rule.

The multiplier in (90) has the same maximal-information form as the modern anytime calibration of Chowdhury and Gopalan (2017). The same construction also covers the exploration schedule in the original RKHS GP–UCB theorem of Srinivas et al. (2010). To state this precisely, fix any $\delta \in (0, 1)$ and define its unit-ball specialization, in the present multiplier notation, by

$$\Gamma_t^{\mathrm{SKKS}}(k) := \sup_{\substack{A \subset \mathcal{X} \\ |A| = t}} \frac{1}{2} \log\det(I + K_A),$$

and

$$b_t^{\mathrm{SKKS}} := \left[2 + 300\, \Gamma_t^{\mathrm{SKKS}}(k) \log^3\left(\frac{t}{\delta}\right)\right]^{1/2}, \qquad X_t \in \arg\max_{x \in \mathcal{X}} \{m_t(x) + b_t^{\mathrm{SKKS}} s_t(x)\}, \tag{91}$$

with the same fixed tie-breaking convention. Srinivas et al. write the acquisition bonus as $\sqrt{\beta_t}\, s_t$, assume $\|f\|_k^2 \le B$, and set $\beta_t = 2B + 300\gamma_t \log^3(t/\delta)$; equation (91) is exactly the specialization

$B = 1$, $\gamma_t = \Gamma_t^{\mathrm{SKKS}}(k)$, and unit noise variance. Their frequentist confidence theorem assumes bounded martingale noise. Here we analyze the same acquisition rule under independent Gaussian noise, so the lower bound uses the original exploration schedule but does not invoke that confidence theorem.

The two fixed-kernel rules use the same zero-mean GP posterior and the same UCB acquisition map $m_t(x) + b_t s_t(x)$; they differ in their global confidence calibration, information-gain convention, and logarithmic factors. The finite-horizon rule (100) in Appendix E.1 is a transparent prototype of the anytime maximal-information form, not either fixed-kernel rule verbatim. Appendix H.20 verifies the anytime and original schedules separately.

Theorem 7.3 (Fixed-kernel separation: GP–UCB versus AIR and minimax)**.** *For every $0 < \alpha < 1/4$, there exist a compact countable action space $\mathcal{X}$, a bounded continuous kernel k on $\mathcal{X}$, its fixed unit RKHS ball $\mathcal{F}$, a sequence $T_m \to \infty$, and a single epochwise finite-marginal action-index AIR Bellman policy such that, under observations $Y_t = f(X_t) + \xi_t$ with $\xi_t \overset{\text{i.i.d.}}{\sim} N(0,1)$,*

$$\inf_{\mathsf{Alg}} \sup_{f \in \mathcal{F}} \mathbb{E}_f \operatorname{Reg}_{T_m}(\mathsf{Alg}) = \Theta(T_m^{1-\alpha}), \tag{92}$$

and the AIR Bellman policy, fixed in advance and independent of the evaluation horizon, satisfies

$$\sup_{f \in \mathcal{F}} \mathbb{E}_f \operatorname{Reg}_{T_m}(\text{AIR–Bellman}) = O(T_m^{1-\alpha}). \tag{93}$$

In contrast, both the single anytime GP–UCB rule (90) and, for every fixed $\delta \in (0,1)$, the original-schedule rule (91) have

$$\sup_{f \in \mathcal{F}} \mathbb{E}_f \operatorname{Reg}_{T_m}(\mathsf{R}) = \Omega(T_m), \qquad \mathsf{R} \in \{\text{GP–UCB}_{\text{any}}, \text{GP–UCB}_{\text{SKKS}}\}. \tag{94}$$

For each of these GP–UCB rules, both the minimax ratio and the worst-case regret ratio relative to the AIR Bellman policy are $\Omega(T_m^{\alpha})$ along an infinite subsequence.

Proof. See Appendix H.20.

Remark 7.4 (What the example disproves)**.** *Theorem 7.3 concerns one bounded continuous kernel and one full RKHS ball. It gives a negative resolution to a canonical and widely studied form of the GP–UCB minimax-optimality question: the failure is algorithmic for both a modern anytime maximal-information calibration and the unit-ball exploration schedule of the influential original RKHS GP–UCB rule. This question is not manufactured by the construction; the literature explicitly asks whether the suboptimal rates of GP–UCB reflect loose confidence analysis or a limitation of the algorithm (Vakili et al., 2021; Whitehouse et al., 2023; Wang and Zhang, 2026). The kernel is spatially inhomogeneous—its diagonal values vary across the hub and cloud blocks—and its compact domain is countable, so the theorem does not settle a conjecture restricted to translation-invariant kernels on Euclidean domains. It also does not rule out every localized, predictable, or data-dependent exploration multiplier.*

Remark 7.5 (Bellman and hub–cloud interpretation)**.** *The direct hub–cloud embedding of Xu (2026) has cloud scale larger than hub scale, and therefore its cloud makes the full RKHS ball minimax-hard; it gives a sharp pointwise GP–UCB trap but not a minimax separation. Theorem E.1 reverses the amplitude comparison while retaining the multiplicity comparison. The pointwise RKHS envelope $|f(c_j)| \le a$ makes the entire cloud worth at most aT to an unrestricted algorithm. Nevertheless, its maximal information gain is of order Ta^2, so the global confidence* *multiplier is of order $a\sqrt{T}$ and the acquisition width of each unqueried cloud point is of order $a^2\sqrt{T}$. When $a = T^{-\alpha}$ and $\alpha < 1/4$, that width diverges even though the decision value of the cloud vanishes.*

In Bellman terms, the upper analysis compares retaining the cloud in the continuation problem with truncating it at a cost of at most a per remaining round. The policy used in the theorem implements the resulting finite-marginal truncation on a predetermined epoch schedule and pays only for the optimal-action index on the retained set. The GP–UCB relaxation replaces that decision-aligned comparison by a global uniform-optimism coefficient. Maximal information gain is therefore not merely paid after the trajectory is generated; through $\overline{\beta}_t$, it generates the bad trajectory. The relevant localization is the elementary pointwise envelope in this example, and in more structured problems it may be a posterior region, a norm shell, or an optimal-action index.

Remark 7.6 (AIR Bellman, AMS/EBO, and GP–UCB)**.** *The roles of the three algorithms should be distinguished. On each predetermined retained marginal $\mathcal{R}_m$, the robust AIR/AMS saddle (equivalently, its EBO dual under the stated minimax conditions) supplies an admissible control for the exact AIR Bellman program. It adapts its candidate belief, reference marginal, and action distribution as observations arrive, and Theorem 7.1 shows that the resulting robust AIR/AMS control and an approximate selector of the exact Bellman recursion satisfy the same regret bound; under the stated minimax-duality identification, so does its EBO formulation. Thus AIR Bellman and AMS/EBO are not separated from one another in the present construction: the latter provides the statewise control used to build a supersolution for the former. An additional discussion of a GP–E2D algorithm is provided in Appendix E.2.*

The comparison with GP–UCB in Theorem 7.3 is a minimax comparison on the same fixed kernel, full RKHS ball, action domain, and observation model. All policies may use the known problem class, but none knows the true reward function or the evaluation horizon. The retained sets and epoch schedule used by the localized AIR/AMS/EBO policy are fixed in advance from this public class geometry, whereas the specified GP–UCB rule acts on the full domain with a multiplier calibrated by the full maximal information gain. The resulting separation is therefore between a class-aware localized policy and this particular global GP–UCB calibration, which is the appropriate comparison for its claimed minimax optimality.

The theorem does not isolate a benefit arising from a nontrivial across-horizon Bellman value recursion: after the predetermined truncation, a state-independent supersolution already gives the desired rate. Nor does it show that an unlocalized AMS/EBO rule must fail, that a GP–UCB rule localized to the same retained marginal must fail, or that either AIR Bellman or AMS/EBO automatically discovers the truncation scale. Endogenous localization would require placing the active marginal in the Bellman state, allowing controls to move between scales, and accounting for the corresponding approximation and information costs.

7.2.1 Fixed Matérn kernels and the remaining GP–UCB question

The fixed-kernel theorem gives a spatially inhomogeneous counterexample. A complementary result is available for Matérn RKHSs. To align notation, write the GP–UCB acquisition as

$$x_t \in \arg\max_x \{\mu_{t-1}(x;\rho_t) + b_t\sigma_{t-1}(x;\rho_t)\}, \qquad L_t := \rho_t b_t^2.$$

The quantity L_t is the effective optimism of Wang and Zhang (2026); their paper writes $b_t = \sqrt{\beta_t}$.

Corollary 7.7 (Polynomial effective optimism is minimax-suboptimal)**.** *Fix d, ν, the Matérn length scale, the RKHS radius $B > 0$, and the Gaussian noise level $\sigma > 0$. Consider the radius-B Matérn-ν RKHS ball on $[0,1]^d$. Assume the conditions of Wang and Zhang (2026, Theorem 1): b_t^2 and ρ_t*

are positive and nondecreasing, $b_t^2 \ge 1$, and, for every $3 \le t \le T$ with constants independent of T,

$$b_t^2 \asymp t^{\theta_1}(\log t)^{\theta_3}, \qquad \rho_t \asymp t^{\theta_2}(\log t)^{\theta_4}, \qquad \theta_1,\theta_2,\theta_3,\theta_4 \ge 0, \qquad \frac{4\nu+d}{2\nu}\theta_1 + \theta_2 < 1,$$

together with their simultaneous uniform-confidence assumption: for constants $\zeta \in (0,1)$ and $p_0 > 0$ independent of T, the event

$$|\mu_{t-1}(x;\rho_t) - f(x)| \le \zeta b_t \sigma_{t-1}(x;\rho_t) \quad \textit{for every } x \textit{ and } t$$

has probability at least p_0. Then

$$\sup_{\|f\|_{\mathcal{H}_k} \le B} \mathbb{E}_f \operatorname{Reg}_T(\text{GP–UCB}) = \Omega\Big(T^{\frac{\nu+d}{2\nu+d}} L_T^{\frac{\nu}{2\nu+d}}\Big). \tag{95}$$

The minimax value on the same standard Matérn class is, up to polylogarithmic factors,

$$\inf_{\mathsf{Alg}} \sup_{\|f\|_{\mathcal{H}_k} \le B} \mathbb{E}_f \operatorname{Reg}_T(\mathsf{Alg}) = \widetilde{\Theta}\Big(T^{\frac{\nu+d}{2\nu+d}}\Big). \tag{96}$$

Consequently, any admissible schedule with $L_T = T^{\vartheta}\,\mathrm{polylog}(T)$, $\vartheta > 0$, is polynomially minimax-suboptimal by the factor

$$\widetilde{\Omega}\Big(T^{\vartheta\nu/(2\nu+d)}\Big).$$

Proof. See Appendix H.21.

The comparison with the Whitehouse–Wu–Ramdas effective-optimism scale is given in Appendix E; the upper and lower statements there concern different schedules and are not asserted to form a single matching theorem.

Position relative to the GP–UCB optimality debate. There is no single formal conjecture covering every kernel and every tuning rule. There is, however, a well-documented and commonly studied question: whether the gap between minimax rates and the regret guarantees for the GP–UCB acquisition rule is an artifact of confidence analysis or an algorithmic limitation (Vakili et al., 2021; Whitehouse et al., 2023; Wang and Zhang, 2026). Theorem 7.3 gives an algorithmic negative answer for two canonical maximal-information forms on a fixed problem, including the unit-ball exploration schedule of the original RKHS GP–UCB theorem. Corollary 7.7 shows that polynomial-effective-optimism schedules covered by the Wang–Zhang assumptions are suboptimal even for fixed Matérn kernels. It remains open whether a GP–UCB rule with localized, polylogarithmic, nonmonotone, or data-dependent effective optimism can maintain valid frequentist control and attain the Matérn minimax rate. Neither a gap between known bounds nor the offline hub–cloud theorem alone resolves that stronger question.

The indexed view explains the mechanism. The exact finite-marginal action-index AIR Bellman program permits a state-dependent continuation and may be computationally demanding. On this construction, however, the predetermined truncation and the state-independent robust AIR/AMS supersolution of Theorem 7.1 already attain the minimax order. Thus the result establishes a successful localized indexed policy; it does not show that AMS/EBO discovers the localization or that a nontrivial continuation value is necessary. The action index supplies the clean finite code length used by the proof, but a separately justified localized model-index or PAC–Bayes analysis is not ruled out. GP–UCB has favorable realized-information geometry but does not optimize the coefficient converting uncertainty into regret. GP–E2D optimizes a one-step offset, whose comparison set and fixed-truth information bound must refer to the same trajectory. The debate is therefore not settled by a single universal coefficient: it separates realized information, uniform calibration, decision localization, and algorithm-specific acquisition dynamics.

7.3 Finite-scale illustration of the hub–cloud mechanism

The preceding results are nonasymptotic; this experiment illustrates only their finite-scale mechanism. For each displayed horizon, take the small-cloud instance of Appendix E.1 with $\alpha = 1/5$, $a = T^{-1/5}$, $N = 4T$, and hub gap $\Delta = 1/4$.

On each finite block we evaluate three GP–UCB calibrations: the horizon-aware rule (100), the finite-block specialization of the anytime rule (90), and the unit-ball original schedule (91), with $\delta = 0.1$. We compare them under the same positive hub truth with a numerical implementation of the robust action-index AIR/AMS control on $\{x_0, x_h\}$. This control is a direct two-model Gaussian specialization of the AIR/AMS saddle of Xu and Zeevi (2025); its relation to Lattimore and György (2021) is through the EBO optimization principle and the corresponding minimax/KL duality, not through a literal implementation of the functional-estimator mirror-descent algorithm.

The scalar saddle (116) is a Bellman-supersolution control for this two-pair class, not a solution of the full continuous-RKHS finite-horizon program. Figure 1 shows two complementary comparisons. Under the hub truth $f^\dagger(x_h) = \Delta$, the left panel displays cumulative pseudoregret; the dashed line is a reference for the deterministic at cost of ignoring a one-hot cloud alternative. The three GP–UCB curves differ only through their exploration multipliers and nearly coincide at the displayed scale. At $T = 16384$, their 200-run Monte Carlo mean regrets under the common positive hub truth are 4074.8, 4072.5, and 4083.8 for the horizon-aware, anytime, and original-schedule rules, respectively, compared with 15.6 for AIR/AMS. Thus each GP–UCB rule incurs approximately 261–262 times as much mean regret. The AIR/AMS mean under the negative retained hub truth, not displayed, is 56.7.

The right panel compares the normalized terminal regret of all four implemented controls under the common positive hub truth and includes the deterministic aT truncation benchmark. For the exact two-pair saddle, the retained-truth bound is $2\sqrt{(1+\Delta^2)T\log 2} < aT$ at every displayed horizon, while any policy confined to $\{x_0, x_h\}$ has regret exactly aT under a positive one-hot cloud truth. The one-hot truths are not part of the binary saddle; this is an external truncation calculation. Thus the algorithmic curves are directly comparable at a common truth, whereas the deterministic aT curve displays the separate approximation cost of truncating a one-hot cloud truth. The plot illustrates both ingredients of the construction: the GP–UCB trajectory spends a constant fraction of its rounds in the zero-mean cloud, while the truncation benchmark decreases as $T^{-1/5}$. It does not establish the fixed-kernel separation: across the displayed horizons, the finite kernel changes with T, whereas Theorem 7.3 concerns one fixed kernel and proves linear GP–UCB regret along an infinite subsequence. Nor does the experiment supply a uniform numerical guarantee over the full RKHS ball. The protocol and a numerical check of the two fixed-truth Bellman brackets are in Appendix E.3.

7.4 Further matching examples

Gaussian multi-armed bandits give a finite-index Bellman–Fano entropy calculation and a matching minimax regret rate $\Theta(\sqrt{KT})$. Hypercube linear bandits give a high-dimensional indexed lower bound of order $d\sqrt{T}$, matched by standard upper bounds up to logarithmic factors. These examples do not claim that MOSS or OFUL is itself the optimizer of the exact log-potential Bellman program. The complete calculations are in Appendices F and G.

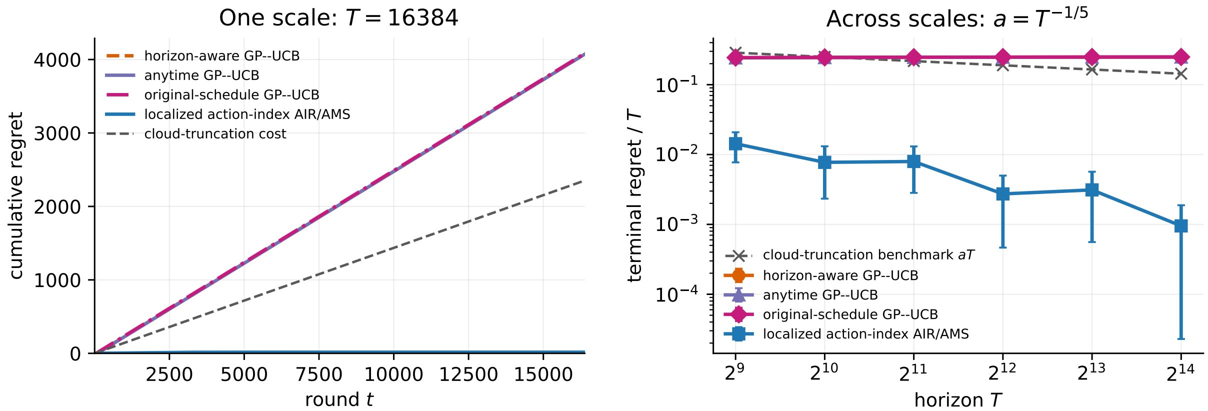

Figure 1: Finite-scale hub–cloud experiment. Monte Carlo curves show means and two standard errors over 200 independent runs; the truncation curves are deterministic. All algorithmic comparisons use the same positive hub truth. Left: cumulative pseudoregret at $T = 16384$ for the horizon-aware, anytime, and original-schedule GP–UCB rules and the localized robust AIR/AMS control; the dashed line is the counterfactual one-hot-cloud truncation cost. Right: terminal regret divided by T across horizons, together with the deterministic aT truncation benchmark. Across scales, the finite kernel, $a = T^{-1/5}$, and the horizon-tuned AIR coefficient change with T. The experiment illustrates the finite-block mechanism rather than simulating the glued fixed kernel.

8 Conclusion

This paper is organized around the principle that the statistical difficulty of an interactive sequential decision problem can be characterized only after identifying the appropriate sufficient state and information index. The environment remains the fixed frequentist truth that generates observations and losses. The Bellman-sufficient state is the part of the history on which prediction, loss evaluation, updating, and future control can be continued. The index $Y = \chi(\Omega)$ is the object whose identification is charged. It may be the optimal action, the optimal policy, a value object, an active finite marginal, or the full model; choosing it is part of the representation. This separation is the main conceptual point. It prevents an upper bound from paying for irrelevant model estimation, and it prevents a lower bound from proving hardness for information that the decision problem never requires.

On the upper side, the basic object is not a one-step coefficient, but a Bellman supersolution with a logarithmic information coordinate. For a fixed truth $\omega^\star$ and $y^\star = \chi(\omega^\star)$, the coordinate potential

$$\gamma \log \frac{1}{q_t(y^\star)}$$

turns indexed learning into a dynamic program: regret is controlled by immediate loss, continuation value, and the fixed-truth log gain of the maintained reference marginal. The exact log-penalized Bellman program is therefore the canonical information-theoretic upper algorithm. Its posterior-averaged form recovers Bayesian AIR/MAIR identities, but the fixed-truth form is more fundamental for minimax analysis. UCB, E2D, and AMS/EBO are best understood as tractable ways to control or relax this same Bellman bracket. UCB uses calibration and optimism to upper-bound the fixed-truth bracket; E2D keeps the immediate loss and uses a one-round separation penalty in place of the continuation value; AMS/EBO replaces the exact continuation value by a KL-dual log-partition, or admissible-relaxation, potential over calibrated candidate beliefs.

On the lower side, the same representation supports a Bellman–Fano value comparison. The information term is the exact indexed-information Bellman capacity, or a valid upper bound on it, along posterior-reference trajectories; the entropy term is the exact ghost-good mass, or a valid upper bound on that mass, for reference histories whose target-wise regret is small. Thus the lower bound does not ask how many environments can be distinguished in the full model class. It asks how much indexed information the interactive experiment can reveal before it would make too many low-regret ghost histories possible. Theorem 3.8 expresses the resulting matching principle: when the log-penalized Bellman upper value and the Bellman–Fano ghost-quantile lower value close at the same radius on the same state/index representation, the minimax rate is determined at that radius. In this sense, the paper gives an interactive analogue of information-risk matching in fixed statistical experiments: local prior mass is replaced by ghost-good mass, and fixed-design KL is replaced by a controlled Bellman information telescope.

The kernel-bandit application illustrates why the representation must remain inside the algorithm. The fixed-kernel small-cloud construction has minimax value $\Theta(T^{1-\alpha})$. A predetermined finite-marginal truncation followed by the action-index AIR/AMS Bellman control attains this order and pays only for the retained decision index, while canonical maximal-information GP–UCB rules have linear regret because the cloud's multiplicity inflates their acquisition bonus. The lower bound includes the unit-ball exploration schedule of the influential original RKHS GP–UCB algorithm. Together with the fixed Matérn obstruction for polynomial-effective-optimism schedules satisfying the Wang–Zhang hypotheses, this gives a negative resolution for prominent, commonly studied forms of GP–UCB minimax optimality, while leaving open whether a differently localized or data-dependent GP–UCB rule can attain the Matérn minimax rate. The distinction is exactly the one enforced by the Bellman formulation: realized information, global calibration, and action-index continuation are different mathematical objects.

The conclusion is therefore methodological as much as technical. To analyze an interactive learning problem, one should not begin by asking for the dimension of the model, or for a universal conversion mechanism between estimation and decision. One should ask: what state representation makes the Bellman recursion close, what index is actually worth learning, what logarithmic potential pays for that index, and what ghost quantile supplies the matching lower entropy? When these four objects align, learning becomes a special dynamic program whose value coordinate is information itself. The statistical complexity of the problem is then no longer a property of the raw model class alone; it is a property of the Bellman-sufficient representation through which the learner, the upper bound, and the lower bound all see the problem.

A Supplementary scope and state representations

Remark A.1 (Adversarial and estimated losses)**.** *The sequential decision-making framework naturally accommodates general adversarial losses beyond the stochastic regret setting (Cesa-Bianchi and Lugosi, 2006; Abernethy et al., 2011). A nonstochastic or adaptive adversary may be treated as part of the fixed environment ω, provided its rule is nonanticipating. Classical online learning often permits the adversary to observe the history as well as the learner's announced mixed action. Then the mixed action is the current control, not an additional component of the history being optimized over, and the general primitives in (1) are*

$$\ell_t(\omega, H_{t-1}, p, a), \qquad P_{\omega,t}(\cdot \mid H_{t-1}, p, a).$$

Equivalently, after p is chosen one may view (H_{t-1}, p) as the post-announcement pre-sampling information; the Bellman operator must still optimize over p. With the full history as state and p as a control argument, this representation is automatic for any fixed nonanticipating adversary rule. Estimated-loss upper bounds can be handled by replacing the true loss in the Bellman bracket with a state-measurable surrogate or domination bound, while carrying the resulting approximation error. Lower bounds require the target cumulative indexed loss used in the ghost event to be lower bounded, usually nonnegative after a harmless shift.

Remark A.2 (Sublinear cumulative loss against the optimal state-dependent policy)**.** *Related finite-time minimax and information-theoretic phenomena appear in sequential decision-making domains where the cumulative loss, measured against an optimal state-dependent policy, can be sublinear without an explicit episodic restarting structure. Examples include inventory control (Jagannathan, 1978), stochastic optimization and programming (Nemirovski and Yudin, 1983; Shapiro and Nemirovski, 2005; Agarwal et al., 2012; Raginsky and Rakhlin, 2011), linear quadratic control (Lai, 1986; Mania et al., 2019; Simchowitz and Foster, 2020; Jedra and Proutiere, 2022), stochastic systems (González-Trejo, Hernández-Lerma, and Hoyos-Reyes, 2002), blind network revenue management (Besbes and Zeevi, 2012), bandits with knapsacks (Badanidiyuru et al., 2018), pathwise state–action learning through value functions (Xu and Zeevi, 2025), and queueing control (Liu et al., 2025; Liang et al., 2026). These settings are important in their own right and are different in flavor from standard finite-horizon or discounted reinforcement-learning formulations, where the planning horizon or discount factor often appears explicitly in regret bounds (Azar et al., 2017; Shah et al., 2022). Existing results provide valuable model-specific upper and lower bounds, but they have not yet been organized into a general Bellman-recursive information-complexity theory for across-episode reinforcement learning. A central direction is to understand whether the representation-level machinery developed here can recover and extend these bounds to broader dynamic programming, planning, search, and reasoning problems under the partial information formalism.*

The following basic examples verify Definition 2.1 directly. They also fix the interpretation of the state used later: the fixed truth remains the environment, while sufficient statistics, posteriors, and finite marginals are retained Bellman states or reference objects.

A.1 Basic examples of Bellman-sufficient representations

Structured bandit problems offer particularly transparent examples of the Bellman-sufficiency framework for sequential decision making, which is guided by the classical intuition of sufficient statistics.

Example A.3 (Classical sufficient statistics as one-step Bellman states)**.** Consider a nonadaptive dominated experiment with observations $X_1, \ldots, X_n$ from P_θ, and no control. Suppose that for every prefix t, $T_t = T_t(X_1, \ldots, X_t)$ is sufficient for the prefix experiment and admits a recursive update $T_t = u_t(T_{t-1}, X_t)$. If the loss and index depend on θ only through quantities whose posterior distribution is determined by T_t, then the process $S_t = (t, T_{t-1})$ is Bellman sufficient. Predictive sufficiency follows from the prefix factorization; posterior predictive and index sufficiency follow because the posterior or conditional reference law depends on the history only through T_t. In a regular exponential family,

$$p_\theta(x) = h(x) \exp\{\vartheta(\theta)^\top T(x) - A(\theta)\},$$

with conjugate or otherwise statistic-measurable reference law, the cumulative statistic $\sum_{i<t} T(X_i)$ gives the corresponding state. This is the static prototype for all examples below.

Example A.4 (Finite stochastic multi-armed bandits)**.** Let $\Omega \subseteq \Theta^K$ and let arm a produce an observation from P_{θ_a}. For unit-variance Gaussian arms, $P_{\theta_a} = N(\theta_a, 1)$. A reference Gaussian prior with independent coordinates gives the posterior state

$$S_t = \big(t, (n_{t,a}, \bar{X}_{t,a}, v_{t,a})_{a=1}^K\big),$$

where $n_{t,a}$ is the number of previous pulls of arm a, $\bar{X}_{t,a}$ is the empirical mean when $n_{t,a} > 0$, and $v_{t,a}$ is the posterior variance. Equivalently one may store the posterior hyperparameters. For a fixed truth $\omega = \theta$,

$$P_{\omega,s,a} = N(\theta_a, 1), \qquad \ell_\omega(s, a) = \max_b \theta_b - \theta_a,$$

and the update of $(n, \bar{X}, v)$ after observing arm a is a function of (s, a, o). The posterior predictive law is $N(m_{t,a}, 1 + v_{t,a})$, and for the action index $Y = A^\star(\theta)$ the conditional predictive law $P^y_{s,a}$ and conditional loss $\ell^\chi_y(s, a)$ are obtained by integrating the same Gaussian posterior over the event $A^\star(\theta) = y$. Hence Definition 2.1 is satisfied. The empirical mean alone is not sufficient: the count or posterior variance is needed for prediction, confidence, and information gain.

Example A.5 (Finite-dimensional Gaussian linear bandits)**.** Let $\Omega \subseteq \mathbb{R}^d$, actions satisfy $\|a\|_2 \le 1$, and

$$O_t = \langle \theta, A_t \rangle + \xi_t, \qquad \xi_t \sim N(0, 1).$$

Under a Gaussian reference prior, the exact Bayesian state is $S_t = (t, m_t, \Sigma_t)$, where

$$\Sigma_{t+1}^{-1} = \Sigma_t^{-1} + A_t A_t^\top, \qquad m_{t+1} = \Sigma_{t+1}\{\Sigma_t^{-1} m_t + A_t O_t\}.$$

For fixed θ, $P_{\theta,s,a} = N(\langle \theta, a \rangle, 1)$ and the regret loss is $\sup_{\|b\|_2 \le 1} \langle \theta, b \rangle - \langle \theta, a \rangle$. The posterior predictive law is $N(\langle m_t, a \rangle, 1 + a^\top \Sigma_t a)$, and conditioning the Gaussian posterior on an index event such as $\chi(\theta) = \theta$, $\chi(\theta) = \theta/\|\theta\|_2$ for $\theta \neq 0$, with an arbitrary fixed value at $\theta = 0$, or $\chi(\theta) = A^\star(\theta)$ determines $P^y_{s,a}$ and $\ell^\chi_y(s, a)$. Therefore the Gaussian mean–covariance pair is an exact Bellman-sufficient state. A frequentist UCB state $(\hat{\theta}_t, V_t, \beta_t)$ is different: it can control a fixed-truth bracket on the calibration event $\|\hat{\theta}_t - \theta^\star\|_{V_t} \le \beta_t$, but it is not an exact posterior-reference state unless a reference belief or confidence-to-belief map is added.

Example A.6 (Finite active marginals in kernel bandits)**.** Let $f^\star$ belong to an RKHS on a possibly infinite domain, but suppose the learner is evaluated on a finite active set $\mathcal{X} = \{x_1, \ldots, x_n\}$. Under the GP reference law on the active reward vector $F = (f(x_1), \ldots, f(x_n))$, the finite marginal posterior $(m_t, \Sigma_t) \in \mathbb{R}^n \times \mathbb{R}^{n \times n}$ is Bellman sufficient for the finite experiment. For fixed $f^\star$, pulling x_i has law $N(f^\star(x_i), \lambda)$ in the Gaussian reference calculation; the posterior predictive law is $N(m_{t,i}, \lambda + \Sigma_{t,ii})$; and the usual rank-one Gaussian update is a function of (m_t, Σ_t, i, O_t). If the index is the optimal active action $Y = A^\star(F)$, then index sufficiency follows by conditioning the finite Gaussian posterior on $A^\star(F) = y$, using the fixed tie-breaking rule. The infinite-dimensional RKHS function is still the frequentist truth; the sufficient state is finite only because the decisions and observations use the active marginal.

A.2 The full environment posterior as a fallback sufficient state

For sequential decision making, a natural Bellman-sufficient representation is obtained by retaining the full environment posterior. For suitably defined Markovian model classes over the state space, which include standard episodic model-based reinforcement learning settings, this posterior serves as a fallback sufficient state.

Example A.7 (Full environment posterior as a fallback sufficient state)**.** *Consider a realizable model class in which each environment $\omega \in \Omega$ specifies, for every retained physical state or context z, action a, and stage t, an observation kernel $P_{\omega,t}(\cdot \mid z, a)$, a loss $\ell_{\omega,t}(z, a)$, and the physical-state transition rule. Let Z_t denote the physical state, context, stage, or episode coordinate needed to make these kernels Markov. Fix a reference prior μ and set*

$$\Pi_t = \mathcal{L}_\mu(\Omega \mid H_{t-1}), \qquad S_t = (Z_t, \Pi_t).$$

Assume that the observation laws are dominated on the relevant support and that the Bayesian update denominator is positive along the reference experiment. Fixed-truth coordinate identities additionally require the evaluated truth's predictive laws to be absolutely continuous with respect to the corresponding reference predictive laws. Then S_t gives an exact Bellman-sufficient state. Indeed, for each fixed truth ω,

$$P_{\omega,S_t,a} = P_{\omega,t}(\cdot \mid Z_t, a), \qquad \ell_\omega(S_t, a) = \ell_{\omega,t}(Z_t, a),$$

and the next retained state is obtained by updating the physical coordinate and the posterior

$$\Pi_{t+1}(d\omega) = \frac{p_{\omega,t}(O_t \mid Z_t, A_t)\Pi_t(d\omega)}{\int p_{\omega',t}(O_t \mid Z_t, A_t)\Pi_t(d\omega')}.$$

For an index $\chi : \Omega \to \mathcal{Y}$, the indexed marginal and conditional predictive components are

$$q_t = \chi_\# \Pi_t, \qquad \Pi_t^y = \mathcal{L}_{\Pi_t}(\Omega \mid \chi(\Omega) = y),$$

$$P^y_{S_t,a} = \int P_{\omega,t}(\cdot \mid Z_t, a)\, \Pi_t^y(d\omega), \qquad \ell^\chi_y(S_t, a) = \int \ell_{\omega,t}(Z_t, a)\, \Pi_t^y(d\omega).$$

Thus prediction, loss evaluation, posterior/reference updating, and indexed information accounting all close on S_t. This verifies Definition 2.1.

In this sense, sequential decision making with Bellman-sufficient representations contains the Decision Making with Structured Observations (DMSO) framework (Foster et al., 2021) as a special case when DMSO is interpreted as an independent episodic model-learning formalism and the full environment posterior is retained as the state. At the same time, the present framework also accommodates Markovian dependence across physical states and inner horizon stages, as in computational dynamic programming. The more important point is not formal inclusion, but the possibility of running dynamic programming on sufficient states and of charging a decision-relevant index rather than the full model label.

Example A.8 (DMSO as a posterior-state specialization)**.** *In the independent-episode DMSO formalism (Foster et al., 2021), an environment M specifies the observation law and loss for each decision rule within an episode, and episodes are conditionally independent given M. Under the observation-packet convention, the episode history H_{k-1} already contains the public information available before episode k. For a fixed prior and likelihood, the posterior*

$$\Pi_k = \mathcal{L}(M \mid H_{k-1})$$

is a deterministic compression of that history, not additional oracle information. Taking Ω to be the DMSO model class and retaining Π_k, together with any public episode context or admissible-decision constraints, gives a literal Bellman-sufficient state for the reference experiment. The posterior predictive observation law, the posterior update after the episode, and every index marginal $\chi_{\#}\Pi_k$ are functions of this state. Thus DMSO fits the present framework as the special case in which the Bellman clock is the outer episode and the posterior is the retained state coordinate. More compressed representations require an additional sufficiency proof; they do not follow from the DMSO notation alone.

Since the general one-step primitives in (1) allow the environment, or an adaptive adversary, to depend on both the history H_{k-1} and the announced mixed action p_k as a current control, the adversarial DMSO setting (Foster et al., 2022) is also subsumed; see Remark A.1. In this case, one may take the full environment to be the entire nonanticipating model sequence, with the posterior over this environment serving as a valid posterior state. The adversarial DMSO framework is still episodic rather than across-episode dynamic. It does not, by itself, capture across-episode, state-dependent planning problems in which the comparator is dynamic and current decisions influence future states across episodes. Extending the framework to such settings would require further development along the lines pursued in this paper.

B Posterior scoring and the AIR/MAIR variational calculation

Danskin, posterior scoring, and why the AIR bracket is not accidental The short algebraic proofs above are the fixed-truth version of the variational argument behind AIR. In the Bellman-sufficient notation, fix a state s, an action law p, and the current state marginal $q_s \in \Delta(\mathcal{Y})$ for the index. Let

$$\nu \in \Delta(\Omega \times \mathcal{Y})$$

be an admissible pair belief, usually with $Y = \chi(\Omega)$ and Y-marginal q_s. Given a prediction rule

$$Q : (s, p, a, o) \mapsto \Delta(\mathcal{Y}),$$

consider the posterior-scoring objective

$$\mathcal{J}_\gamma(s, p, \nu, Q; q_s) := \mathbb{E}_{\substack{(\Omega,Y)\sim\nu \\ A\sim p \\ O\sim P_{\Omega,s,p,A}}} \left[\ell_{\Omega,Y}(s, p) - \gamma \log \frac{Q(s, p, A, O)(Y)}{q_s(Y)} \right].$$

Here q_s is held fixed as part of the current state. If the analysis also optimizes over the index marginal itself, then that marginal is part of the Bellman control or enlarged state; the local Danskin statement is applied after fixing its current value.

For fixed (s, p, ν, q_s), the logarithmic score is convex in Q and is minimized, pointwise in (s, p, a, o), by the posterior index marginal generated by the same pair belief ν:

$$Q_\nu(s, p, a, o) = q_\nu^+(\cdot \mid s, p, a, o) := \nu(Y \in \cdot \mid s, p, a, O = o)\,,$$

with the usual regular-conditional interpretation, or equivalently the Bayes formula under a dominated observation model. The optimized value

$$\mathcal{A}_\gamma(s,p,\nu;q_s) := \inf_Q \mathcal{J}_\gamma(s,p,\nu,Q;q_s)$$

is exactly the indexed AIR one-step objective: it is immediate indexed loss minus γ times the posterior log gain of the index.

For fixed Q and fixed q_s, the unoptimized objective is affine in the pair belief ν. Hence the optimized value is concave in ν, being the infimum of affine functions. Under the standard compactness, support, and differentiability conditions needed to apply Danskin's theorem, the directional derivative of $\mathcal{A}_\gamma$ is obtained by freezing the optimizer $Q_\nu = q_\nu^+$. Therefore the pointwise gradient in the direction of a fixed truth $(\omega^\star, y^\star)$ is precisely

$$\ell_{\omega^\star,y^\star}(s,p) - \gamma\, \mathbb{E}_{\substack{A\sim p \\ O\sim P_{\omega^\star,s,p,A}}} \log \frac{q_\nu^+(y^\star \mid s,p,A,O)}{q_s(y^\star)},$$

which is the AIR gradient bracket used above. Thus the bracket is not an accidental cancellation: posterior scoring selects the Bayes index predictor, Danskin differentiates through that selected predictor, and the coordinate log score telescopes along the maintained reference update.

The notation in Xu and Zeevi (2025, Section 5) uses the action index $Y = A^\star(\Omega)$. Taking $Y = \Omega$ gives the MAIR or model-index form. In the stationary-posterior case, ν is the current posterior over (Ω, Y) and $q_s = \nu_Y$. In robust AIR, AMS, or EBO, ν may be a selected or optimized admissible belief, and validity comes from the corresponding Bellman bracket or calibration bound. The same concavity–convexity structure also explains the Nash-equilibrium statement in Xu and Zeevi (2025, Lemma 5.2), although the formal proof still has to check compactness, support, and boundary conditions.

C Relation to interactive Fano

Theorem 6.1 is an adaptation of the interactive Fano method of Chen et al. (2024) to the Bellman-sufficient representation framework. The binary-KL step is the same ghost-data argument: one compares the true experiment with a reference experiment, applies data processing to the small-regret event, and obtains a Bayes risk lower bound when the trajectory information is smaller than the logarithmic inverse ghost-good probability. The additional point here is representational. We choose the reference experiment to be the Bayesian posterior-predictive trajectory and choose an information index $Y = \chi(\Omega)$. With this choice, and with an initial packet that is deterministic or independent of Y, the trajectory information admits the exact stepwise identity through Bellman states:

$$I_\mu(Y; H_T) = \mathbb{E}_{H'_T \sim \bar{\mathbb{P}}_\mu^{\mathsf{Alg}}} \sum_{t=1}^T \mathcal{I}_\chi(S'_t, p'_t).$$

If O_0 is informative, the right-hand side is the information acquired after conditioning on O_0, and the unconditional identity includes $I_\mu(Y; O_0)$. Thus the same reference history H'_T generates both the ghost quantile and the indexed information telescope. This is the lower-bound counterpart of the AIR/MAIR upper-bound identity. In particular, $\chi(\Omega) = \Omega$ gives the model-index MAIR form, while $\chi(\Omega) = \pi^\star(\Omega)$ gives the action-index AIR form.

Remark C.1 (Why average notation is useful)**.** *The theorem could be written entirely with cumulative quantities. Writing $\bar{C}_\chi = T^{-1}\mathbb{E}\sum_t \mathcal{I}_\chi$ and $\bar{L} = T^{-1}L$ makes the scaling visible:*

$$\textit{information side} = T\bar{C}_\chi, \qquad \textit{regret side} = Tr.$$

Thus T appears exactly as in one-round reductions, but no posterior trajectory has been collapsed into a static action distribution.

D DEC and one-step information complexity

Offset DEC gives a powerful one-step specialization of the fixed-truth log-potential bracket (41), as illustrated in Lemma 4.5. It retains the immediate decision loss and represents statistical separation by a one-step penalty relative to a reference predictive law, in place of the dynamic continuation term.

In the model-index case $\chi(\omega) = \omega$, let $\mathfrak{C}_t(s)$ be the local comparison set at the current time-state pair and let $\bar{\nu}$ be a reference belief at state s. We write

$$P_{\bar{\nu},s,a} := \int P_{\omega,s,a}\, \bar{\nu}(d\omega)$$

for its predictive law. The corresponding pointwise KL offset is

$$\mathrm{dec}_{\gamma}^{\mathrm{KL}}(s;\bar{\nu}) := \inf_{p\in\Delta(\mathcal{A}_t(s))} \sup_{\omega\in\mathfrak{C}_t(s)} \left\{\mathbb{E}_{a\sim p}\ell_\omega(s,a) - \gamma \mathbb{E}_{a\sim p} D_{\mathrm{KL}}(P_{\omega,s,a} \,\|\, P_{\bar{\nu},s,a})\right\}. \tag{one-step model-index offset}$$

Thus the usual offset DEC may be viewed as a one-step specialization of the Bellman program with a fixed reference law: the learner chooses the one-step decision distribution p, and the local adversary then chooses the hardest model relative to the fixed reference predictive law.

On the lower-bound side, passing from a dynamic posterior-reference experiment to a one-step DEC entails the corresponding reduction. Theorem 6.1 and Theorem 6.2 retain the dynamic reference posteriors; a one-step formulation fixes a reference model or predictive law, decomposes trajectory divergence into one-step divergences, and summarizes the adaptive reference history by a one-round decision distribution. The quantile PAC–DEC of Chen et al. (2024, Section 3.2.2) makes this connection explicit: a reference model $\bar{M}$ generates the ghost experiment, an adversarial model M supplies the hard alternative, and the loss criterion is a quantile of the one-step risk rather than its expectation. Localization, regularity, and problem-specific assumptions enter in the subsequent passage to constrained regret-DEC.

The passage from the dynamic theorem to a one-step coefficient has substantive consequences. On both the upper- and lower-bound sides, DEC may be viewed as a one-step specialization that represents the posterior-reference process through reference models or predictive laws. This interpretation is primarily conceptual: even the distinction between quantile and expected loss raises nontrivial technical questions when deriving regret-DEC lower bounds under regularity conditions. It nevertheless clarifies how one-step DEC values arise from the dynamic indexed Bellman picture.

From this perspective, DEC can be viewed as a one-step information complexity. It is sharp in many settings, particularly when the localized reference geometry is fully represented by the one-round game. The Bellman formulation is complementary in problems where the evolution of the reference trajectory itself carries useful localization information. In such problems, a one-round calculation may separately charge an entropy or estimation term already represented by the posterior-reference lower bound, or may not retain all of the trajectory geometry. The examples below identify conditions under which the two calculations coincide and conditions under which further localization or a dynamic treatment gives a sharper description. This provides a complementary perspective on when one-step DEC calculations are sharp and when retaining the reference trajectory may help.

A global RKHS DEC and the role of localization. Consider a Gaussian RKHS bandit with noise variance $\sigma^2 > 0$, feature map $\phi(x) \in \mathcal{H}$ satisfying $\|\phi(x)\|_{\mathcal{H}} \le 1$, and an action set containing points $\{x_j : j \ge 1\}$ with $\phi(x_j) = e_j$, where $\{e_j\}_{j\ge1}$ is an orthonormal sequence in $\mathcal{H}$. Let the reference model be $f_0 \equiv 0$, and let

$$\mathcal{F}_B = \{f \in \mathcal{H} : \|f\|_{\mathcal{H}} \le B\}.$$

For any one-round decision distribution p, define

$$q_j := \mathbb{E}_{x\sim p}\langle e_j, \phi(x)\rangle_{\mathcal{H}}^2.$$

By Parseval's inequality,

$$\sum_{j\ge1} q_j \le \mathbb{E}_{x\sim p}\|\phi(x)\|_{\mathcal{H}}^2 \le 1.$$

Hence $q_j \to 0$ along a subsequence. For the alternative $f_j = Be_j$, the action x_j is optimal because

$$f_j(x_j) = B, \qquad f_j(x) = B\langle e_j, \phi(x)\rangle_{\mathcal{H}} \le B$$

for every action x. Its expected one-step regret under p is

$$\mathbb{E}_{x\sim p}\Delta_{f_j}(x) = B - B\,\mathbb{E}_{x\sim p}\langle e_j, \phi(x)\rangle_{\mathcal{H}} \ge B(1 - \sqrt{q_j}),$$

where the last step uses Cauchy–Schwarz. The one-step KL information against f_0 is

$$\begin{aligned}\mathbb{E}_{x\sim p}D_{\mathrm{KL}}\big(N(f_j(x), \sigma^2)\big\|N(0, \sigma^2)\big) &= \frac{1}{2\sigma^2}\mathbb{E}_{x\sim p}f_j(x)^2 \\ &= \frac{B^2}{2\sigma^2}q_j.\end{aligned}$$

Therefore, for every finite offset coefficient γ, one may choose j with

$$\sqrt{q_j} + \frac{\gamma B}{2\sigma^2}q_j \le \frac{1}{2},$$

and obtain

$$\mathbb{E}_{x\sim p}\Delta_{f_j}(x) - \gamma\mathbb{E}_{x\sim p}D_{\mathrm{KL}}\big(N(f_j(x), \sigma^2)\big\|N(0, \sigma^2)\big) \ge \frac{B}{2}.$$

Since this holds for every p,

$$\inf_p \sup_{f\in\mathcal{F}_B} \{\mathbb{E}_{x\sim p}\Delta_f(x) - \gamma\mathbb{E}_{x\sim p}D_{\mathrm{KL}}(P_{f,x}\|P_{0,x})\} \ge \frac{B}{2}.$$

The constrained form exhibits the same obstruction at any positive information radius. If the constraint allows alternatives with one-step information at most $\varepsilon > 0$, choose j such that

$$q_j \le \min\left\{\frac{1}{16}, \frac{2\sigma^2\varepsilon}{B^2}\right\}.$$

Then the information of f_j against f_0 is at most ε, while its expected regret under p is at least $3B/4$.

Thus the fixed-reference DEC at $f_0 = 0$ remains at least $B/2$ for every finite γ. The same lower bound applies to any global worst-reference DEC whose reference class includes f_0. This does not by itself yield an effective finite-dimensional rate in the infinite orthogonal example, reflecting the genuine geometry of the full RKHS ball: without localization or additional structure, it contains infinitely many nearly untested possible best-arm directions. Effective-dimension analyses therefore naturally combine DEC with posterior localization, spectral decay, compactness, finite active support, or calibrated confidence restrictions, in the spirit of constrained and localized DEC (Foster et al., 2023).

E Kernel-bandit details

E.1 Finite-scale small-cloud construction

We will show that maximal information gain can be more than a loose final step in the analysis: when it is used to calibrate the acquisition rule, it can create a self-fulfilling exploration cost. The example is a sequential, small-amplitude version of the hub–cloud construction of Xu (2026). Unlike the direct large-cloud embedding of that construction, the cloud below does not make the full class linearly hard. Its height is chosen so that an unrestricted minimax algorithm may ignore it at cost $a_T T$, while its multiplicity still inflates the maximal-information confidence multiplier.

Fix a horizon T, a number $N \geq 4T$, and $a \in (0, 1]$. Let

$$\mathcal{X}_T = \{x_0, x_h, c_1, \ldots, c_N\}$$

and define the diagonal kernel

$$k_T(x_0, x) = 0, \qquad k_T(x_h, x_h) = 1, \qquad k_T(c_i, c_j) = a^2 \mathbf{1}\{i = j\}, \tag{97}$$

with all remaining covariances zero. Thus $k_T(x, x) \leq 1$. Its unit RKHS ball is

$$\mathcal{F}_T = \left\{ f : f(x_0) = 0, \quad f(x_h)^2 + a^{-2} \sum_{j=1}^{N} f(c_j)^2 \leq 1 \right\}. \tag{98}$$

Observations are $Y_t = f(X_t) + \xi_t$ with $\xi_t \overset{\text{i.i.d.}}{\sim} N(0, 1)$.

Let

$$\Gamma_s(T) := \sup_{x_{1:s} \in \mathcal{X}_T^s} \frac{1}{2} \log \det(I + K_{x_{1:s}, x_{1:s}})$$

be the maximal information gain at noise variance one. Consider the following horizon-aware maximal-information calibration

$$\beta_t^{\max} := 1 + \sqrt{2\{\Gamma_{t-1}(T) + 2 \log T\}}, \tag{99}$$

and the zero-mean, unit-ridge GP–UCB rule

$$X_t \in \arg\max_{x \in \mathcal{X}_T} \{m_t(x) + \beta_t^{\max} s_t(x)\}. \tag{100}$$

The conclusion is insensitive to tie-breaking.

Equation (100) is the finite-horizon prototype of the anytime rule (90): it uses the same posterior-UCB acquisition map and the same maximal-information mechanism, with the horizon-dependent kernel and logarithmic calibration made explicit. It is not the original Srinivas–Krause–Kakade–Seeger schedule (91), whose multiplier uses distinct-set information gain and an additional $\log^{3/2}(t/\delta)$ factor. The fixed-kernel proof verifies the small-cloud argument separately for both calibrations.

Theorem E.1 (Small-cloud separation)**.** *Fix $0 < \alpha < 1/4$ and put $a = T^{-\alpha}$. There are constants $0 < c_\alpha < C_\alpha < \infty$ such that, for all sufficiently large T and every $N \geq 4T$,*

$$c_\alpha a T \leq \inf_{\mathsf{Alg}} \sup_{f \in \mathcal{F}_T} \mathbb{E}_f \operatorname{Reg}_T(\mathsf{Alg}) \leq C_\alpha a T. \tag{101}$$

In contrast, the GP–UCB rule (100) satisfies

$$\sup_{f\in\mathcal{F}_T} \mathbb{E}_f \operatorname{Reg}_T(\text{GP–UCB}) \geq c_\alpha T. \tag{102}$$

Consequently,

$$\frac{\sup_{f\in\mathcal{F}_T} \mathbb{E}_f \operatorname{Reg}_T(\text{GP–UCB})}{\inf_{\mathsf{Alg}} \sup_{f\in\mathcal{F}_T} \mathbb{E}_f \operatorname{Reg}_T(\mathsf{Alg})} \geq c_\alpha T^\alpha. \tag{103}$$

Thus maximal-information-calibrated GP–UCB is polynomially minimax-suboptimal on a family of bounded finite kernels, even though the comparison is made over the entire unit RKHS ball.

Proof. We prove the minimax lower bound, the minimax upper bound, and the GP–UCB lower bound separately.

Minimax lower bound. For $j \in [N]$, let $f^j(c_j) = a$ and let f^j vanish on every other action. By (98), $f^j \in \mathcal{F}_T$. Fix an arbitrary possibly randomized nonanticipating algorithm, let U denote its independent random seed, draw J uniformly from $[N]$, and pre-generate independent Gaussian noise stacks at every action. Couple the experiments under f^J and $f \equiv 0$ using the same U and noise stacks, and let $\tau_J = \inf\{t \geq 1 : X_t = c_J\}$. The two histories agree strictly before τ_J, so τ_J is identical in the coupled experiments. Under the null experiment, if S_T^0 is the set of distinct cloud actions queried by time T, then J is independent of S_T^0 and

$$\mathbb{P}(\tau_J \leq T) = \mathbb{P}(J \in S_T^0) = \frac{\mathbb{E}|S_T^0|}{N} \leq \frac{T}{N} \leq \frac{1}{4}.$$

On $\{\tau_J > T\}$ every selected action has mean zero under f^J, whereas the optimal mean is a. Thus the Bayes regret is at least $(1 - T/N)aT \geq 3aT/4$. This proves the first inequality in (101).

Minimax upper bound. Ignore the cloud and run a horizon-T minimax policy for two unit-sub-Gaussian arms on $\{x_0, x_h\}$; a sub-Gaussian version of MOSS is one example (Audibert and Bubeck, 2009; Lattimore and Szepesvári, 2020). Put

$$g_f = \max\{0, f(x_h)\}, \qquad f^\star_{\max} = \max\{g_f, \max_{j\leq N} f(c_j)\}.$$

Since $|f(x_h)| \leq 1$, $\max_j f(c_j) \leq a$, and $f(x_0) = 0$, we have $0 \leq f^\star_{\max} - g_f \leq a$. The restricted policy has expected regret at most $C\sqrt{T}$ relative to g_f, so its regret relative to the full set is at most

$$C\sqrt{T} + aT \leq C_\alpha aT,$$

where the last inequality uses $\alpha < 1/2$. This proves the second inequality in (101).

GP–UCB lower bound. Take the hub truth

$$f^\dagger(x_h) = \Delta := \frac{1}{4}, \qquad f^\dagger(x_0) = f^\dagger(c_j) = 0, \quad j \leq N. \tag{104}$$

It belongs to $\mathcal{F}_T$. For every $s \leq T$, querying s distinct cloud actions is feasible, while for an arbitrary design the hub contributes at most $\frac{1}{2}\log(1+s)$ and the cloud contributes at most $a^2s/2$. Since $\log(1+u) \geq u/2$ for $u \in [0,1]$,

$$\frac{sa^2}{4} \leq \Gamma_s(T) \leq \frac{1}{2}\log(1+s) + \frac{sa^2}{2}. \tag{105}$$

Indeed, if $n_h, n_1, \ldots, n_N$ are the design counts, then $n_h + \sum_j n_j \leq s$, pulls of x_0 contribute zero, and diagonalization gives

$$\frac{1}{2}\log(1+n_h) + \frac{1}{2}\sum_{j=1}^N \log(1 + a^2 n_j) \leq \frac{1}{2}\log(1+s) + \frac{a^2 s}{2}.$$

An unqueried cloud action has posterior mean zero and posterior standard deviation a, independently of all observations at other actions. By the lower bound in (105), at every $t \geq T/2$ its acquisition score is at least

$$a\beta_t^{\max} \geq a^2\sqrt{\frac{t-1}{2}} \geq cT^{1/2-2\alpha}. \tag{106}$$

This diverges because $\alpha < 1/4$. The upper bound in (105) also gives, uniformly for $t \leq T$,

$$\beta_t^{\max} \leq C\{a\sqrt{T} + \sqrt{\log T}\}. \tag{107}$$

Realize observations through the independent action-specific noise stacks: the rth pull of x reveals $f(x) + \xi_{x,r}$. Let $\xi_{h,r} = \xi_{x_h,r}$ and define

$$\mathcal{E}_T := \left\{\sum_{r=1}^{n} \xi_{h,r} \leq 2\sqrt{n\log T} \text{ for every } 1 \leq n \leq T\right\}.$$

A Gaussian tail bound and a union bound give

$$\mathbb{P}(\mathcal{E}_T^c) \leq \sum_{n=1}^{T} \exp\left\{-\frac{(2\sqrt{n\log T})^2}{2n}\right\} \leq T^{-1}.$$

After n hub pulls, diagonal conjugacy gives

$$m_t(x_h) = \frac{n\Delta + \sum_{r=1}^{n}\xi_{h,r}}{1+n}, \qquad s_t(x_h) = \frac{1}{\sqrt{1+n}}.$$

Suppose, toward a contradiction, that fewer than $T/4$ cloud pulls occur by time T. The anchor x_0 is never selected: its acquisition score is zero, while an unqueried cloud has strictly positive score. Hence, for every $t \in [\lceil T/2\rceil, T]$, the number n of preceding hub pulls is at least $T/4 - 1$. On $\mathcal{E}_T$, (107) yields

$$\begin{aligned} m_t(x_h) + \beta_t^{\max} s_t(x_h) &\leq \Delta + C\sqrt{\frac{\log T}{T}} + C\left(a + \sqrt{\frac{\log T}{T}}\right) \\ &\leq \frac{1}{2} \end{aligned}$$

for all sufficiently large T. On the other hand, (106) is then larger than one. Since $N \geq 4T$, an unqueried cloud is always available. GP–UCB must therefore select a cloud action on every round in the last half of the horizon, contradicting the supposition that it makes fewer than $T/4$ cloud pulls. Thus, on $\mathcal{E}_T$, it makes at least $T/4$ cloud pulls. Each such pull has regret Δ, and consequently

$$\mathbb{E}_{f^\dagger}\operatorname{Reg}_T(\text{GP–UCB}) \geq \frac{\Delta T}{4}(1 - T^{-1}),$$

which proves (102). Combining this lower bound with the minimax upper bound in (101) proves (103). $\square$

The Whitehouse effective-optimism scale. For comparison, Whitehouse et al. (2023, Corollary 2) prove that, for $\nu > 1/2$, their horizon-tuned GP–UCB construction with $\rho \asymp T^{d/(2\nu+2d)}$ satisfies

$$\sup_{\|f\|_{\mathcal{H}_k} \le B} \mathbb{E}_f \operatorname{Reg}_T = \widetilde{O}\Big(T^{\frac{\nu+2d}{2\nu+2d}}\Big). \tag{108}$$

Here their high-probability statement implies the expectation bound after taking the failure probability polynomially small and using the uniform RKHS envelope. Example 2 of Wang and Zhang (2026) identifies the corresponding Whitehouse-inspired effective-optimism scale. For every time-indexed schedule satisfying their assumptions and

$$L_T = \widetilde{\Theta}\Big(T^{\frac{d}{2\nu+2d}}\Big),$$

their Theorem 1 gives

$$\sup_{\|f\|_{\mathcal{H}_k} \le B} \mathbb{E}_f \operatorname{Reg}_T = \widetilde{\Omega}\Big(T^{\frac{\nu+2d}{2\nu+2d}}\Big), \tag{109}$$

because

$$\frac{\nu+d}{2\nu+d} + \frac{d}{2\nu+2d}\,\frac{\nu}{2\nu+d} = \frac{\nu+2d}{2\nu+2d}.$$

The common exponent exceeds the minimax exponent by

$$\frac{\nu d}{2(\nu+d)(2\nu+d)}. \tag{110}$$

Thus the Whitehouse–Wu–Ramdas upper exponent and the Wang–Zhang lower exponent coincide at this effective-optimism scale. The two statements are not a matching $\widetilde{\Theta}$ theorem for one identical implementation: the former holds a horizon-dependent regularizer fixed during the run, whereas the latter assumes time-indexed, two-sided polynomial-times-log schedules.

E.2 Two further algorithmic specializations

Algorithm 3: GP–E2D. A GP–E2D rule keeps the same GP posterior reference but chooses a distribution over actions by solving a localized DEC offset. Given a localized comparison set $\mathcal{F}_t$ containing plausible functions, define

$$\Delta_f(p) := \max_{x\in\mathcal{X}} f(x) - \mathbb{E}_{x\sim p} f(x),$$

and let $P_{f,x} = N(f(x), \lambda)$ while $P_{t,x} = N(m_t(x), \lambda + s_t^2(x))$ is the posterior predictive. A schematic GP–E2D decision is

$$p_t \in \arg\min_{p\in\Delta(\mathcal{X})} \sup_{f\in\mathcal{F}_t} \left\{\Delta_f(p) - \gamma_t \mathbb{E}_{x\sim p} D_{\mathrm{KL}}(P_{f,x}\|P_{t,x})\right\}. \tag{111}$$

If $f^\star \in \mathcal{F}_t$ and the optimized value in (111) is at most ε_t, then

$$\Delta_{f^\star}(p_t) \le \gamma_t \mathbb{E}_{x\sim p_t} D_{\mathrm{KL}}(P_{f^\star,x}\|P_{t,x}) + \varepsilon_t. \tag{112}$$

Summing (112) gives a regret bound once the weighted fixed-truth sum

$$\sum_{t=1}^{T} \gamma_t \mathbb{E}_{x\sim p_t} D_{\mathrm{KL}}(P_{f^\star,x}\|P_{t,x})$$

is controlled by a compatible posterior-ratio, redundancy, PAC–Bayes, or other sequential argument. When $\gamma_t \equiv \gamma$, it is enough to control the corresponding unweighted KL sum. Under a well-specified Bayesian GP reference, the ordinary MAIR chain rule supplies the unweighted control after prior averaging; it does not by itself justify arbitrary time-varying weights. At a fixed frequentist truth, even the unweighted statement requires a sequential bound along the same trajectory.

This is the technical point that a GP–E2D analysis must address: the one-round DEC is optimized, but the compatibility between that coefficient and the realized GP information process is not automatic. Boundedness of the RKHS ball ensures only that the one-step offset is finite; it does not force the offset to decrease with the information multiplier. The global infinite-orthogonal example in Appendix D makes this distinction explicit: although $\mathcal{F}_B$ is a bounded radius-B ball, its fixed-reference DEC remains at least $B/2$ for every finite γ, and hence the standard upper bound obtained by summing these offsets cannot be sublinear. That example uses the fixed reference $f_0 = 0$ and does not, by itself, establish the same lower bound for a GP–E2D rule with an evolving posterior-predictive reference. In the latter case, one must additionally control the fixed-truth predictive KL terms by a redundancy, posterior-ratio, PAC–Bayes, or other sequential argument along the trajectory. Localization can address either obstruction, but the present paper does not prove failure of every unlocalized GP–E2D rule.

Algorithm 4: AMS/EBO on the action index. AMS/EBO attacks the same specialized log-potential bracket from the robust posterior side. In the finite-action kernel marginal, the natural AIR index is $A^\star$. Let q_t be the current reference distribution on $A^\star$, let $\mathfrak{B}_t(q_t)$ be a calibrated local set of candidate beliefs ν on the active reward vector F, and fix a constant coefficient $\gamma > 0$. For each ν, define $\Delta_\nu^{\mathrm{AIR}}(p)$ and $\mathcal{I}_\nu^{\mathrm{AIR}}(p)$ by the analogues of (83) and (82), using the conditional laws induced by ν. A robust AIR/EBO decision is

$$p_t \in \operatorname*{arg\,min}_{p \in \Delta(\mathcal{X})} \sup_{\nu \in \mathfrak{B}_t(q_t)} \left\{ \Delta_\nu^{\mathrm{AIR}}(p) - \gamma \mathcal{I}_\nu^{\mathrm{AIR}}(p) - \gamma D_{\mathrm{KL}}(\nu_{A^\star} \| q_t) \right\}. \tag{113}$$

The maximization over ν is essential: without it the display is only the posterior AIR bracket for a chosen belief, not the robust AMS/EBO relaxation. In a finite-index abstraction, the last two terms are the KL-dual form of the log-partition potential (46). If the robust value in (113) is at most ε_t, the required concavity/first-order conditions in Lemma 5.6 hold, and the belief set is surely calibrated so that the fixed truth $f^\star$ is an admissible comparison direction at every reached state, then Theorem 5.1 gives

$$\mathbb{E}_{f^\star} \mathrm{Reg}_T \le \gamma \log \frac{1}{q_1(a^\star)} + \mathbb{E}_{f^\star} \sum_{t=1}^{T} \varepsilon_t, \tag{114}$$

where $a^\star = A^\star(f^\star)$. The constant coefficient avoids the false shortcut of replacing a weighted fixed-truth log-gain sum by a single telescope. If a varying coefficient is used, the corresponding weighted telescope must be proved separately. Under a Bayesian reference prior, posterior averaging of the same coordinate bound gives

$$\mathbb{E}\, \mathrm{Reg}_T \le \gamma H(A^\star) + \mathbb{E} \sum_{t=1}^{T} \varepsilon_t.$$

A ratio form gives the familiar Cauchy–Schwarz variant after replacing the fixed coefficient by the appropriate sequential AIR ratio and applying the action-index chain rule. Since $H(A^\star) \le \log n$, this route scales with the finite decision index and the sequential AIR coefficient.

Two forms of localization. GP–E2D and AMS/EBO localize different variables. In (111), GP–E2D restricts the fixed-truth comparison to a model set $\mathcal{F}_t$ and measures its elements against the posterior predictive law $P_{t,x}$. Its one-step inequality becomes a sequential frequentist guarantee only when the truth remains admissible and the resulting KL sum has a compatible fixed-truth bound. In (113), AMS/EBO instead optimizes a robust class of pair beliefs and propagates a finite optimal-action marginal. Calibration of the belief class and compatibility of its generated reference update are separate requirements; posterior concentration alone supplies neither one.

For Theorem 7.3, the retained sets $\mathcal{R}_m$ and the epoch schedule are fixed in advance. The robust action-index AIR/AMS saddle adapts its belief and action law *within* each retained marginal, but it is not shown to discover the spatial truncation. Nor does the rate proof require a nonconstant continuation value: after truncation, the state-independent supersolution $\overline{W}_t = (T-t+1)c_\gamma$ and the across-time coordinate log-score telescope suffice. The action index is crucial to this clean proof because it gives a finite initial code length, but the theorem does not establish that action indexing or nontrivial Bellman continuation is necessary. It therefore leaves open whether a model-index GP–E2D or MAIR construction with a valid cover, neighborhood, redundancy, or PAC–Bayes argument can attain the same rate.

E.3 Numerical protocol for the hub–cloud experiment

For each $T \in \{512, 1024, 2048, 4096, 8192, 16384\}$, the experiment takes $\alpha = 1/5$, $a = T^{-1/5}$, $N = 4T$, unit observation variance, and $\Delta = 1/4$. Because $N \geq T$, the maximal information gain of the diagonal kernel is computed exactly as

$$\Gamma_s(T) = \frac{1}{2} \max_{0 \leq h \leq s} \left\{ \log(1+h) + (s-h)\log(1+a^2) \right\}. \tag{115}$$

The integer maximizer lies among the clipped integers adjacent to $1/\log(1+a^2) - 1$. The horizon-aware rule uses (99) without approximation, while the finite-block anytime rule replaces $2\log T$ by $2\log(t \vee 2)$. For the original schedule with $\delta = 0.1$, the exact distinct-set information gain is

$$\Gamma_s^{\mathrm{SKKS}}(T) = \frac{1}{2}\{\log 2 + (s-1)\log(1+a^2)\}, \qquad 1 \leq s \leq T,$$

because an optimal size-s set contains the hub and $s-1$ distinct cloud points. For a diagonal coordinate of prior variance d, after n pulls with observation sum S, the implementation uses posterior mean $dS/(1+nd)$ and variance $d/(1+nd)$. If a sampled cloud coordinate has posterior mean m and standard deviation $s < a$, its score gap from an unused cloud coordinate is $m - \beta_t(a-s)$, which is nonincreasing because β_t is nondecreasing. Once this gap is nonpositive, the coordinate cannot be selected again. Ties between sampled and unused cloud coordinates are resolved in favor of the unused one; equivalently, under Gaussian noise the pruning gives almost surely the same choices as direct maximization over all $N+2$ actions.

For the localized comparator, let f^+ and f^- have hub means $+\Delta$ and $-\Delta$, respectively, and value zero at the anchor. Their optimal-action indices are the hub and the anchor. Write q for the current reference mass on f^+, r for the candidate pair-belief mass on f^+, and p for the probability of selecting the hub. The following control is a direct two-model Gaussian specialization of the AIR/AMS saddle of Xu and Zeevi (2025). Its relation to Lattimore and György (2021) is through the EBO principle and minimax/KL duality: the numerical code solves the belief-space AIR/AMS saddle, not EBO's functional-estimator mirror-descent implementation. The robust AIR/AMS saddle is

$$\inf_{0 \leq p \leq 1} \sup_{0 \leq r \leq 1} \left[\Delta\{r(1-p) + (1-r)p\} - \gamma p I_r - \gamma \,\mathrm{kl}(r,q) \right], \qquad \gamma = \sqrt{\frac{(1+\Delta^2)T}{\log 2}}, \tag{116}$$

where I_r is the mutual information of the binary Gaussian channel $Z \mid f^{\pm} \sim N(\pm\Delta, 1)$:

$$I_r = r\mathbb{E}_{Z\sim N(\Delta,1)} \left[-\log\{r + (1-r)e^{-2\Delta Z}\}\right] \\ + (1-r)\mathbb{E}_{Z\sim N(-\Delta,1)} \left[-\log\{(1-r) + re^{2\Delta Z}\}\right]. \tag{117}$$

This is a finite instance of (113). The exact saddle is calibrated only to the displayed pair class $\{f^+, f^-\}$, not to the full projected RKHS ball. Sion's identity reduces it to

$$\sup_{0\le r\le 1} \min\{\Delta r - \gamma\,\mathrm{kl}(r,q), \Delta(1-r) - \gamma I_r - \gamma\,\mathrm{kl}(r,q)\},$$

which is a scalar concave maximization. For completeness, let $v_c = \mathrm{logit}(r_c)$, where r_c solves

$$\Delta(2r_c - 1) + \gamma I_{r_c} = 0,$$

and write $I'(r) = dI_r/dr$. If $u = \mathrm{logit}(q)$, define

$$u_0 = v_c - \Delta/\gamma, \qquad u_1 = v_c + \Delta/\gamma + I'(r_c).$$

The saddle has $p = 0$ and $\mathrm{logit}(r) = u + \Delta/\gamma$ for $u \le u_0$; it has $r = r_c$ and $p = (u - u_0)/(u_1 - u_0)$ for $u_0 < u < u_1$; and it has $p = 1$ for $u \ge u_1$, with $v = \mathrm{logit}(r)$ determined by

$$v - u + \Delta/\gamma + I'\big(\mathrm{logit}^{-1}(v)\big) = 0.$$

After an anchor pull, $u^+ = v$. After a hub observation $Z = z$, Gaussian Bayes updating gives $u^+ = v + 2\Delta z$.

The numerical policy evaluates (117) by 48-node Gauss–Hermite quadrature. Scalar roots are solved to absolute tolerance 2×10^{-13}; the smooth $p = 1$ branch is tabulated on 5001 log-odds points and linearly interpolated, with its limiting shift used beyond the tabulated range. Thus the code approximates the exact scalar saddle. As a diagnostic, it reevaluates both fixed-truth brackets by 96-node quadrature,

$$\Delta_\sigma(p) - \gamma\mathbb{E}_\sigma \log\frac{q^+(Y_\sigma \mid A, Z)}{q(Y_\sigma)} - \frac{1+\Delta^2}{\gamma}$$

on a grid of more than ten thousand reference log-odds and at every reference state reached in the Monte Carlo runs. The largest static-grid and realized-state values over all displayed horizons are approximately -3.584×10^{-3}, and the largest tabulated first-order residual is 6.81×10^{-8}. Thus no positive fixed-truth Bellman-bracket violation is observed at the reported numerical precision. This numerical check is diagnostic, not an interval-arithmetic proof of a uniform inequality.

Each algorithm–truth pair uses 200 independent NumPy streams initialized from seed 20260717; the streams of the original horizon-aware/AIR experiment are retained unchanged, and separate streams are used for the two added GP–UCB schedules. Shaded regions are twice the Monte Carlo standard error. The right-panel localization curve is the exact value aT, not a Monte Carlo estimate. The one-hot cloud truths are external to the binary saddle class; their role is only to evaluate the deterministic loss incurred by the predetermined truncation. The script `kernel_air_simulation.py`, included with the source, reproduces the figure and prints the parameters, package versions, all three GP–UCB regrets, retained-truth regrets, first-order residual, and bracket checks.

F Gaussian bandits: finite-index matching

In this subsection, we apply our lower-bound approach to multi-armed bandits (MAB). Consider $K \geq 3$ Gaussian arms with unit noise variance. Let the ambient model space be a class $\Omega_{\mathrm{MAB}} \subseteq \mathbb{R}^K$ of mean vectors, and write a model environment as $\omega = (\theta_\omega(1), \ldots, \theta_\omega(K))$. Pulling arm a in environment ω produces an observation distributed as $N(\theta_\omega(a), 1)$. Let

$$A^\star(\omega) \in \arg\max_{a \in [K]} \theta_\omega(a)$$

with an arbitrary fixed tie-breaking rule, and define the information index

$$\chi(\omega) = A^\star(\omega) \in [K].$$

Thus the random environment is the full mean vector Ω, while the information target used in the lower bound is only

$$Y = \chi(\Omega) = A^\star(\Omega).$$

For the lower bound, fix $\Delta > 0$ and use the K-point prior μ_Δ supported on the spike environments

$$\omega^j = \Delta e_j, \qquad j \in [K],$$

where e_j is the jth coordinate vector. Equivalently, draw $J \sim \mathrm{Unif}([K])$ and set $\Omega = \omega^J$. On this prior support the optimal arm is unique and

$$Y = \chi(\Omega) = J.$$

The equality $Y = J$ is only a property of this least-favorable prior; it should not be read as identifying the information target with the full model environment. The one-step regret in environment ω^j is

$$\ell_{\omega^j}(a) = \max_b \theta_{\omega^j}(b) - \theta_{\omega^j}(a) = \Delta \mathbf{1}\{a \neq j\}.$$

For brevity write $\ell_j = \ell_{\omega^j}$ on this subfamily.

Lemma F.1 (MAB ghost entropy)**.** *Let $Y = \chi(\Omega)$ under the prior $\Omega \sim \mu_\Delta$. For every algorithm and every reference history H'_T independent of $Y \sim \mathrm{Unif}([K])$,*

$$\mathbb{P}\left(\bar{L}_Y(H'_T) \leq \frac{\Delta}{2}\right) \leq \frac{2}{K},$$

where $\bar{L}_y(H'_T) = T^{-1} \sum_{t=1}^T \ell_y(A'_t)$ and A'_t denotes the arm selected in the reference history at time t.

Proof. For a fixed reference action sequence $a'_1, \ldots, a'_T$, let

$$N'_y = \sum_{t=1}^T \mathbf{1}\{a'_t = y\}.$$

The average regret against the spike environment ω^y is

$$\bar{L}_y(H'_T) = \Delta(1 - N'_y/T).$$

The event $\bar{L}_y(H'_T) \le \Delta/2$ implies $N'_y \ge T/2$. Since Y is uniform and independent of the reference sequence,

$$\mathbb{E}[N'_Y \mid H'_T] = \frac{1}{K}\sum_{y=1}^{K} N'_y = \frac{T}{K}.$$

Markov's inequality gives

$$\mathbb{P}(N'_Y \ge T/2 \mid H'_T) \le \frac{2}{K}.$$

Averaging over H'_T proves the claim. □

Lemma F.2 (MAB information bound)**.** *There is a universal constant $c_0 > 0$ such that if $\Delta^2 T/K \le c_0$, then every adaptive algorithm satisfies*

$$I(Y; H_T) \le \frac{1}{4}\log(K/2),$$

where $Y = \chi(\Omega)$ and $\Omega \sim \mu_\Delta$.

Proof. Let $\mathbb{P}_0$ be the law of the history under the zero-mean reference model, let $\mathbb{P}_j$ be the law under the spike environment ω^j, and let

$$\bar{\mathbb{P}} = \frac{1}{K}\sum_{j=1}^{K} \mathbb{P}_j$$

be the marginal law of H_T under the prior μ_Δ. Under this prior, $Y = j$ is equivalent to $\Omega = \omega^j$, but the mutual information is still the action-index information $I(Y; H_T)$, not information about an ambient full mean vector outside the prior support. Since $\bar{\mathbb{P}} \ge K^{-1}\mathbb{P}_j$, the likelihood ratio $d\mathbb{P}_j/d\bar{\mathbb{P}}$ is bounded by K. The inequality

$$D_{\mathrm{KL}}(P\|Q) \le (2 + \log K) H^2(P, Q), \qquad \text{whenever } Q \ge K^{-1}P,$$

where

$$H^2(P, Q) := \int (\sqrt{dP} - \sqrt{dQ})^2$$

is the standard squared Hellinger distance (twice the h^2 normalization used earlier), together with the triangle inequality and convexity for squared Hellinger distance, gives

$$I(Y; H_T) = \frac{1}{K}\sum_{j=1}^{K} D_{\mathrm{KL}}(\mathbb{P}_j\|\bar{\mathbb{P}}) \le 4(2 + \log K)\frac{1}{K}\sum_{j=1}^{K} H^2(\mathbb{P}_j, \mathbb{P}_0).$$

Using $H^2(P, Q) \le D_{\mathrm{KL}}(Q\|P)$ and the adaptive Gaussian KL chain rule under the zero model,

$$\frac{1}{K}\sum_{j=1}^{K} H^2(\mathbb{P}_j, \mathbb{P}_0) \le \frac{1}{K}\sum_{j=1}^{K} D_{\mathrm{KL}}(\mathbb{P}_0\|\mathbb{P}_j) = \frac{\Delta^2}{2K}\mathbb{E}_0\sum_{t=1}^{T}\sum_{j=1}^{K} \mathbf{1}\{A_t = j\} = \frac{\Delta^2 T}{2K}.$$

Therefore

$$I(Y; H_T) \le 2(2 + \log K)\frac{\Delta^2 T}{K}.$$

Choosing, for example,

$$c_0 \le \inf_{K \ge 3} \frac{\log(K/2)}{8(2 + \log K)}$$

which is a strictly positive universal constant, proves the displayed bound. □

Theorem F.3 (MAB lower bound)**.** *For Gaussian K-armed bandits with unit noise variance, over any ambient mean-vector class Ω_{MAB} that contains the spike instances $\{\Delta e_j : j \in [K]\}$ used below,*

$$\mathfrak{R}_T^\star \geq c\sqrt{KT}$$

for a universal constant $c > 0$ whenever $T \geq K \geq 3$. In particular, this applies to any bounded class such as $[0,1]^K$ once the universal constant in the choice of Δ is small enough.

Proof. Choose

$$\Delta = c_1\sqrt{K/T}$$

with c_1 a sufficiently small universal constant, and let $r = \Delta/2$. Lemma F.1 gives $p_r \leq 2/K \leq 2/3$. The proof of Lemma F.2 gives the sharper bound

$$T\bar{C} = I(Y; H_T) \leq 2(2 + \log K)c_1^2.$$

Choose c_1 so small that, for every $K \geq 3$,

$$2(2 + \log K)c_1^2 \leq \mathrm{kl}\left(\frac{3}{4}, \frac{2}{K}\right).$$

This is possible because

$$\inf_{K \geq 3} \frac{\mathrm{kl}\left(\frac{3}{4}, \frac{2}{K}\right)}{2(2 + \log K)} > 0.$$

Since $p_r \leq 2/K \leq 2/3 < 3/4$, the exact quantile bound (65) gives

$$\psi(p_r, T\bar{C}) \leq \frac{3}{4}.$$

Therefore

$$\mathfrak{R}_T^\star \geq Tr\left(1 - \frac{3}{4}\right) = \frac{T\Delta}{8} = c\sqrt{KT},$$

where $c > 0$ is a universal constant. □

Proposition F.4 (MAB upper bound and minimax-rate matching)**.** *For Gaussian K-armed bandits with unit noise variance and mean vectors in $[0,1]^K$, there exists a universal constant $C > 0$ and a nonanticipating algorithm, for example a sub-Gaussian version of MOSS, whose regret satisfies*

$$\sup_\omega \mathbb{E}_\omega^{\mathsf{Alg}} L_\omega(H_T) \leq C\sqrt{KT}$$

for all $T \geq K \geq 3$. For another fixed bounded cube, the constant may depend on its mean range. Consequently, together with Theorem F.3,

$$\mathfrak{R}_T^\star(\Omega_{\mathrm{MAB}}) = \Theta(\sqrt{KT})$$

on every ambient class containing the spike hard family and contained in $[0,1]^K$. In the notation of Section 3, the hard prior has $M = K$ index values, the upper code length on that same spike subfamily is $L_0 = \log K$, and Lemma F.1 gives

$$E_r = \log(1/p_r) \geq \log(K/2)$$

at radius $r = \Delta/2 \asymp \sqrt{K/T}$. =Hence the hard-subfamily entropy ratio L_0/E_r is bounded by a universal constant for $K \geq 3$, while the standard minimax upper bound matches the lower rate on the ambient class. This rate comparison does not by itself verify upper-at-lower-entropy calibration of the specific Bellman program for MOSS.

Proof. The lower bound is Theorem F.3. The upper bound is the standard distribution-free minimax upper bound for sub-Gaussian stochastic multi-armed bandits, achieved by a sub-Gaussian MOSS variant and recorded in modern bandit texts (Audibert and Bubeck, 2009; Lattimore and Szepesvári, 2020). The entropy comparison follows directly from the explicit hard prior: under the uniform optimal-arm index prior $q_1(j) = 1/K$, the upper coordinate code length is $\log K$, while Lemma F.1 gives $p_r \leq 2/K$. Therefore $L_0/E_r \leq \log K/\log(K/2)$, which is bounded for $K \geq 3$. ☐

This is the cleanest finite-index example of the Bellman–Fano entropy calculation together with a matching minimax rate. The full environment is the whole mean vector, but the relevant index is the optimal arm. The hard prior, ghost entropy, and information bound operate at the same radius, and a standard minimax algorithm attains the resulting rate; a formal Bellman sandwich for that particular algorithm would additionally identify its upper budget value and verify Definition 3.6.

G Hypercube linear bandits: indexed matching

Consider stochastic linear bandits with action set

$$\mathcal{A} = \{a \in \mathbb{R}^d : \|a\|_2 \leq 1\}.$$

Let the ambient model class be the Euclidean unit parameter ball

$$\Omega_{\text{lin}} = \{\omega \in \mathbb{R}^d : \|\omega\|_2 \leq 1\},$$

and regard an environment $\omega \in \Omega_{\text{lin}}$ as the full mean parameter vector. At time t, after choosing $A_t \in \mathcal{A}$, the learner observes

$$O_t = \langle \omega, A_t \rangle + \xi_t, \qquad \xi_t \sim N(0, 1),$$

where the noises are independent over time and independent of the learner's external randomization. For $\omega \neq 0$, the unique optimal action over $\mathcal{A}$ is

$$A^\star(\omega) = \frac{\omega}{\|\omega\|_2}.$$

At $\omega = 0$, fix any deterministic tie-breaking action; this case is not used by the lower-bound prior. The information index is the optimal action

$$\chi(\omega) = A^\star(\omega), \qquad Y = \chi(\Omega),$$

not the full environment Ω. Thus two environments that have the same optimal action are not distinguished by the index unless the prior support itself makes them distinguishable.

We prove the standard $\Omega(d\sqrt{T})$ minimax lower bound through the quantile indexed information theorem. The proof uses a hypercube prior together with an adaptive information bound. The role of the hypercube variable below is only to code the finitely many optimal actions in the prior support. It should not be read as replacing the ambient environment Ω by a model label. Note that for a non-Gaussian prior, one cannot condition on the realized adaptive design matrix and then apply the fixed-design Gaussian-channel capacity formula, because the design itself is a statistic of the past observations and may carry information about the unknown environment and hence about Y. Chen et al. (2024, Section 3.3) prove a closely related lower bound by a different route: they use the interactive Fano method with a Gaussian prior and a unit-sphere calculation. Both approaches show that action-indexed information arguments can recover the sharp linear-bandit order, whereas model-indexed relaxations such as DEC can lose factors when they charge for estimating more of the model than is needed to identify the optimal action.

Let

$$V \sim \mathrm{Unif}(\{\pm 1\}^d), \qquad \Omega = \Delta V.$$

Assume $\Delta\sqrt{d} \le 1$, so that this prior is supported on Ω_{lin}. On this prior support,

$$Y = \chi(\Omega) = A^\star(\Omega) = \frac{V}{\sqrt{d}}. \tag{118}$$

The map $v \mapsto v/\sqrt{d}$ is one-to-one on $\{\pm 1\}^d$. Hence, for this prior only, V may be used as a code for the action index Y, and

$$I(Y; H_T) = I(V; H_T).$$

For an index value $y = v/\sqrt{d}$, write ℓ_v for the regret under the corresponding environment $\omega = \Delta v$. The one-step regret of action $a \in \mathbb{R}^d$, $\|a\|_2 \le 1$, is

$$\ell_v(a) = \Delta\sqrt{d} - \Delta\langle v, a\rangle.$$

Assume $T \ge d^2$ and choose $\Delta \le 1/\sqrt{d}$, so that $\|\Omega\|_2 = \Delta\sqrt{d} \le 1$.

Lemma G.1 (Linear ghost entropy)**.** *There is a universal constant $c_{\mathrm{gh}} > 0$ such that every deterministic reference action sequence $a'_1, \ldots, a'_T$, with $\|a'_t\|_2 \le 1$, satisfies*

$$\mathbb{P}_{V \sim \mathrm{Unif}(\{\pm 1\}^d)}\left(\frac{1}{T}\sum_{t=1}^{T}\left(\Delta\sqrt{d} - \Delta\langle V, a'_t\rangle\right) \le \frac{\Delta\sqrt{d}}{4}\right) \le \exp(-c_{\mathrm{gh}} d).$$

Consequently, the same bound holds after averaging over any random reference history H'_T whose induced action sequence is independent of the action index Y, equivalently independent of V under the hypercube prior.

Proof. Let

$$\bar{a} := \frac{1}{T}\sum_{t=1}^{T} a'_t.$$

Since $\|\bar{a}\|_2 \le T^{-1}\sum_t \|a'_t\|_2 \le 1$, the event in the display implies

$$\Delta\sqrt{d} - \Delta\langle V, \bar{a}\rangle \le \frac{\Delta\sqrt{d}}{4},$$

or equivalently,

$$\langle V, \bar{a}\rangle \ge \frac{3\sqrt{d}}{4}.$$

The random variable $\langle V, \bar{a}\rangle = \sum_{i=1}^d V_i \bar{a}_i$ is a centered Rademacher sum with variance proxy $\|\bar{a}\|_2^2 \le 1$. Hoeffding's inequality therefore gives

$$\mathbb{P}\left(\langle V, \bar{a}\rangle \ge \frac{3\sqrt{d}}{4}\right) \le \exp\left(-\frac{9d}{32}\right).$$

Thus the first claim holds, for example with $c_{\mathrm{gh}} = 9/32$. If the reference action sequence is random but independent of Y, then because $Y = V/\sqrt{d}$ is a bijective function of V on the prior support, the reference sequence is independent of V. Conditioning on the reference history and applying the deterministic bound gives the second claim. $\square$

Lemma G.2 (Adaptive hypercube information capacity)**.** *For the hypercube prior above, every possibly randomized adaptive algorithm satisfies*

$$I(Y; H_T) = I(V; H_T) \leq \Delta^2 T,$$

where $H_T = (A_1, O_1, \ldots, A_T, O_T)$.

Proof. Since $Y = V/\sqrt{d}$ is a bijective function of V on the hypercube support, $I(Y; H_T) = I(V; H_T)$. It remains to bound $I(V; H_T)$. Let U denote the learner's internal random seed, independent of V. Then

$$I(V; H_T) \leq I(V; H_T, U) = \mathbb{E}_U I(V; H_T \mid U).$$

It is therefore enough to prove the claim after conditioning on U, so that the algorithm is deterministic.

For $v \in \{\pm 1\}^d$, let $\mathbb{P}_v$ denote the trajectory law when $\Omega = \Delta v$. We use the chain rule over coordinates:

$$I(V; H_T) = \sum_{i=1}^{d} I(V_i; H_T \mid V_1, \ldots, V_{i-1}).$$

Fix a coordinate i and a prefix $u = (v_1, \ldots, v_{i-1})$. Let $\mathbb{P}_{u,+}$ and $\mathbb{P}_{u,-}$ be the trajectory laws after averaging uniformly over the remaining coordinates $V_{i+1}, \ldots, V_d$, conditional respectively on $V_i = +1$ and $V_i = -1$. Then

$$I(V_i; H_T \mid V_{<i} = u) = D_{\mathrm{JS}}(\mathbb{P}_{u,+}, \mathbb{P}_{u,-}),$$

where D_{JS} denotes Jensen–Shannon divergence with equal weights. For any two probability measures P, Q,

$$D_{\mathrm{JS}}(P, Q) \leq \frac{1}{4}\Big(D_{\mathrm{KL}}(P \| Q) + D_{\mathrm{KL}}(Q \| P)\Big).$$

Next couple the two mixtures by using the same suffix. If $w \in \{\pm 1\}^{d-i}$ and

$$v^+(u, w) = (u, +1, w), \qquad v^-(u, w) = (u, -1, w),$$

then joint convexity of relative entropy gives

$$D_{\mathrm{KL}}(\mathbb{P}_{u,+} \| \mathbb{P}_{u,-}) \leq \mathbb{E}_w D_{\mathrm{KL}}\left(\mathbb{P}_{v^+(u,w)} \,\middle\|\, \mathbb{P}_{v^-(u,w)}\right),$$

and the same argument gives the analogous reverse inequality.

For two fixed vertices v^+ and v^- that differ only in coordinate i, the adaptive KL chain rule gives

$$\begin{aligned} D_{\mathrm{KL}}(\mathbb{P}_{v^+} \| \mathbb{P}_{v^-}) &= \frac{1}{2}\mathbb{E}_{v^+} \sum_{t=1}^{T} \Big(\Delta \langle v^+ - v^-, A_t \rangle\Big)^2 \\ &= 2\Delta^2 \mathbb{E}_{v^+} \sum_{t=1}^{T} A_{t,i}^2. \end{aligned}$$

Here the action kernel contributes no KL because, after conditioning on the learner's internal randomness, the algorithm uses the same decision rule under both environments; only the Gaussian observation kernel changes. Similarly,

$$D_{\mathrm{KL}}(\mathbb{P}_{v^-} \| \mathbb{P}_{v^+}) = 2\Delta^2 \mathbb{E}_{v^-} \sum_{t=1}^{T} A_{t,i}^2.$$

Combining the preceding displays yields

$$
\begin{aligned}
I(V_i; H_T \mid V_{<i} = u) &\le \frac{\Delta^2}{2}\mathbb{E}_w\left[\mathbb{E}_{v^+(u,w)}\sum_{t=1}^{T} A_{t,i}^2 + \mathbb{E}_{v^-(u,w)}\sum_{t=1}^{T} A_{t,i}^2\right] \\
&= \Delta^2\mathbb{E}\left[\sum_{t=1}^{T} A_{t,i}^2 \,\middle|\, V_{<i} = u\right],
\end{aligned}
$$

where the final expectation is under the original hypercube prior and the algorithm's trajectory law. Averaging over $V_{<i}$ and summing over i gives

$$
\begin{aligned}
I(V; H_T) &\le \Delta^2\mathbb{E}\sum_{t=1}^{T}\sum_{i=1}^{d} A_{t,i}^2 \\
&= \Delta^2\mathbb{E}\sum_{t=1}^{T}\|A_t\|_2^2 \\
&\le \Delta^2 T.
\end{aligned}
$$

This proves the claim. $\square$

Remark G.3 (Why the fixed-design log-determinant argument is not used)**.** *For a Gaussian prior on a full parameter vector, posterior conjugacy gives the adaptive information identity*

$$
I(\Theta; H_T) = \frac{1}{2}\mathbb{E}\log\det\left(I + \Sigma_0^{1/2}\Big(\sum_{t=1}^{T} A_t A_t^\top\Big)\Sigma_0^{1/2}\right),
$$

and this is bounded by a trace-constrained log determinant. For the hypercube prior, however, the posterior is not Gaussian and the realized design matrix is itself a statistic of the observations. Conditioning only on that realized design matrix controls the conditional observation information given the design; it does not by itself control the information carried by the adaptive design. Lemma G.2 therefore uses a coordinatewise adaptive KL argument to control $I(Y; H_T)$ directly.

Theorem G.4 (Linear bandit lower bound)**.** *For stochastic linear bandits with action set $\{a \in \mathbb{R}^d : \|a\|_2 \le 1\}$, unit Gaussian noise, and unit-ball parameter class $\Omega_{\text{lin}} = \{\omega \in \mathbb{R}^d : \|\omega\|_2 \le 1\}$,*

$$
\mathfrak{R}_T^\star \ge c\, d\sqrt{T}
$$

for a universal constant $c > 0$, whenever $T \ge d^2$.

Proof. Assume first that d is larger than a sufficiently large universal constant d_0 to be fixed below. Choose

$$
\Delta = c_1\sqrt{\frac{d}{T}},
$$

where $c_1 > 0$ is a sufficiently small universal constant. Since $T \ge d^2$, choosing $c_1 \le 1$ ensures

$$
\Delta\sqrt{d} = c_1\frac{d}{\sqrt{T}} \le c_1 \le 1,
$$

so the hypercube prior is supported on the unit parameter ball.

Let

$$r = \frac{\Delta\sqrt{d}}{4}.$$

Lemma G.1 gives the reference ghost-good probability bound

$$p_r \le \exp(-c_{\mathrm{gh}} d).$$

Lemma G.2 gives the adaptive action-index information bound

$$T\bar{C} = I(Y; H_T) = I(V; H_T) \le \Delta^2 T = c_1^2 d.$$

Choose c_1 so small that $c_1^2 \le c_{\mathrm{gh}}/4$, and then choose d_0 so large that $\log 2 \le c_{\mathrm{gh}} d/4$ for all $d \ge d_0$. Then, for all $d \ge d_0$,

$$T\bar{C} + \log 2 \le \frac{1}{2}\log\frac{1}{p_r}.$$

Theorem 6.1, applied to the action index $Y = \chi(\Omega) = A^\star(\Omega)$, gives a Bayes regret lower bound under the hypercube prior of at least

$$\frac{Tr}{2} = \frac{T\Delta\sqrt{d}}{8} = \frac{c_1}{8} d\sqrt{T}.$$

Since this prior is supported on Ω_{lin}, the minimax regret over Ω_{lin} is at least this Bayes risk.

It remains to handle the finitely many dimensions $1 \le d < d_0$. For these dimensions, use the two-point prior $\Omega = \sigma\delta e_1$, where $\sigma \sim \mathrm{Unif}(\{\pm 1\})$, e_1 is the first coordinate vector, and $\delta = (2\sqrt{T})^{-1}$. This prior is also supported on the unit parameter ball. Under $\sigma = +1$, the optimal action is e_1; under $\sigma = -1$, the optimal action is $-e_1$. Let $\mathbb{P}_+$ and $\mathbb{P}_-$ be the laws of the history under these two environments, and let $\mathbb{P}_\pm^{t-1}$ denote the law of the history before action t. For any algorithm,

$$\begin{aligned} \frac{1}{2}\mathbb{E}_+\left[\delta(1 - A_{t,1})\right] + \frac{1}{2}\mathbb{E}_-\left[\delta(1 + A_{t,1})\right] &= \delta\left(1 - \frac{1}{2}(\mathbb{E}_+ A_{t,1} - \mathbb{E}_- A_{t,1})\right) \\ &\ge \delta\left(1 - \mathrm{TV}(\mathbb{P}_+^{t-1}, \mathbb{P}_-^{t-1})\right). \end{aligned}$$

By Pinsker's inequality and the adaptive Gaussian KL chain rule,

$$\mathrm{TV}(\mathbb{P}_+^{t-1}, \mathbb{P}_-^{t-1}) \le \sqrt{\frac{1}{2} D_{\mathrm{KL}}(\mathbb{P}_+^{t-1} \| \mathbb{P}_-^{t-1})} \le \delta\sqrt{t-1} \le \frac{1}{2}.$$

Therefore each round has Bayes regret at least $\delta/2$, and the total Bayes regret is at least $T\delta/2 = \sqrt{T}/4$. Since $d < d_0$, this is at least $(4d_0)^{-1} d\sqrt{T}$. Reducing the universal constant c, if necessary, completes the proof for all d. □

Proposition G.5 (Linear-bandit upper bound and near matching)**.** *For stochastic linear bandits on the Euclidean unit ball with unit Gaussian noise and unit-ball parameter class, there are algorithms based on confidence ellipsoids, such as OFUL, with regret*

$$\sup_{\omega \in \Omega_{\mathrm{lin}}} \mathbb{E}_\omega^{\mathsf{Alg}} L_\omega(H_T) \le C d\sqrt{T}\,\mathrm{polylog}(T, d)$$

for a universal constant C and all $T \ge d$. Consequently Theorem G.4 is matched in the polynomial order $d\sqrt{T}$, with only logarithmic factors belonging to the particular tractable upper bound. On the hard hypercube prior, the index support has $M = 2^d$, the initial code length on that same hypercube subfamily is $L_0 = d\log 2$, and Lemma G.1 gives $E_r \ge c_{\mathrm{gh}} d$ at radius $r = \Delta\sqrt{d}/4 \asymp d/\sqrt{T}$. Thus the Bellman–Fano hard-subfamily entropy gap is constant, while OFUL supplies minimax-rate matching up to logarithmic factors. This does not by itself verify the upper-at-lower-entropy calibration of the exact log-potential Bellman program for the full unit-ball class.

Proof. The standard self-normalized analysis for stochastic linear bandits gives the upper bound (Abbasi-Yadkori et al., 2011; Lattimore and Szepesvári, 2020). The lower bound and entropy calculation are exactly the proof of Theorem G.4: the hard index is $Y = V/\sqrt{d}$ with $V \in \{\pm 1\}^d$, so the hypercube-subfamily code length is $L_0 = d \log 2$, while Lemma G.1 yields $p_r \le e^{-c_{\mathrm{gh}} d}$. □

This example is the high-dimensional analogue of the finite-arm calculation. The hypercube prior is not a model-identification target; it is a finite action-index packing for the optimal direction. The lower proof shows that ghost entropy and information capacity meet at the scale $d\sqrt{T}$. Standard tractable upper algorithms achieve the same scale up to logarithmic terms; proving that the exact log-potential Bellman upper value closes at this entropy scale would give the corresponding formal sandwich theorem. An interesting open question is to consider more irregular action sets in linear bandits, such as the disjoint-union example of Maiti et al. (2026), and to identify new rates that require a more algorithmic lower-bound approach, beyond the classical Assouad, Fano, and Le Cam methods (Chen et al., 2024; Xu, 2026).

H Proofs of results stated in the main text

The remaining proofs are collected here.

H.1 Proof of Proposition 2.2

Proof. The Bellman-operator identity follows directly from the state-measurability of the action set, comparison set, one-step cost, transition law, and continuation value. Histories in the same fiber of S_t induce the same local optimization problem. The information inequality is the data-processing inequality applied to the measurable map $S_t = \phi_t(H_{t-1})$. The equality condition is the standard equality case for conditional mutual information, $I(Y; H_{t-1} \mid S_t) = 0$. □

H.2 Proof of Lemma 3.2

Proof. Since $C_\mu \ge 0$, condition (10) implies $\underline{E}_{\mu,r} \ge 2\log 2$. Because $\underline{E}_{\mu,r} \le E^\star_{\mu,r} = -\log p^\star_{\mu,r}$, one has $p^\star_{\mu,r} \le e^{-\underline{E}_{\mu,r}} \le 1/4$. The converse statement is only the exact entropy bound $E^\star_{\mu,r} \ge \log(1/\varrho)$; it does not control C_μ. □

H.3 Proof of Proposition 3.3

Proof. The uniform representative prior gives $q_1(y) = 1/M$, hence the upper code length is $\log M$. If μ_h is Bellman–Fano admissible, Lemma 3.2 gives $E^\star_{\mu_h,r} \ge 2\log 2$, proving (11). Under the multiplicity condition, every algorithm satisfies

$$p^{\mathsf{Alg}}_{\mu_h,r} = \mathbb{E}_{H'_T \sim \bar{\mathbb{P}}^{\mathsf{Alg}}_{\mu_h}} \frac{|G_r(H'_T)|}{M} \le \frac{m_r}{M}.$$

Taking the supremum over Alg gives $p^\star_{\mu_h,r} \le m_r/M$, and hence $E^\star_{\mu_h,r} \ge \log(M/m_r)$. This proves (12). The final statement follows from $E^\star_{\mu_h,r} \ge \log(1/\varrho)$. □

H.4 Proof of Lemma 3.5

Proof. Set $\mathsf{U}(x) = 1$ for $x < L$ and $\mathsf{U}(x) = \max\{1, B\}$ for $x \ge L$. Then U is nondecreasing, $\mathsf{U}(E) = 1$, and $\mathsf{U}(L) = \max\{1, B\} \ge B$. Thus a logarithmic entropy gap, even $L/E = O(\log M)$, does not by itself imply any controlled regret gap. Assumption 3.4, or an equivalent fixed-point/rate condition, is the additional structure that rules out such jumps. □

H.5 Proof of Proposition 3.7

Proof. Condition (i) is exactly (14). Condition (ii) implies (i) because $\mathsf{U}_T(E^\star_{\mu,r})$ is the infimum over $\gamma > 0$ of the left side in (13). Condition (iii) is a restatement of (i) with the constant displayed after dividing by T. □

H.6 Proof of Theorem 3.8

Proof. The upper inequality is Theorem 5.2, optimized over γ, with the uniform coordinate code length bounded by L_0 and the remaining errors included in Δ_T^γ. The lower inequality is Theorem 6.2 applied to the Bellman–Fano admissible prior, followed by Yao's principle because μ is supported on Ω_0. For the final comparison, set

$$a = \max\left\{1, \frac{L_0}{E^\star_{\mu,r}}\right\}.$$

Since U_T is nondecreasing and $aE^\star_{\mu,r} \geq L_0$, Assumption 3.4 and (14) give

$$\mathsf{U}_T(L_0) \leq \mathsf{U}_T(aE^\star_{\mu,r}) \leq C_{\mathrm{gr}} a^\beta \mathsf{U}_T(E^\star_{\mu,r}) \leq C_{\mathrm{gr}} C_{\mathrm{base}} a^\beta Tr.$$

Combining this with $\mathfrak{R}^\star_T(\Omega_0) \geq Tr/2$ gives the regret-ratio bound. The finite-index statements are Proposition 3.3. □

H.7 Proof of Proposition 4.2

Proof. Under the stated assumption on O_0, the mutual-information chain rule gives

$$I_\mu(Y; H_T) = \sum_{t=1}^{T} I_\mu(Y; A_t, O_t \mid H_{t-1}).$$

Given H_{t-1}, the action A_t is sampled by the algorithm using only randomization independent of the environment and is conditionally independent of Y. Therefore

$$I_\mu(Y; A_t, O_t \mid H_{t-1}) = I_\mu(Y; O_t \mid H_{t-1}, A_t).$$

Conditioning further on $A_t = a$, the law of O_t given $Y = y$ is $P^y_{S_t,p_t,a}$, while the posterior predictive law is $P_{S_t,p_t,a}$. Hence

$$I_\mu(Y; O_t \mid H_{t-1}, A_t = a) = \int D_{\mathrm{KL}}(P^y_{S_t,p_t,a} \| P_{S_t,p_t,a}) q_{S_t}(dy).$$

Averaging over $a \sim p_t(\cdot \mid H_{t-1})$ and over the marginal law of the history under the Bayesian mixture law proves the result. Writing this marginal law as the reference history law gives the displayed expression. □

H.8 Proof of Lemma 4.3

Proof. For fixed s, p, a, write the joint mixture of (Y, O) under ν as $w_{y,o}$ and the observation mixture as $w_o = \sum_y w_{y,o}$. In the dominated case, the sums below are interpreted as integrals with respect to the fixed dominating measures. The information and reference term can be written as

$$I_\nu(Y; O \mid s, p, a) + D_{\mathrm{KL}}(q_\nu \| q) = \sum_{y,o} w_{y,o} \log \frac{w_{y,o}}{w_o q(y)}.$$

Differentiating with respect to the mass at (ω, y) gives

$$\mathbb{E}_{O\sim P_{\omega,t}(\cdot|s,p,a)} \log \frac{q^+_\nu(y \mid s,p,a,O)}{q(y)},$$

because the $+1$ terms from the numerator and denominator cancel. Averaging over $a \sim p$ and adding the derivative of the linear loss term proves (24). Pairing the gradient with $\delta_{(\omega,y)} - \nu$ cancels the posterior-averaged terms and gives (25). □

H.9 Proof of Theorem 4.4

Proof. Condition on H_{t-1}. By the update rule (26),

$$\mathbb{E}\left[\log \frac{q_{t+1}(y^\star)}{q_t(y^\star)}\middle| H_{t-1}\right] = \mathbb{E}_{a\sim p_t,\, O\sim P_{\omega^\star,t}(\cdot|S_t,p_t,a)} \log \frac{q^+_{\nu_t}(y^\star \mid S_t,p_t,a,O)}{q_t(y^\star)}.$$

Combining this equality with (25) gives the one-step identity between expected loss, bracket, and log-ratio increment. Summing over t telescopes the logarithm. □

H.10 Proof of Lemma 4.5

Proof. The derivative and the first two specializations are exactly the environment-index specialization of Lemma 4.3. For the square-root claim, fix $\bar{a}$ and write

$$Z_{\bar{a}}(\bar{o}) := \int \sqrt{p_{\omega',\bar{a}}(\bar{o})}\,\rho(d\omega').$$

Then

$$\log \frac{d\mu^{1/2}_{\bar{a},\bar{o}}}{d\rho}(\omega^\star) = \frac{1}{2}\log p_{\omega^\star,\bar{a}}(\bar{o}) - \log Z_{\bar{a}}(\bar{o}).$$

Under $\bar{o} \sim P_{\omega^\star,s,p,\bar{a}}$, Jensen's inequality gives

$$\begin{aligned}
\mathbb{E}\log \frac{d\mu^{1/2}_{\bar{a},\bar{o}}}{d\rho}(\omega^\star) &= -\mathbb{E}\log \frac{Z_{\bar{a}}(\bar{o})}{\sqrt{p_{\omega^\star,\bar{a}}(\bar{o})}} \\
&\geq -\log \int \sqrt{p_{\omega^\star,\bar{a}}(o)}Z_{\bar{a}}(o)d\nu_{s,p,\bar{a}}(o) \\
&= -\log\left(1 - \mathbb{E}_{\omega\sim\rho} h^2(P_{\omega^\star,s,p,\bar{a}}, P_{\omega,s,p,\bar{a}})\right) \\
&\geq \mathbb{E}_{\omega\sim\rho} h^2(P_{\omega^\star,s,p,\bar{a}}, P_{\omega,s,p,\bar{a}}).
\end{aligned}$$

Averaging over $\bar{a} \sim p$, substituting into (34), and dropping the nonnegative KL term proves (38). □

H.11 Proof of Theorem 5.1

Proof. Taking the conditional expectation of $\gamma\log(1/q_{t+1}(y^\star)) - \gamma\log(1/q_t(y^\star))$ under $A \sim p$ and $O \sim P_{\omega^\star,t}(\cdot \mid s,p,A)$, with $q_{t+1} = q^+_\nu(\cdot \mid s,p,A,O)$, gives $-\gamma\mathsf{G}_\chi(s,p,\nu;\omega^\star)$. Substituting the bracket inequality into the telescoping identity (42) proves (43).

□

H.12 Proof of Theorem 5.2

Proof. By sure calibration, the fixed truth is one of the environments in the robust supremum at every reached state. Therefore the displayed approximate minimization implies almost surely

$$\ell_{\omega^\star}(S_t, p_{u_t}) + \mathbb{E}_{\omega^\star}[W^\gamma_{t+1}(S^+) \mid S_t, u_t] - W^\gamma_t(S_t) - \gamma \mathsf{G}_\chi(S_t, u_t; \omega^\star) \le \varepsilon_t.$$

Apply the log-potential telescope (42) with $V_t = W^\gamma_t$. Since $W^\gamma_{T+1} = 0$ and the terminal index log loss is nonnegative for a finite or countable index, the result follows. $\square$

H.13 Proof of Theorem 5.4

Proof. Rearrange the displayed inequality and telescope U^γ_t. If O_0 is deterministic or independent of Y, Proposition 4.2 identifies $\mathbb{E}\sum_t \mathcal{I}_\chi(S_t, p_t)$ with $I_\mu(Y; H_T)$. If O_0 is informative, the sum is instead $I_\mu(Y; H_T \mid O_0) \le I_\mu(Y; H_T)$. In either case $I_\mu(Y; H_T) \le H_\mu(Y)$ gives the displayed bound. The equivalence with (52) follows from the posterior entropy-drop identity. $\square$

H.14 Proof of Lemma 5.5

Proof. The optimism/calibration hypothesis (54) and Young's inequality give

$$c_{\mathrm{opt}} \beta_t \sigma_t(a_t) \le \frac{c^2_{\mathrm{opt}} \beta^2_t}{4\gamma} + \gamma \sigma^2_t(a_t).$$

This proves (55). Subtracting $\gamma J_\chi(S_t, \delta_{a_t}; \omega^\star)$ from both sides and using the definition of $B_{\chi,\gamma}$ proves (56). The final sentence is the result of summing the displayed inequality over time. $\square$

H.15 Proof of Lemma 5.6

Proof. The AIR gradient calculation in Lemma 4.3 identifies the fixed-truth bracket as the first-order value of $\mathfrak{A}_{q_t,\gamma}$ in the direction $\delta_{(\omega^\star, y^\star)} - \bar{\nu}_t$. Concavity and optimality of $\bar{\nu}_t$ make that first-order correction nonpositive for every admissible comparison direction. Hence the bracket is no larger than $\mathfrak{A}_{q_t,\gamma}(s, p_t, \bar{\nu}_t)$, which is at most the robust value and therefore at most ε_t. The regret bound is the fixed-truth log-potential telescope. $\square$

H.16 Proof of Theorem 6.1

Proof. Consider the true joint law of (Y, H_T) and the ghost law of (Y, H'_T). Both have the same marginal law of Y, while Y and H'_T are independent under the ghost law. Hence

$$D_{\mathrm{KL}}\big(\mathcal{L}(Y, H_T) \| \mathcal{L}(Y, H'_T)\big) = I_\mu(Y; H_T).$$

Data processing applied to the indicator of the event $\{\bar{L}_\chi(Y, \cdot) \le r\}$ gives

$$\mathrm{kl}(q^{\mathsf{Alg}}_r, p^{\mathsf{Alg}}_r) \le I_\mu(Y; H_T).$$

Proposition 4.2 identifies the right-hand side with $C^{\mathsf{Alg}}_\chi(\mu) = T\bar{C}^{\mathsf{Alg}}_\chi(\mu)$, proving (64). The definition of ψ gives $q^{\mathsf{Alg}}_r \le \psi(p^{\mathsf{Alg}}_r, T\bar{C}^{\mathsf{Alg}}_\chi)$. By nonnegativity of the cumulative indexed loss and by (4),

$$\begin{aligned}
\mathbb{E}_{\Omega, H_T} L_\Omega(H_T) &= \mathbb{E}_{Y, H_T} L_\chi(Y, H_T) \\
&\ge Tr\, \mathbb{P}\{\bar{L}_\chi(Y, H_T) > r\} \\
&= Tr(1 - q^{\mathsf{Alg}}_r),
\end{aligned}$$

which proves (65). If (66) holds, then $q_r^{\mathsf{Alg}} \le 1/2$, giving (67). Finally,

$$\mathrm{kl}\left(\frac{1}{2}, p\right) = \frac{1}{2}\log\frac{1}{4p(1-p)} \ge \frac{1}{2}\log\frac{1}{p} - \log 2,$$

so (68) implies (66). □

H.17 Proof of Theorem 6.2

Proof. For every algorithm, (69) gives $T\bar{C}_\chi^{\mathsf{Alg}}(\mu) = C_\chi^{\mathsf{Alg}}(\mu) \le \overline{C}_1(s_1)$, and (70) gives $p_r^{\mathsf{Alg}}(\mu,\chi) \le e^{-\underline{\mathcal{E}}_1^r(\mu,\chi)}$. Substitute these two inequalities into Theorem 6.1, using the monotonicity of ψ in both arguments. The minimax statement follows from Yao's principle. □

H.18 Proof of Theorem 7.2

Proof. Let $x^\star \in \arg\max_{x\in\mathcal{X}} f^\star(x)$. On $\mathcal{E}_{\mathrm{GP}}$, calibration at $x^\star$ and the optimistic choice of X_t give

$$f^\star(x^\star) \le m_t(x^\star) + w_t(x^\star) \le m_t(X_t) + w_t(X_t).$$

Calibration at X_t gives $m_t(X_t) \le f^\star(X_t) + w_t(X_t)$. Hence

$$f^\star(x^\star) - f^\star(X_t) \le 2w_t(X_t) = 2\beta_t s_t(X_t) \le 2\beta_T s_t(X_t).$$

Summing and applying Cauchy–Schwarz yields

$$\mathrm{Reg}_T(f^\star) \le 2\beta_T\sqrt{T\sum_{t=1}^{T} s_t^2(X_t)}.$$

For $u \in [0, \kappa^2/\lambda]$, monotonicity of $u/\log(1+u)$ on a bounded interval gives

$$\lambda u \le \frac{\kappa^2}{\log(1+\kappa^2/\lambda)}\log(1+u).$$

Applying this with $u = s_t^2(X_t)/\lambda$ and using (81) proves (88). The expectation bound adds the trivial regret bound T on $\mathcal{E}_{\mathrm{GP}}^c$ and uses Jensen's inequality. □

H.19 Proof of Theorem 7.1

Proof. Let $\Theta = \Theta_{\mathcal{R}} \subset [-L, L]^{\mathcal{R}}$ be the compact class of retained reward vectors. To avoid a discontinuous tie-breaking graph in the saddle argument, use the closed optimal relation

$$\widetilde{\Theta} := \{(\theta, y) : \theta \in \Theta,\ y \in \arg\max_{x\in\mathcal{R}} \theta_x\}.$$

This is a compact subset of $\Theta \times \mathcal{R}$. Let $\mathcal{Y}_\Theta$ be the set of actions that occur as the second coordinate of some pair in $\widetilde{\Theta}$, and put $K_0 = |\mathcal{Y}_\Theta| \le K$. The fixed tie-breaking rule used for the true index selects an element of this relation. Candidate pair beliefs range over the weakly compact convex set $\Delta(\widetilde{\Theta})$.

Fix $q \in \operatorname{int}\Delta(\mathcal{Y}_\Theta)$ and $\gamma > 0$. For $\nu \in \Delta(\widetilde{\Theta})$ and $p \in \Delta(\mathcal{R})$, write

$$\mathfrak{A}_{q,\gamma}(p,\nu) = \Delta_\nu(p) - \gamma\mathcal{I}_\nu(Y; Z_A \mid A) - \gamma D_{\mathrm{KL}}(\nu_Y \| q),$$

where $A \sim p$ is independent of $(\theta, Y) \sim \nu$, $Z_A = \theta_A + \xi$ with $\xi \sim N(0, \lambda)$, and

$$\Delta_\nu(p) := \mathbb{E}_{\nu,p}[\theta_Y - \theta_A].$$

The loss is affine in each of p and ν separately. Moreover,

$$\mathcal{I}_\nu(Y; Z_A \mid A) + D_{\mathrm{KL}}(\nu_Y \| q) = \sum_{a \in \mathcal{R}} p(a) D_{\mathrm{KL}}\big(P^\nu_{Y,Z_a} \,\big\|\, q \otimes P^\nu_{Z_a}\big)\,.$$

The map $\theta \mapsto N(\theta_a, \lambda)$ is weakly continuous, and all its means range over a compact interval. Joint convexity and lower semicontinuity of relative entropy therefore make $\mathfrak{A}_{q,\gamma}$ concave and upper semicontinuous in ν and affine and continuous in p. Both feasible sets are compact, so Sion's theorem gives

$$\inf_p \sup_\nu \mathfrak{A}_{q,\gamma}(p, \nu) = \sup_\nu \inf_p \mathfrak{A}_{q,\gamma}(p, \nu). \tag{119}$$

We bound the right-hand side. Fix ν, let $r = \nu_Y$, take $p = r$ on $\mathcal{Y}_\Theta$ and zero on the remaining actions, and define

$$d_i := \mathbb{E}_\nu[\theta_i \mid Y = i] - \mathbb{E}_\nu \theta_i, \qquad i \in \mathcal{Y}_\Theta,$$

with $d_i = 0$ when $r(i) = 0$. The marginal law of Z_i is $(\lambda + L^2)$-sub-Gaussian about its mean: the Gaussian noise contributes variance proxy λ, while Hoeffding's lemma contributes L^2 for the random mean in $[-L, L]$. The entropy variational formula consequently gives

$$D_{\mathrm{KL}}(P^\nu_{Z_i|Y=i} \| P^\nu_{Z_i}) \geq \frac{d_i^2}{2(\lambda + L^2)}.$$

It follows that

$$\mathcal{I}_\nu(Y; Z_A \mid A, p = r) \geq \frac{1}{2(\lambda + L^2)} \sum_{i \in \mathcal{Y}_\Theta} r(i)^2 d_i^2,$$

$$\Delta_\nu(r) = \sum_{i \in \mathcal{Y}_\Theta} r(i) d_i.$$

By Cauchy–Schwarz,

$$\Delta_\nu(r)^2 \leq 2(\lambda + L^2) K_0 \, \mathcal{I}_\nu(Y; Z_A \mid A, p = r). \tag{120}$$

Dropping the nonpositive term $-\gamma D_{\mathrm{KL}}(r \| q)$ and maximizing $\sqrt{2(\lambda + L^2) K_0 z} - \gamma z$ over $z \geq 0$ now yields

$$\inf_p \sup_\nu \mathfrak{A}_{q,\gamma}(p, \nu) \leq \frac{(\lambda + L^2) K_0}{2\gamma} \leq \frac{(\lambda + L^2) K}{2\gamma} =: c_\gamma. \tag{121}$$

Take a saddle pair $(p_q, \bar{\nu}_q)$ in (119) and update q by the posterior optimal-action marginal generated by $\bar{\nu}_q$. The identity

$$-\mathcal{I}_\nu(Y; Z_A \mid A) - D_{\mathrm{KL}}(\nu_Y \| q) = H_\nu(Y \mid A, Z_A) + \mathbb{E}_\nu \log q(Y)$$

shows that a maximizing belief assigns positive mass to every attainable index. Indeed, suppose $\bar{\nu}_{q,Y}(y) = 0$, choose $(\theta^y, y) \in \widetilde{\Theta}$, and put

$$\nu_\epsilon = (1 - \epsilon)\bar{\nu}_q + \epsilon \delta_{(\theta^y, y)}.$$

Let B indicate the new mixture component and let $h(\epsilon)$ be binary entropy. Since the new label did not occur under $\bar{\nu}_q$,

$$H_{\nu_\epsilon}(Y \mid A, Z_A) = (1 - \epsilon) H_{\bar{\nu}_q}(Y \mid A, Z_A) + H(B \mid A, Z_A),$$

$$
\begin{aligned}
H(B \mid A, Z_A) &\geq h(\epsilon) - \epsilon \sup_{a \in \mathcal{R}} D_{\mathrm{KL}}\Big(N(\theta_a^y, \lambda) \,\Big\|\, P_{Z_a}^{\bar{\nu}_q}\Big) \\
&\geq h(\epsilon) - \frac{2L^2}{\lambda}\epsilon.
\end{aligned}
$$

The last step uses convexity in the second argument of KL and the uniform pairwise Gaussian bound. Thus the conditional-entropy contribution increases by $\epsilon \log(1/\epsilon) - O(\epsilon)$, while the loss and $\mathbb{E} \log q(Y)$ change by only $O(\epsilon)$. This contradicts maximality. Hence the generated posterior coordinate remains positive for every feasible fixed-truth index after every observation.

Because $\lambda > 0$, the predictive mixture densities are strictly positive. Bounded Gaussian means and the preceding KL bound also make the relevant log likelihood ratios integrable. The one-sided Gateaux derivative toward every $\delta_{(\theta,y)} - \bar{\nu}_q$ therefore exists, so Lemma 4.3 applies to this dominated continuous-belief problem.

At the maximizing belief, the directional derivative of the concave AIR functional toward any $(\theta, y) \in \widetilde{\Theta}$ is nonpositive. Lemma 4.3 and (121) therefore give

$$
\Delta_\theta(p_q) - \gamma \mathbb{E}_\theta \log \frac{q^+(y \mid A, Z_A)}{q(y)} \leq c_\gamma \qquad \text{for every } (\theta, y) \in \widetilde{\Theta}. \tag{122}
$$

Thus $\overline{W}_t = (T - t + 1)c_\gamma$ is a supersolution of the finite-marginal action-index Bellman recursion: at every state, insert the robust AIR saddle control in the Bellman bracket. Take the comparison set to be the full class Θ at every state and restrict the admissible AIR updates to those that remain positive on every attainable index. Backward induction gives $W_t^\gamma \leq \overline{W}_t$. A measurable approximate selector of the exact W recursion and Theorem 5.2 therefore give the bound below, up to arbitrarily small optimization error. The robust saddle policy itself satisfies the same estimate: with continuation $V_t = \overline{W}_t$, inequality (122) makes its Bellman bracket nonpositive, so Theorem 5.1 applies directly. In either case, the uniform initial law on $\mathcal{Y}_\Theta$ gives

$$
\sup_{\theta \in \Theta} \mathbb{E}_\theta \operatorname{Reg}_T^{\mathcal{R}} \leq T c_\gamma + \gamma \log K_0 \leq \frac{(\lambda + L^2)KT}{2\gamma} + \gamma \log K.
$$

Standard measurable-selection results apply to the compact saddle problem; equivalently, measurable approximate saddles may be used and their arbitrarily small optimization errors added to the display.

Finally, for a policy confined to $\mathcal{R}$,

$$
\sup_{x \in \mathcal{X}_{\mathrm{all}}} f(x) - f(X_t) \leq \varepsilon_{\mathcal{R}} + \max_{x \in \mathcal{R}} f(x) - f(X_t).
$$

Summing proves (84). Optimizing in γ proves (85). □

H.20 Proof of Theorem 7.3

Proof. Choose increasing integers T_m recursively, starting with T_1 sufficiently large. Put

$$
a_m = T_m^{-\alpha}, \qquad N_m = \lceil 4T_m \rceil, \qquad M_{m-1} = \sum_{\ell < m} N_\ell,
$$

and choose T_m sufficiently large that

$$
(M_{m-1} + 2)\log(1 + T_m) \leq a_m^2 T_m = T_m^{1-2\alpha}. \tag{123}
$$

This is possible because $1-2\alpha>0$. Let

$$\mathcal{X}=\{x_0,x_h\}\cup\bigcup_{m\ge 1}B_m,\qquad B_m=\{c_{m,1},\dots,c_{m,N_m}\},$$

and define a diagonal kernel by

$$k(x_h,x_h)=1,\qquad k(c_{m,j},c_{m,j})=a_m^2,$$

with every other entry zero. Let $D=\bigcup_{m\ge 1}B_m$ have the discrete topology and endow $D\cup\{x_0\}$ with its one-point compactification: a neighborhood base at x_0 is

$$U_F=\{x_0\}\cup(D\setminus F),\qquad F\subset D \text{ finite}.$$

Adjoin x_h as an isolated point. The resulting $\mathcal{X}$ is compact, metrizable, and countable, with x_0 as its only accumulation point. Since T_m is strictly increasing, $a_m\downarrow 0$. Every point other than x_0 is isolated. If $z_r\to x_0$, then the block index of z_r tends to infinity and hence $k(z_r,z_r)\to 0$. All off-diagonal values are zero, so the same observation verifies continuity at (x_0,x_0) and at every (x_0,x) and (x,x_0). Thus k is jointly continuous on $\mathcal{X}\times\mathcal{X}$ and bounded by one. Its unit RKHS ball is

$$\mathcal{F}=\left\{f: f(x_0)=0,\quad f(x_h)^2+\sum_{m\ge 1}a_m^{-2}\sum_{j=1}^{N_m}f(c_{m,j})^2\le 1\right\}. \tag{124}$$

Fix the horizon T_m and put

$$\mathcal{R}_m:=\{x_0,x_h\}\cup\bigcup_{\ell<m}B_\ell,\qquad K_m:=|\mathcal{R}_m|=M_{m-1}+2.$$

The one-hot alternatives supported on B_m and the coupling argument from Theorem E.1 give the lower bound $3a_mT_m/4$. For the upper bound, run a minimax finite-armed sub-Gaussian policy on $\mathcal{R}_m$. Its regret relative to the best retained action is at most $C\sqrt{K_mT_m}$. Every point in the ignored blocks B_ℓ, $\ell\ge m$, has $|f(x)|\le a_\ell\le a_m$, while the retained anchor has value zero. By (123), the full regret is therefore at most

$$C\sqrt{K_mT_m}+a_mT_m\le C'a_mT_m.$$

This proves (92).

We next define one AIR Bellman policy for the entire time axis. Set $T_0=0$ and $L_m=T_m-T_{m-1}$. During the predetermined epoch $(T_{m-1},T_m]$, the policy discards data from earlier epochs, restricts its actions to $\mathcal{R}_m$, initializes the optimal-action reference uniformly, and runs the L_m-round finite-marginal action-index AIR Bellman selector from Theorem 7.1 with

$$\gamma_m=\sqrt{\frac{K_mL_m}{\log K_m}}.$$

The control invoked here is the finite-marginal AIR/AMS saddle of Xu and Zeevi (2025). Its connection to Lattimore and György (2021) is the EBO optimization principle described after Theorem 7.1; the proof does not use EBO's functional-estimator implementation. The epoch schedule and all its controls are fixed in advance, so this is one nonanticipating policy and does not take the evaluation horizon as input. The projection of the unit RKHS ball onto $\mathcal{R}_m$ is a compact ellipsoid contained in $[-1,1]^{K_m}$, and the approximation error of $\mathcal{R}_m$ is at most a_m. Therefore the regret accumulated during epoch m is at most

$$a_m L_m + 2\sqrt{K_m L_m \log K_m} \le 3 a_m T_m,$$

where the last inequality uses

$$K_m \log K_m \le K_m \log(1+T_m) \le a_m^2 T_m.$$

Here the first inequality follows because the scale condition itself implies $K_m \le T_m$ for all sufficiently large m. The RKHS envelope bounds per-round regret by two. For $m \ge 2$, moreover, $K_m \ge N_{m-1} \ge 4T_{m-1}$, so the regret before the current epoch is at most

$$2T_{m-1} \le \frac{K_m}{2} \le \frac{a_m^2 T_m}{2\log(1+T_m)} \le a_m T_m.$$

Consequently, the single epochwise AIR Bellman policy has worst-case regret at most $4a_m T_m = O(T_m^{1-\alpha})$ at time T_m. This proves (93).

Let $\overline{\Gamma}_s$ denote the maximal information gain of this fixed kernel. For $s \le T_m$, selecting distinct points of B_m and using $\log(1+u) \ge u/2$ gives

$$\frac{s a_m^2}{4} \le \overline{\Gamma}_s \le \frac{M_{m-1}+1}{2}\log(1+s) + \frac{a_m^2 s}{2} \le a_m^2 T_m. \tag{125}$$

Indeed, the hub and each coordinate in the earlier blocks contribute at most $\frac{1}{2}\log(1+s)$, while all coordinates in B_ℓ, $\ell \ge m$, contribute in total at most $a_m^2 s/2$. Use the anytime multiplier

$$\overline{\beta}_t = 1 + \sqrt{2\{\overline{\Gamma}_{t-1} + 2\log(t \vee 2)\}}.$$

At round t, the fixed-kernel GP–UCB rule selects a maximizer of $m_t(x) + \overline{\beta}_t s_t(x)$. This is one policy, independent of the evaluation horizon. After every finite history, $m_t(x)$ and $s_t^2(x)$ are finite algebraic combinations of the continuous kernel sections $k(x, X_r)$ and $k(x,x)$. Hence $m_t + \overline{\beta}_t s_t$ is continuous on compact $\mathcal{X}$ and attains its maximum. The lower bound in (125) makes the score of an unqueried point of B_m at least $a_m^2\sqrt{(t-1)/2}$, which diverges uniformly for $t \ge T_m/2$. For $t \le T_m$, the upper bound and (123) give $\overline{\beta}_t \le C a_m \sqrt{T_m}$.

Finally take the fixed hub truth $f^\dagger(x_h) = 1/4$ and zero elsewhere. Repeat the Gaussian-noise-stack event and last-half argument in the proof of Theorem E.1. For every $t \le T_m$, an unqueried point of B_m remains available because $N_m \ge 4T_m$. Its score is $a_m \overline{\beta}_t > 0$, whereas the anchor x_0 has score zero; consequently, x_0 is never selected. Hence, if fewer than $T_m/4$ nonanchor, nonhub pulls occur, the hub has been sampled at least $T_m/4 - 1$ times before every round in the last half. On an event of probability at least $1 - T_m^{-1}$ its acquisition score is at most $1/4 + O(a_m) + O(\sqrt{\log T_m / T_m}) < 1/2$. An unqueried point of B_m has score larger than one for all sufficiently large m. Thus every round in the last half must select a cloud action, a contradiction. At least $T_m/4$ pulls therefore incur regret $1/4$, proving (94) and the ratio claim for (90).

For the original schedule (91), the same argument applies with only two scale estimates changed. The distinct-set information gain obeys the same bounds needed here:

$$\frac{s a_m^2}{4} \le \Gamma_s^{\mathrm{SKKS}}(k) \le \frac{M_{m-1}+1}{2}\log 2 + \frac{a_m^2 s}{2} \le a_m^2 T_m, \qquad s \le T_m.$$

The lower bound selects s distinct points of B_m; the upper bound follows because a set uses every earlier coordinate at most once and all later diagonal values are at most a_m^2. Hence, uniformly for $T_m/2 \le t \le T_m$,

$$a_m b_t^{\mathrm{SKKS}} \ge c\, a_m^2 \sqrt{T_m}\, \log^{3/2} T_m \longrightarrow \infty,$$

because $\alpha < 1/4$. Uniformly for $t \le T_m$, it also gives

$$b_t^{\mathrm{SKKS}} \le C\{1 + a_m \sqrt{T_m}\, \log^{3/2} T_m\}.$$

After at least $T_m/4 - 1$ hub pulls, the corresponding hub bonus is

$$O\Big(T_m^{-1/2} + a_m \log^{3/2} T_m\Big) = o(1),$$

while an unqueried point of B_m has score larger than one. The same noise-stack event and contradiction force at least $T_m/4$ cloud pulls. This proves (94) and the ratio claim for (91) as well. □

H.21 Proof of Corollary 7.7

Proof. Equation (95) is Wang and Zhang (2026, Theorem 1) after translating their squared exploration coefficient into b_t. The minimax lower bound is due to Scarlett et al. (2017); matching upper bounds under the standard compact-domain regularity assumptions are achieved by localized kernel-bandit algorithms such as GP–ThreDS (Salgia et al., 2021). For fixed kernel parameters, the auxiliary Hölder condition in the latter result follows uniformly from the reproducing property and the Matérn kernel increment bound, while its range condition follows from the uniform RKHS envelope. Its high-probability guarantee gives the displayed expectation rate by taking the failure probability polynomially small and bounding regret on failure by that envelope. Dividing (95) by (96) proves the polynomial ratio. □

Acknowledgements

The author used ChatGPT and Codex as research and editorial aids to generate technical suggestions, check derivations for internal consistency, implement simulations, and improve the clarity of the exposition. All mathematical ideas, claims, proofs, and final wording were independently proposed, examined, and revised by the author, who assumes full responsibility for the final form and validity of the results.

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